

Sobolev Extremal Polynomials and Applications

by

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A dissertation submitted in partial fulfillment of the requirements for
the degree of Doctor of Philosophy in

Mathematical Engineering

Universidad Carlos III de Madrid

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Leganés, March 2020

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a mi abuela América y mi prima Mabel

Agradecimientos

Quiero agradecer en primer lugar a Héctor Pijeira Cabrera por dirigir esta investigación, por los conocimientos compartidos, sus consejos, su ayuda incondicional y sobre todo por acogerme -junto a Marcela- como un hijo.

A Ignacio Pérez Yzquierdo, mi profesor universitario más admirado, primer tutor y la primera persona que me impulsó en el mundo de la investigación.

Mi más sincero agradecimiento al departamento de Matemáticas de la Universidad Carlos III de Madrid, especialmente a los profesores Guillermo López Lagomasino y Francisco Marcellán Español, con los que tuve el placer de trabajar y son también responsables de los resultados de esta investigación.

Al programa de ayudas del Departamento de Matemáticas y al del Ministerio de Economía Industria y Competitividad por financiarme durante mis dos años de master y estos tres de doctorado respectivamente.

Quiero agradecer además al Grupo de Análisis Matemático Aplicado (GAMA) por hacerme parte de este excelente equipo.

A los profesores Yamilet Quintana y Wilfredo Urbina por la colaboración, la revisión y sugerencias al manuscrito.

A los profesores Walter Van Assche y Arno Kuijlaars por los conocimientos compartidos y su cálida acogida durante mis tres meses de estancia en la Universidad Católica de Lovaina (UK Leuven).

A mi madre y a Laura por su incondicional e ilimitado apoyo. A mi padre por su eterna complicidad. Al resto de la familia: Anita, Amalia, José Julián, Ana, Ariel, Migue, Mayito, Cristy, Elisa y Mayi porque la distancia no ha mellado su afecto y cariño en estos 5 años.

A todos mi amigos de dentro y fuera de la Matemática, especialmente Vity, Marcos y Javier quienes me ayudaron con las correcciones del manuscrito.

Abel Díaz González
Leganés, November 27, 2019



Published and submitted content

The content of this doctoral thesis is based on the following articles:

1. A. Díaz-González, G. López Lagomasino, H. Pijeira. Asymptotic zero distribution for a class of extremal polynomials, accepted in Bulletin of Mathematical Sciences. <https://doi.org/10.1142/S166436071950019X>.

I am the author of this article and it is included in Chapter 2 of this work.

2. A. Díaz-González, I. Pérez-Yzquierdo, H. Pijeira Cabrera. Rational Approximation and Sobolev-type Orthogonality, submitted to Journal of Approximation Theory. arXiv: 1907.12243.

I am the author of this article and it is included in Chapter 3 of this work.

3. A. Díaz-González, F. Marcellán Español, H. Pijeira-Cabrera and W. Urbina. Discrete-Continuous Jacobi-Sobolev Spaces and Fourier Series, submitted to Bulletin of the Malaysian Mathematical Sciences Society.

I am the author of this article and it is included in Chapter 4 of this work.

Resumen y aportaciones

Los polinomios ortogonales de Sobolev han sido estudiados en profundidad a lo largo de las últimas tres décadas como extensión natural de los polinomios ortogonales estándar. Existen tres grandes clasificaciones de los polinomios de Sobolev según las medidas involucradas en el producto interno que los origina: caso continuo, caso discreto, y el caso discreto-continuo. En nuestro trabajo se abordan cada uno de ellos en los Capítulos II, III y IV respectivamente.

1. El Capítulo I está dedicado a la introducción de la teoría general de los polinomios ortogonales de Sobolev, así como presentar el estado del arte de los principales resultados que preceden nuestro trabajo.
2. En el Capítulo II se considera, para $1 < p < \infty$, la clase general de polinomios extremales con respecto a la p -ésima norma de Sobolev

$$\|Q\|_{s,p} = \left(\sum_{k=0}^{m_b} \|Q^{(k)}\|_{\mathbf{L}^p(\mu_k)}^p \right)^{\frac{1}{p}}, \quad Q \in \mathbb{P}.$$

El n -ésimo polinomio extremal de Sobolev L_n se define como el polinomio mónico de grado n que minimiza la norma. Es decir

$$\|L_n\|_{s,p} = \min_{Q=x^n+\dots} \|Q\|_{s,p}.$$

La extremalidad es una extensión de la ortogonalidad a espacios donde la norma no es inducida por un producto interno. Para $p \neq 2$, $\|\cdot\|_{s,p}$ no proviene de un producto interior por lo que la inmensa mayoría de los resultados conocidos para polinomios ortogonales de Sobolev no estarán disponibles. En general, son pocos los resultados que se tienen hasta el momento del caso extremal. Las principales referencias son [93] para el caso de normas secuencialmente dominadas y [89] para medidas con soporte no acotado sobre la recta real.

En el Capítulo II se caracteriza la extremalidad en el caso Sobolev con $1 < p < \infty$ por medio del operador

$$\langle P, Q \rangle_{s,p} := \sum_{k=0}^{m_b} \int P^{(k)} \operatorname{sgn}(Q^{(k)}) |Q^{(k)}|^{p-1} d\mu_k, \quad P, Q \in \mathbb{P},$$

el cual viene a suplir la ausencia, para $p \neq 2$, de un producto escalar. Esta idea queda totalmente clara en el primer resultado de este capítulo. Un polinomio

mónico de grado n es el n -ésimo polinomio extremal si y solo si

$$\langle x^k, L_n \rangle_{s,p} = 0, \quad k = 0, 1, \dots, n-1.$$

El segundo resultado está relacionado con la localización de los ceros y puntos críticos de los polinomios extremales de Sobolev en el caso $\mathfrak{M} = 1$ y medidas μ_0 y μ_1 soportadas en conjuntos cuyas envolturas convexas tienen interiores disjuntos. Bajo estas condiciones se demuestra que los ceros de L'_n son simples y están contenidos en el interior de la envoltura convexa de la unión de los soportes. Como consecuencia de esto, se describe un conjunto acotado que contiene todos los ceros de los polinomios extremales $\{L_n\}_{n \geq 0}$. Este resultado extiende a [92, Th. 1.3], obtenido en el caso ortogonal, y que a su vez fue conjeturado por W. Gautschi y A.B.J. Kuijlaars [55, Conj. 1 and Conj. 2] para configuraciones arbitrarias de los soportes de las medidas.

Posteriormente se consideran medidas continuas μ_0 y μ_1 con ciertas condiciones habituales de regularidad. Con estos supuestos, similares a los que se requiere en el caso ortogonal, se deduce el comportamiento asintótico de la raíz n -ésima de las derivadas de L_n . Además, se demuestra que las derivadas de los polinomios extremales de Sobolev tienen una distribución regular de sus ceros y la distribución de los ceros de L_n está gobernada por el balayage de la medida de equilibrio de la unión de los soportes de las medidas en una cierta región, la cual es descrita explícitamente. Estos resultados extienden [55, Th. 1] y [55, Th. 2] para el caso extremal.

3. En el Capítulo III se estudian los polinomios discretos de Sobolev, los cuales son ortogonales con respecto al producto interno

$$\langle f, g \rangle_s = \int f^{(k)}(x) g^{(k)}(x) d\mu(x) + \sum_{j=1}^{\mathcal{N}} \sum_{k=0}^{\ell_j} M_{j,k} f^{(k)}(c_j) g^{(k)}(c_j).$$

Una nueva clase de productos internos discretos es presentada, los productos secuencialmente ordenados, la cual establece un tipo orden en la parte discreta del producto interno. Se demuestra que el n -ésimo polinomio de Sobolev ortogonal con respecto a un producto interno secuencialmente ordenado tiene al menos $n - \ell^*$ ceros simples ubicados en el interior de la envoltura convexa de la medida μ , donde ℓ^* denota el numero de términos en la parte discreta de $\langle \cdot, \cdot \rangle_s$ ($M_{j,k} \neq 0$). Se muestran, además, dos ejemplos de productos que no son secuencialmente ordenados donde la tesis de este resultado no se tiene. Esto hace pensar que estas restricciones de orden parecen ser óptimas para tener garantizado que, para todo valor de n , al menos $n - \ell^*$ ceros son simples se encuentran en el interior del soporte de μ . En este sentido, este resultado es el primero que admite más de un punto de masa ($\mathcal{N} > 1$) en el producto interno y generaliza además los resultados previos para polinomios de Sobolev discretos con medidas generales [6, Prop. 3.2], [34, Th. 4.1] [117, Th. 5.1], [119, Th. 4.1], [3, Th. 2.2], [68, Prop. 4.11] y [107, Th. 2].

El segundo resultado de este capítulo es una extensión para polinomios discretos de Sobolev del teorema de Markov clásico de 1895. Para enunciar y demostrar este teorema la sucesión de polinomios de Sobolev asociados $\{S_n^{[k]}\}_{n \geq 0}$ es definida y varias de sus propiedades son demostradas.

Previamente, el teorema de Markov había sido extendido en varias direcciones, en [88] se pueden ver dos de ellas. La primera añadiendo funciones racionales a la función de Markov y la segunda considerando aproximantes multipuntuales de Padé de tipo $(n-1, n)$. El teorema de Markov y sus extensiones en la teoría de aproximación son discutidos en [58, 59, 38, 63] y las referencias que allí aparecen.

Hasta donde sabemos la única extensión del Teorema de Markov para polinomios de Sobolev aparece en [96, Th. 5.5], donde el producto interno $\langle \cdot, \cdot \rangle_s$ considerado es el caso en que $\mathcal{N} = 1$, $\ell_1 = 1$, $c_1 = 0$, $M_{1,0} = 0$, $M_{1,1} > 0$ y μ está soportada en $[a, b]$ con $b < 0$. En este capítulo se extiende el teorema de Markov para polinomios ortogonales de Sobolev en el caso en que $\langle \cdot, \cdot \rangle_s$ es secuencialmente ordenado, la medida μ está soportada en $[-1, 1]$, pertenece a la clase de Nevai ($\mu \in \mathbf{M}(0, 1)$) y los puntos de masa c_j están fuera de $[-1, 1]$. Nuestra extensión también extiende la mencionada anteriormente pues la condición $\mu \in \mathbf{M}(0, 1)$, en este caso puede ser reemplazada por el uso de [96, Lem. 3.1] en lugar del resultado de convergencia de ceros que se obtiene como consecuencia de la asintótica relativa demostrada en [91].

4. En el Capítulo IV se estudia el espacio de Sobolev $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$ asociado a la p -ésima norma discreto-continua de Jacobi-Sobolev

$$\|f\|_{s, p} = \left(\sum_{k=0}^{\mathfrak{m}-1} \left| f^{(k)}(c_k) \right|^p + \int_{-1}^1 \left| f^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha, \beta}(x) \right)^{\frac{1}{p}}, \quad p \geq 1,$$

donde $d\mu^{\alpha, \beta}(x) = (1-x)^\alpha(1+x)^\beta dx$ denota la medida clásica de Jacobi, $c_0, c_1, \dots, c_{\mathfrak{m}} \in \mathbb{R}$ y $\alpha, \beta > -1$. Dicho espacio se define de manera natural como el completamiento mediante la norma $\|\cdot\|_{s, p}$ del espacio vectorial de los polinomios. Siguiendo las ideas de [149, 150] se define el espacio $\mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$ de funciones f más amplio para el cual $\|f\|_{s, p}$ está bien definido y es un valor finito no negativo. Se prueba que $\mathbf{U}_{\vec{c}, p}^{\alpha, \beta} \subset \mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$ y se hallan las combinaciones de parámetros $c_0, c_1, \dots, c_{\mathfrak{m}}$, α, β y p para los cuales se tiene la igualdad. Es decir, cuando todos los elementos del espacio de Sobolev son funciones.

Las series de Fourier son definidas para todo elemento del espacio de Sobolev $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$, sean funciones o no, y se hallan las combinaciones de parámetros, en este caso solo dependen de α, β y p para los cuales la serie de Fourier de una función converge a ella.

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Introduction

This chapter is dedicated to introduce the Sobolev orthogonal polynomials, from the point of view of mathematical outcomes required to formulate the theory and the historical context that motivated its study which preceded this work. The chapter is also developed to collect the results that will be used in the rest of the manuscript as well as to present the necessary definitions and terminologies.

1.1 Preliminaries

The limit process plays a main role in every analytic problem. To define a “notion of limit” in some space, we primarily need to have a notion of “closeness”. Although there are more general ways to define “closeness” consistent with the structure of vector space (such as topological vector space and metric vector space) the simplest and most used way is through the concept of norm.

Definition 1.1.1 (Normed vector space)

A vector space X over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, is said to be a normed vector space if it is endowed with a function $\|\cdot\| : X \rightarrow \mathbb{R}_+$ called norm, satisfying:

1. If $x \in X$ and $x \neq 0$ then $\|x\| > 0$.
2. $\|\alpha x\| = |\alpha| \|x\|$, $\alpha \in \mathbb{K}$, $x \in X$.
3. $\|x + y\| \leq \|x\| + \|y\|$, $x, y \in X$.

The normed space is denoted by the order pair $(X, \|\cdot\|)$ or simply X when no confusion can arise.

In a natural way, a norm function induces a metric over X by defining the distance from a vector x to a vector y as $d(x, y) = \|x - y\|$. In addition, both vector space operations (the operations of vector addition and scalar multiplication) and the function norm itself, with this distance, become continuous. The normed space is called Banach space if its induced metric is complete. Although metric and normed spaces as well as its properties are very basic concepts, the references [84, Chap. 8 and Chap. 14] and [13, Chap. 1] are included for the sake of completeness.

Maybe the most widely used example of complete normed space is the space of p -th power integrable functions and we do not want to be the exception.

Example 1.1.1 (The normed space $\mathbf{L}^p(\mu)$)

Consider a fixed value $1 \leq p \leq \infty$. Let μ be a positive measure over an arbitrary set X . For each measurable complex-valued function f we denote the $\mathbf{L}^p$ -norm of f by

$$\|f\|_{\mathbf{L}^p(\mu)} = \begin{cases} (\int_X |f|^p d\mu)^{\frac{1}{p}}, & \text{if } p \in [1, \infty), \\ \inf\{\alpha > 0 : \mu(\{|f| > \alpha\}) = 0\}, & \text{if } p = \infty. \end{cases}$$

The space $\mathbf{L}^p(\mu)$ consists of all functions whose norm is finite ($\|f\|_{\mathbf{L}^p(\mu)} < \infty$) under the equivalence relation

$$f \equiv g \Leftrightarrow f = g \quad \mu - a.e. \quad [\mu(\{x \in X : f(x) \neq g(x)\}) = 0].$$

When μ is the Lebesgue measure over a subset X of $\mathbb{R}$ or $\mathbb{C}$ we simply write $\mathbf{L}^p(X)$ for short. In addition, in the case $p = 2$ we just write $\|\cdot\|_\mu$ instead of $\|\cdot\|_{\mathbf{L}^2(\mu)}$ for brevity.

It is not difficult to check that the equivalence relation we have described above is compatible with the vector space structure so $\mathbf{L}^p(\mu)$ becomes a vector space. It can be also proved that $\mathbf{L}^p(\mu)$ endowed with the norm $\|\cdot\|_{\mathbf{L}^p(\mu)}$ is a Banach normed space (see [153, Th. 3.9 and Th 3.11]). Note that the elements of $\mathbf{L}^p(\mu)$ are not really functions, but they are equivalence classes given by this equivalence relation. For more information about this type of spaces we refer to [153, Chap. 3]. Because we are interested in one special type of functions, the polynomials, an additional requirement over the measure μ must be considered in order to obtain the norm $\|\cdot\|_{\mathbf{L}^p(\mu)}$ well defined over the space of polynomials.

Definition 1.1.2 (Nontrivial measure)

Let μ be a Borel positive measure on the complex plane. We call μ trivial if μ is supported on a finite set. Otherwise we say that μ is a nontrivial measure.

The fact that the support of the measure contains infinitely many points is necessary to different polynomials belong to different equivalence classes. Therefore, the norm of a polynomial can be interpreted as the norm of the equivalence class to which it belongs.

The orthogonality is a relation between vectors which is not sufficiently well described by the normed space environment. The inner product provides the vector space with a geometric structure which allows to define the orthogonality in a right way.

Definition 1.1.3 (Inner product)

Let H be a vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, we will say that a function $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{K}$ is an inner product if has the following properties

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}, \quad x, y \in H.$
2. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle, \quad x, y, z \in H.$
3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle, \quad x, y \in H, \alpha \in \mathbb{K}.$
4. $\langle x, x \rangle > 0$ if $x \in H \setminus \{0\}.$

where $\bar{z} = a - bi$ denotes the complex conjugate of a complex number $z = a + bi$ with $a, b \in \mathbb{R}$. The ordered pair $(H, \langle \cdot, \cdot \rangle)$ is called an inner product space and will be denoted simply by H if no confusion arises (can be also found in the literature with the names of Euclidean and pre-Hilbert space).

In a natural way, an inner product space induces a normed space taking the function $\|x\| = \sqrt{\langle x, x \rangle}$ as its norm.

Proposition 1.1.1 (Induced norm)

Let H be an inner product space and consider the function $\|\cdot\| : H \rightarrow \mathbb{R}_+$ given by $\|x\| = \sqrt{\langle x, x \rangle}$. Then the function $\|\cdot\|$ is a norm. In addition, the identity

$$\|x + y\| = \|x\| + \|y\|$$

holds only when the vectors x and y are linearly dependent (i.e. $y = \alpha x$ for some $\alpha \in \mathbb{K}$ or any of them is zero).

Proof. We refer the reader to [13, Th. 1.9] for a proof. □

By agreement, the induced norm will be adopted as the norm of the inner product space, unless another one is specified. An inner product space, is called a Hilbert space if endowed with the induced norm becomes a Banach space.

In the case $p = 2$ the norm $\|\cdot\|_{\mathbf{L}^2(\mu)}$ is induced by the inner product

$$\langle f, g \rangle_\mu := \int f(x) \overline{g(x)} d\mu(x), \quad f, g \in \mathbf{L}^2(\mu). \quad (1.1)$$

So, the space $\mathbf{L}^2(\mu)$ is a Hilbert space.

To continue we will enunciate an essential property of the inner product spaces, the Cauchy-Schwarz inequality.

Proposition 1.1.2 (Cauchy-Schwarz inequality)

Every inner product space H satisfies

$$|\langle x, y \rangle| \leq \|x\| \|y\|, \quad x, y \in H.$$

In addition, $|\langle x, y \rangle| = \|x\| \|y\|$ holds if and only if x and y are linearly dependent.

Proof. A proof can be found in [13, Th. 8.1] □

When $\mathbb{K} = \mathbb{R}$, the inner product can be used to define the angle between two vectors x and y as the quantity θ such that

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|}, \quad 0 \leq \theta < \pi.$$

Note that by Cauchy-Schwarz inequality (see Proposition 1.1.2) the value θ is well defined. Thus, two vectors determine a right angle if and only if $\langle x, y \rangle = 0$ and in this case these vectors are orthogonal to each other.

This concept can be generalized to an arbitrary family of vectors, but only the numerable families, the sequences, are of interest to this work.

Definition 1.1.4 (Orthogonal sequence)

Let H be an inner product space. A sequence of vectors $\{e_n\}_{n \geq 0} \subset H$ is called an orthogonal vector sequence if satisfies

$$\langle e_n, e_m \rangle \begin{cases} \neq 0, & \text{if } n = m, \\ = 0, & \text{if } n \neq m. \end{cases}$$

In the case that $\|e_n\| = 1$, for $n \geq 0$, we will say that $\{e_n\}_{n \geq 0}$ is an orthonormal vector sequence.

As it is expected from the geometrical interpretation of the vectors in $\mathbb{R}^3$ with the usual inner product, orthogonality is a stronger property than linear independence.

Proposition 1.1.3

Let X be a vector space with inner product. If $E = \{e_n\}_{n \geq 0} \subset X$ is an orthogonal vector sequence, then E is a set of linearly independent vectors.

Proof. A proof can be found in [13, Th. 1.10]. □

In the case of inner product spaces, the orthogonality has a main role in the problem of finding the best approximation of a vector by vectors of a fixed linear subspace. This is because the sequences of orthonormal vectors are the most efficient basis to express the best approximation.

Theorem 1.1.1 (Fourier Series)

Let Y be an inner product space, and $\{e_n\}_{n \geq 0} \subset Y$ an orthonormal sequence. Given a vector $y \in Y$ we define the Fourier series of y as the series whose n -partial sums are given by

$$\mathcal{F}_n(y) := \sum_{k=0}^n \langle y, e_k \rangle e_k, \quad n \geq 0.$$

This series satisfies the following minimization property

$$\min_{x \in E_n} \{\|y - x\|\} = \|y - \mathcal{F}_n(y)\|,$$

where E_n is the span of $\{e_k\}_{k=0}^n$, i.e. the space of all linear combinations of the vectors $\{e_k\}_{k=0}^n$.

Proof. A proof can be found in [13, Th. 1.16]. □

It is well known that every vector space has a basis of vectors. If a numerable basis exists, the following theorem provides a method to obtain a new basis consisting of orthogonal vectors.

Theorem 1.1.2 (Gram-Schmidt orthogonalization process)

Let H be an inner product space and $\{v_n\}_{n \geq 0}$ a sequence of vectors linearly independent in H . Then, there exists an orthogonal sequence of vectors $\{e_n\}_{n \geq 0}$ in H such that for every $m \in \mathbb{N}$ the set $\{e_k\}_{k=0}^m$ is a basis for the subspace spanned by $\{v_k\}_{k=0}^m$. In addition, every element e_n , $n \geq 0$, is unique up to constant factor.

Proof. See [13, Th. 1.13] for a proof. $\square$

Now, we are able to introduce orthogonal polynomial sequences.

1.2 Orthogonal Polynomials

The field of orthogonal polynomials is a very broad research area with over a century's intense study and whose origins go back to 1783 when Legendre discovered the Legendre Polynomial by modeling problems of planetary motion. This topic is closely connected to many areas of pure and applied mathematics such as the special functions theory, theory of continued fractions, spectral theory, approximation theory, mathematical physics, and number theory just to name some of them. The aim of this section is to provide the main concepts of the theory that serves as the basis for this thesis.

The notation that will be used later is the following.

1. The $\mathbb{K}$ -linear space of polynomials with coefficients on $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ is denoted by $\mathbb{P}$. The notations $\mathbb{R}[x]$ and $\mathbb{C}[x]$ are also used when it is necessary to specify which $\mathbb{K}$ we refer to. The case of the real linear space of polynomials with complex coefficients is not considered in this manuscript.
2. As usual, $\mathbb{P}_n \subset \mathbb{P}$ denotes the vector subspace of polynomials of degree $\leq n$.
3. Given an arbitrary polynomial of degree n

$$P(x) := \sum_{k=0}^n a_k x^k, \quad a_n \neq 0,$$

we denote:

- a) The *degree* of P by $\deg(P) := n$
- b) The *leading coefficient* of P by $\kappa(P) := a_n$.
- c) If $P \in \mathbb{R}[x]$ and A is a real subset of the real line, then
 - i. $\mathbf{N}_s(P; A)$ denotes the number of zeros with odd multiplicity of the polynomial P on A . (i.e. points of sign change).
 - ii. $\mathbf{N}_o(P; A)$ denotes the number of values on A where the polynomial P vanishes, (i.e. zeros of P on A without counting multiplicities).
 - iii. $\mathbf{N}_z(P; A)$ denotes the total number of zeros (counting multiplicities) of P on A .
4. A polynomial P is said to be *monic* if has 1 as leading coefficient, i.e. $\kappa(P) = 1$.
5. The set of monic polynomials of exact degree n is denoted by $\mathbb{P}_n^*$.
6. We denote by $\mathbf{C}_h(A)$ the *convex hull* of a real set A , or equivalently, the smallest real interval that contains A .

7. For a real set A , we will denote by A° the interior of A with the usual Euclidean topology on $\mathbb{R}$.
8. Given a measure μ , its support will be denoted by $\mathbf{supp}(\mu)$ and the convex hull of $\mathbf{supp}(\mu)$ by Δ or Δ_i in case of the measure is denoted by μ_i .
9. Given a set $B \subset \mathbb{C}$ and a complex function f defined on B we will adopt the notation

$$\|f\|_B := \sup_{z \in B} \{|f(z)|\}.$$

10. $f^{(k)}$ denotes the k th derivative of f , as usual when $k = 0, 1, 2$ we simply write f, f' and f'' respectively.
11. The conformal function which maps the complement of $[-1, 1]$ onto the exterior of the unit circle will be denoted by

$$\varphi(z) := z + \sqrt{z^2 - 1}, \quad \text{with } \sqrt{z^2 - 1} > 0 \text{ for } z > 1.$$

12. We use the notation $f_n \xrightarrow[n]{} f; K \subset U$ when the sequence of functions f_n converges to f uniformly on every compact subset K of the region U .
13. If $a \in \mathbb{C}$, as usual, δ_a denotes the Dirac measure with mass one at the point a , i.e.

$$\delta_a(A) = \begin{cases} 1, & \text{if } a \in A, \\ 0, & \text{if } a \notin A, \end{cases}, \quad A \subset X.$$

Although there are more general definitions of orthogonal polynomials, the definition by means of inner products is a very intuitive one and it will be broadly enough for the purposes of this work.

Definition 1.2.1 (Orthogonal polynomials)

Let $\langle \cdot, \cdot \rangle$ be an inner product defined on the linear space of polynomials $\mathbb{P}$. A polynomial sequence $\{P_n\}_{n \geq 0}$ is said to be an orthogonal polynomial sequence with respect to the inner product $\langle \cdot, \cdot \rangle$ if

1. $\deg(P_n) = n, \quad n \in \mathbb{Z}_+.$
2. $\langle P_n, P_m \rangle \begin{cases} > 0, & \text{if } n = m, \\ = 0, & \text{if } n \neq m. \end{cases} \quad (\text{Orthogonality conditions})$

Through the application of the Gram-Schmidt orthonormalization process (Theorem 1.1.2) to the polynomial sequence $\{x^n\}_{n \geq 0}$ it can be demonstrated the existence and uniqueness, up to a multiplicative constant sequence, of orthogonal polynomials. The following notation will be adopted in the sequel.

Consider $\{P_n\}_{n \geq 0}$ an orthogonal polynomial sequence with respect to the inner product $\langle \cdot, \cdot \rangle$, then

- The n th monic orthogonal polynomial $\frac{1}{\kappa(p_n)}P_n$ is denoted by $\hat{P}_n$.
- The n th orthonormal polynomial $\frac{\hat{P}_n}{\|\hat{P}_n\|}$ is denoted by the corresponding lower letter p_n .
- We just write κ_n to denote the leading coefficient of the n th orthonormal polynomial

$$\kappa_n := \kappa(p_n) = \frac{1}{\|\hat{P}_n\|}$$

when no confusion arises.

Orthogonal polynomials have a number of significant properties. In this section, it will be highlighted some of them to be used later.

Theorem 1.2.1 (Minimum norm theorem)

Let $\langle \cdot, \cdot \rangle$ be an inner product over $\mathbb{P}$ and $\{\hat{P}_n\}_{n \geq 0}$ its monic orthogonal polynomial sequence. If M_n is an arbitrary monic polynomial of degree n , then

$$\|\hat{P}_n\| \leq \|M_n\| \quad (1.2)$$

and the equality holds if and only if $M_n = \hat{P}_n$.

Proof. Consider the polynomial $R_{n-1} = \hat{P}_n - M_n$ of degree at most $n-1$. Then.

$$\begin{aligned} \|M_n\|^2 &= \langle \hat{P}_n - R_{n-1}, \hat{P}_n - R_{n-1} \rangle = \langle \hat{P}_n, \hat{P}_n \rangle + \langle R_{n-1}, R_{n-1} \rangle = \|\hat{P}_n\|^2 + \|R_{n-1}\|^2, \\ &\geq \|\hat{P}_n\|^2, \end{aligned}$$

where in the second equality the orthogonality conditions have been used. In addition, the equality holds if and only if $\|R_{n-1}\| = 0$, which is equivalent to our claim. $\square$

The standard theory of orthogonal polynomials deals with orthogonal polynomials with respect to the inner product $\langle \cdot, \cdot \rangle_\mu$ defined by (1.1), where μ is a nontrivial measure supported on the real line with finite moments

$$\mu_n := \int x^n d\mu(x), \quad n = 0, 1, 2, \dots$$

The orthogonality with respect to this inner product is known in the literature as the standard orthogonality. There are two fundamental characterizations of this special kind of orthogonality.

Theorem 1.2.2 (Characterization of Standard Orthogonality)

Let $\{p_n\}_{n \geq 0}$ be a sequence of polynomials such that $\deg(p_0) = 0$. Then, the following statements are equivalent:

1. There exists a measure μ supported on the real line such that the polynomial sequence $\{p_n\}_{n \geq 0}$ is orthonormal with respect to $\langle \cdot, \cdot \rangle_\mu$.

2. *Three-term recurrence relation.*

There exist real sequences $\{a_n\}_{n \geq 0}$ and $\{b_n\}_{n \geq 0}$ with $a_n > 0$ such that

$$xp_n(x) = a_{n+1}p_{n+1}(x) + b_np_n(x) + a_np_{n-1}(x), \quad n = 0, 1, 2, \dots, \quad (1.3)$$

where $p_{-1} \equiv 0$.

3. *There exists an inner product $\langle \cdot, \cdot \rangle$ defined on $\mathbb{P}$ whose orthonormal polynomial sequence is $\{p_n\}_{n \geq 0}$ and such that the multiplication operator is symmetric, i.e.*

$$\langle xP(x), Q(x) \rangle = \langle P(x), xQ(x) \rangle, \quad P, Q \in \mathbb{P}. \quad (1.4)$$

Notice that by the three-term recurrence relation, every polynomial p_n has real coefficients.

Proof. The proof of (1.) implies (3.) is straightforward. To prove (3.) implies (2.) write $xp_n(x)$ as linear combination of $\{p_k\}_{k=0}^{n+1}$

$$xp_n(x) = \sum_{k=0}^{n+1} a_{n,k} p_k(x),$$

From the orthogonality conditions, for $j = 0, 1, \dots, n-2$, we obtain

$$a_{n,j} = \langle xp_n(x), p_j(x) \rangle = \langle p_n(x), xp_j(x) \rangle = 0.$$

Now, (2.) follows from

$$a_n := \langle xp_n(x), p_{n-1}(x) \rangle = \langle p_n(x), xp_{n-1}(x) \rangle = \langle p_n(x), \kappa_{n-1}x^n \rangle = \frac{\kappa_{n-1}}{\kappa_n} > 0,$$

$$b_n := \langle xp_n(x), p_n(x) \rangle = \overline{\langle p_n(x), xp_n(x) \rangle} = \overline{\langle xp_n(x), p_n(x) \rangle} = \overline{b_n},$$

$$\langle xp_n(x), p_{n+1}(x) \rangle = \langle \kappa_n x^{n+1}, p_{n+1}(x) \rangle = \frac{\kappa_n}{\kappa_{n+1}} = a_{n+1}.$$

To prove (2.) implies (3.), for each fixed real polynomials $P(x) = \sum_{j=0}^{\infty} \alpha_j p_j$ and $Q(x) = \sum_{k=0}^{\infty} \beta_k p_k$ we define

$$\langle P, Q \rangle = \sum_{j=0}^{\infty} \alpha_j \beta_j,$$

where the sums have been represented as infinite ones to simplify the notation. Then, we obtain trivially that $\langle \cdot, \cdot \rangle$ is an inner product having $\{p_n\}_{n \geq 0}$ as its orthonormal polynomial sequence. In order to prove (1.4), note that from the linearity of the inner product (see conditions 1-3 of Definition 1.1.3) it is enough to prove (1.4) for $P, Q \in \{p_n\}_{n \geq 0}$. Finally

$$\begin{aligned} \langle xp_n(x), p_m(x) \rangle &= \langle a_{n+1}p_{n+1}(x) + b_np_n(x) + a_np_{n-1}(x), p_m(x) \rangle, \\ &= \begin{cases} 0, & \text{if } |m-n| \geq 2, \\ a_{n+1}, & \text{if } m = n+1, \\ b_n, & \text{if } m = n, \\ a_n, & \text{if } m = n-1, \end{cases} = \begin{cases} 0, & \text{if } |n-m| \geq 2, \\ a_{m+1}, & \text{if } n = m+1, \\ b_m, & \text{if } n = m, \\ a_m, & \text{if } n = m-1, \end{cases} \\ &= \overline{\langle xp_m(x), p_n(x) \rangle} = \langle p_n(x), xp_m(x) \rangle. \end{aligned}$$

This implication is known in the literature as Favard's Theorem. The remaining implication, (3.) implies (1.), is not trivial at all. In the literature it is known as Representation Theorem, see [27, Th. 2.3.1] taking the positive moment functional $P \mapsto \langle P, 1 \rangle, P \in \mathbb{P}$. $\square$

In the case of the standard orthogonality, when $\langle \cdot, \cdot \rangle \equiv \langle \cdot, \cdot \rangle_\mu$, one usually says that the polynomial sequence $\{P_n\}_{n \geq 0}$ is orthogonal with respect to the measure μ . Using that $a_n = \kappa_{n-1}/\kappa_n$ a formula similar to (1.3) is obtained but for the monic polynomial sequence $\hat{P}_n$

$$\begin{aligned} x\hat{P}_n(x) &= x \frac{p_n(x)}{\kappa_n} = \frac{1}{\kappa_n} \left(\frac{\kappa_n}{\kappa_{n+1}} p_{n+1}(x) + b_n p_n(x) + \frac{\kappa_{n-1}}{\kappa_n} p_{n-1}(x) \right), \\ &= \hat{P}_{n+1}(x) + b_n \hat{P}_n(x) + a_n^2 \hat{P}_{n-1}(x). \end{aligned}$$

So, the following monic version of (1.3)

$$\hat{P}_{n+1}(x) = (x - b_n) \hat{P}_n(x) - a_n^2 \hat{P}_{n-1}(x), \quad n = 0, 1, 2, \dots \quad (1.5)$$

is obtained, where $\hat{P}_{-1} \equiv 0$ and $\hat{P}_0 \equiv 1$. The three-term recurrence relation is a very important property, connected with several fields such as: difference equations [96], numerical analysis [54], Fourier analysis [66, 131, 164], among others. When such a relation is not available most of the standard results are not longer true.

Another important aspect which plays a main role in the theory of orthogonal polynomials, and also in this thesis dissertation, is the location of its zeros. In the case of standard orthogonality the following result is well known.

Theorem 1.2.3 (Zero location of standard orthogonal polynomials)

Let μ be a nontrivial measure with finite moments. Then, the n th polynomial orthogonal with respect to μ has all its zeros simple and located in Δ° .

Proof. Let $\xi_1, \xi_1, \dots, \xi_\eta$ be the points on Δ° where P_n changes sign ($\eta = N_s(P_n; \Delta_0)$). Suppose that $\eta < n$, then the polynomial $Q_\eta(x) = \prod_{i=1}^\eta (x - \xi_i)$ has degree $\eta < n$, and $P_n Q_\eta$ does not change sign on Δ° , so we get to the contradiction

$$0 = \langle P_n, Q_\eta \rangle_\mu = \int P_n(x) Q_\eta(x) dx \neq 0.$$

$\square$

For an in-depth study of standard theory we refer the reader to the books [52], [27] and [163].

Certainly, the most studied sequence of standard polynomials are the classical orthogonal polynomial sequences. They are given when the measure μ is absolutely continuous with respect to the Lebesgue measure i.e. $d\mu = w(x)dx$ and the weight function w belongs to one of the families of Table 1.1.

Family	Weight w	Polynomial
Hermite	$e^{-x^2} \quad x \in \mathbb{R}$	$H_n(x) = 2^n x^n + \dots$
Laguerre	$x^\alpha e^{-x} \quad \alpha > -1, x \in [0, \infty)$	$L_n^{(\alpha)}(x) = \frac{(-1)^n}{n!} x^n + \dots$
Jacobi	$(1-x)^\alpha (1+x)^\beta \quad \alpha, \beta > -1, x \in [-1, 1]$	$P_n^{(\alpha, \beta)}(x) = \frac{1}{2^n} \binom{2n+\alpha+\beta}{n} x^n + \dots$

Table 1.1: Classical Polynomials.

As usual, we have used the notation

$$\binom{a}{b} := \frac{\Gamma(a+1)}{\Gamma(a-b+1)\Gamma(b+1)},$$

where Γ denotes the Gamma function (see [164, §10.1]). These polynomials have the following explicit expressions

$$H_n(x) = n! \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^k}{k!(n-2k)!} (2x)^{n-2k},$$

$$L_n^{(\alpha)}(x) = \sum_{k=0}^n \binom{n+\alpha}{n-k} \frac{(-x)^k}{k!}, \quad (1.6)$$

$$P_n^{(\alpha, \beta)}(x) = \sum_{k=0}^n \binom{n+\alpha}{k} \binom{n+\beta}{n-k} \left(\frac{x-1}{2}\right)^{n-k} \left(\frac{x+1}{2}\right)^k, \quad (1.7)$$

where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x (floor function). Note that expressions (1.6) and (1.7) have sense for every real value of α and β . In general, the polynomials defined by these expressions are called the generalized Laguerre and Jacobi polynomials respectively.

The special interest on these three families of orthogonal polynomials comes from its very good differential properties. It is a well known result, proved by S. Bochner [19] in 1929, that, up to a linear change of variable, these families of classical orthogonal polynomials are the only standard orthogonal polynomial sequences solving linear differential equations of the form

$$\sigma(x)y_n''(x) + \tau(x)y_n'(x) = \lambda_n y_n(x), \quad \lambda_n \neq 0, \quad (1.8)$$

where σ and τ are polynomials independent on n of degree at most 2 and 1 respectively. These families are also characterized by satisfying the following Rodrigues' formula

$$w(x)P_n(x) = C_n \frac{\partial^n}{\partial x^n} (w(x)\sigma^n(x)),$$

where w is the weight of the family, σ is the same as in (1.8) and $C_n \in \mathbb{R}$ is a normalization constant.

We refer the reader to [163, Chap. IV and Chap. VI], [27, §5.2] and [65, Chap. IV] to know more about these three classical families of orthogonal polynomials. In

addition, some of the main properties of the Jacobi polynomials are summarized in Section 4.2.

Otherwise, if an inner product is not symmetric with respect to the multiplication operator, see (1.4), we say that it is non-standard. This is the case of inner products modified by terms involving derivatives, which are at the heart of this work.

1.3 Sobolev orthogonal polynomials

With the goal of solving different problems emerging from the theory of partial differential equations, arose the interest for considering inner products that do not only involve the functions, but also its derivatives. Those inner products are known as Sobolev inner products.

Definition 1.3.1 (*k*-standard vectorial measure)

A vectorial measure $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ is called a *k*-standard vectorial measure $k = 0, 1, \dots, \mathfrak{m}$, if satisfies

1. μ_i is a Borel positive measure supported on the real line, for $i = 0, 1, \dots, \mathfrak{m}$.
2. $\mathbb{P} \subset \mathbf{L}^1(\mu_i)$, for $i = 0, 1, \dots, \mathfrak{m}$.
3. μ_k is nontrivial (see Definition 1.1.2).
4. $\mu_0, \mu_1, \dots, \mu_{k-1}$, and $\mu_{\mathfrak{m}}$ are non-null measures.

When the vectorial measure is a 0-standard vectorial measure we just say that it is a standard vectorial measure for short. We also say that a measure μ is standard if, seen as a vector of one component, is a standard vectorial measure.

Definition 1.3.2 (Sobolev inner product)

Let $0 \leq k \leq \mathfrak{m}$ and let $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ be a *k*-standard vectorial measure. We define the Sobolev inner product of the vectorial measure $\vec{\mu}$ by

$$\langle P, Q \rangle_s := \sum_{k=0}^{\mathfrak{m}} \int P^{(k)} \overline{Q^{(k)}} d\mu_k = \sum_{k=0}^{\mathfrak{m}} \langle P^{(k)}, Q^{(k)} \rangle_{\mu_k}, \quad P, Q \in \mathbb{P} \quad (1.9)$$

In addition, for $1 \leq p < \infty$ we define the *p*-Sobolev norm

$$\|Q\|_{s,p} = \left(\sum_{k=0}^{\mathfrak{m}} \left\| Q^{(k)} \right\|_{\mathbf{L}^p(\mu_k)}^p \right)^{\frac{1}{p}}, \quad Q \in \mathbb{P}. \quad (1.10)$$

Note that $\|\cdot\|_{s,2}$ is induced by $\langle \cdot, \cdot \rangle_s$, in this case we just will write $\|\cdot\|_s$.

It is not hard to prove that $\langle \cdot, \cdot \rangle_s$ defines an inner product on $\mathbb{P}$. The conditions (1-3) in Definition 1.1.3 and $\langle P, P \rangle_s \geq 0$, for $P \in \mathbb{P}$, are easily satisfied. Suppose that $\langle P, P \rangle_s = 0$, then $P^{(i)} \equiv 0$, μ_i - a.e. for $i = 0, 1, \dots, k$. As μ_k is a nontrivial measure and μ_i are non-null we obtain $P^{(k)} \equiv 0$ and there exists c_i , $i = 0, 1, \dots, k-1$, such that $P^{(i)}(c_i) = 0$. So we conclude that $P \equiv 0$.

On the other hand, following the reasoning above it can be proved that (1.10) satisfies condition (1.) in Definition 1.1.1 and it is straightforward to prove (2.). The third one is a consequence of the triangular inequalities of the norms $\|\cdot\|_{\mathbf{L}^p(\mu_k)}$ and the usual p -norm of $\mathbb{R}^{\mathfrak{m}+1}$

$$\begin{aligned} \|P + Q\|_{s,p} &= \left(\sum_{k=0}^{\mathfrak{m}} \|P^{(k)} + Q^{(k)}\|_{\mathbf{L}^p(\mu_k)}^p \right)^{1/p} \\ &\leq \left(\sum_{k=0}^{\mathfrak{m}} \left(\|P^{(k)}\|_{\mathbf{L}^p(\mu_k)} + \|Q^{(k)}\|_{\mathbf{L}^p(\mu_k)} \right)^p \right)^{1/p} \\ &\leq \|P\|_{s,p} + \|Q\|_{s,p}. \end{aligned}$$

Observe that the measures μ_i , $i \neq k$, may be a trivial or a non-trivial measure. Thus, given a k -standard vectorial measure $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$, there are essentially three types of such inner products:

- I.- *Continuous case.* $\vec{\mu}$ is a k -standard vectorial measure for $k = 0, 1, \dots, \mathfrak{m}$, i.e. all the component measures are non-trivial.
- II.- *Discrete case.* $\vec{\mu}$ is a standard vectorial measure ($k = 0$) and $\mu_1, \mu_2, \dots, \mu_{\mathfrak{m}}$ are trivial measures.
- III.- *Discrete-continuous case.* $\vec{\mu}$ is a $\mathfrak{m}$ -standard vectorial measure and $\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}-1}$ are trivial measures.

Although, the study of the Sobolev orthogonal polynomials might seem analogous to the standard case, it has already been noticed in [10] that the Sobolev case is different, and a new approach is required.

Unlike the standard polynomials, in the continuous Sobolev case the number of terms of the recurrence relation increases in the same way as the polynomial index does. In [135, §II.6] it is obtained a relation which describes the structure of the recurrence formula that satisfy the continuous Sobolev polynomials and its connection with the corresponding moment matrix. In the discrete case, a higher order term recurrence relation is satisfied (see (1.18) in the next section) while in the discrete continuous case the only result available in the literature is [9, Th. 5.1]. Here the authors found, for a general class of discrete-continuous Sobolev inner products, the only three families of orthogonal polynomials that satisfy a generalized three term recurrence relation (i.e. (1.3) without the restriction $a_n \neq 0$).

Another essential difference appears in the study of the location of its zeros. While in the standard case all zeros are simple and lie in the interior of the convex hull of the support of the measure (see Theorem 1.2.3), in the Sobolev case this does not remain true. In general, the zeros of Sobolev polynomials are neither simple nor located in the interior of the support of the involved measures, and they may even be located outside of the real line.

Despite of this, some partial results can be stated. In the continuous case, theorems similar to Theorem 1.2.3 were obtained in the framework of coherent pairs, see [35],

[37], and [61]. In the discrete case, previous to this work, the most general result concerning the location of the zeros of S_n was proved in [3] (this result is enunciated as Theorem 3.1.1 in Chapter 3). In Theorem 3.1.2 this result is extended here for a more general class of discrete inner products. Lastly, in the discrete-continuous case the behavior of the zeros is quite different, see [16, Th. 2.1, Th. 2.2 and Th. 2.3] for a study of the zeros of S_n when the nontrivial measure μ_m is the Gegenbauer measure (Jacobian measure with $\alpha = \beta$).

What has been proved in very general cases is that the zeros of $\{S_n\}_{n \geq 0}$ conform a bounded set of the complex plane, whereas the supports of the involved measures are bounded, see [114, Cor. 1], and [47, Th. 1.1] for the continuous case and [91, Th. 4] for the discrete case. When the inner product is discrete-continuous the boundedness of the zeros has not been considered explicitly beyond when the nontrivial measure is the Gegenbauer one, see [16, Th. 2.2]. But in the case considered in [9], see (1.27) in Section 1.4.4, the boundedness of the zeros of $\{S_n\}_{n \geq 0}$ is a consequence of [9, Th. 2.1] and the application of Biernacki's Lemma (see Lemma 2.3.1).

As we have seen, the Sobolev orthogonality is not only different to the standard one, but also the three classifications are very different to each other. These three essential classifications will be addressed in each of the following three Chapters in the same order that they have been presented. In the following section a historical overview is also included with the sake of introducing the state of the art of those results that precede this work. Readers who want to learn more about Sobolev orthogonal polynomials may refer to the surveys [5, 121, 108, 110, 98, 106] and the references therein.

1.4 Historical overview

1.4.1 Earlier results

The origins of Sobolev orthogonal polynomials date back to 1947 in a least square approximation problem studied by D.C. Lewis in [82]. The problem consists in obtaining the polynomial that is the best least square approximation to a function and, simultaneously, to its derivatives. Specifically, given $\alpha_0, \alpha_1, \dots, \alpha_m$ monotonic, non-decreasing functions defined on $[a, b]$ and a sufficiently regular function f , find the polynomial P_n of degree at most n that minimizes

$$\sum_{k=0}^m \int_a^b \left| f^{(k)}(x) - P_n^{(k)}(x) \right|^2 d\alpha_k(x).$$

However, Lewis just focused on the remainder $f - P_n$ and did not say anything else about how to construct the polynomial P_n .

It was P. Althammer [10] who really introduced the Sobolev orthogonal polynomials in 1962 through a new interpretation of Lewis's problem. However, it was not until 1975 that E.A. Cohen [32] began to call them "Orthogonal Polynomials in Sobolev Space". Althammer showed that P_n is the n th partial sum for the Fourier series of

f with respect to $\{s_n\}_{n \geq 0}$, the orthonormal polynomial sequence for the inner product

$$\langle f, g \rangle_s = \sum_{k=0}^{\mathfrak{m}} \int_a^b f^{(k)}(x) g^{(k)}(x) d\alpha_k(x), \quad (1.11)$$

This means (see Theorem 1.1.1)

$$P_n(x) = \mathcal{F}_n(f, x) = \sum_{k=0}^n \langle f, s_k \rangle_s s_k(x).$$

In his paper, Althammer presented Sobolev polynomials as a natural extension of standard orthogonal polynomials ($\mathfrak{m} = 0$) and showed some analogies between them. However, since this early paper, Althammer alerted about one of the essential differences, the zero location.

It is a well known result that the zeros of standard orthogonal polynomials are all simple and lie on the interior of the support interval (a, b) , see Theorem 1.2.3. In his paper, Althammer found the polynomial $S_2(x) = k_2(x^2 + \frac{27}{35}x - \frac{1}{3})$, orthogonal with respect to the Sobolev inner product

$$\langle f, g \rangle_s = \int_{-1}^1 f(x)g(x)dx + 10 \int_{-1}^0 f'(x)g'(x)dx + \int_0^1 f'(x)g'(x)dx$$

(take in (1.11) $a = -1$, $b = 1$, $\mathfrak{m} = 1$, $d\alpha_0(x) = dx$, and $d\alpha_1(x) = 10dx$ if $-1 \leq x < 0$, $d\alpha_1(x) = dx$ if $0 < x \leq 1$), which has a zero $x_1 = -1.08$ outside of the support interval $[-1, 1]$. This example was enough to take aside the general case and just consider maybe one of the simplest Sobolev inner products,

$$\langle f, g \rangle_s = \int_{-1}^1 f(x)g(x)dx + \lambda \int_{-1}^1 f'(x)g'(x)dx, \quad \lambda > 0. \quad (1.12)$$

Althammer studied systematically the Sobolev polynomials $\{s_n\}_{n \geq 0}$ orthonormal with respect to this Legendre-Sobolev inner product and obtained as main results:

1. The polynomial s_n has n simple zeros and located in $(-1, 1)$, for all $n \geq 1$.
2. The polynomials $\{s_n\}_{n \geq 0}$ satisfy a second order linear differential equation in the form

$$\lambda s_n''(x) - s_n(x) = A_n \frac{d}{dx} \hat{P}_{n+1}(x) + B_n \frac{d}{dx} \hat{P}_{n-1}(x),$$

where A_n and B_n are constant sequences calculated explicitly by Althammer in [10] and $\hat{P}_n$ denotes the n th monic Legendre polynomial (i.e. Jacobi polynomial for $\alpha = \beta = 0$).

3. The polynomials s_n satisfy the differential recurrence relation

$$s'_{n+2}(x) = (n+2)\kappa(s_{n+2})\hat{P}_{n+1}(x) + \frac{s_{n+2}(1)}{s_n(1)}s'_n(x),$$

and, as above, $\hat{P}_n$ denotes the n th monic Legendre polynomial .

In 1967, these Legendre-Sobolev polynomials were studied independently by W. Gröbner [60], who also established a generalization of Rodrigues' formula for them. In the subsequent years, more papers arised extending and simplifying the results of Althammer (see [22] [81] [155] [156] and [32]). J. Brenner [22] considered the Laguerre-Sobolev inner product

$$\langle f, g \rangle_s := \int_0^\infty f(x)g(x)e^{-x}dx + \lambda \int_0^\infty f'(x)g'(x)e^{-x}dx, \quad \lambda > 0 \quad (1.13)$$

and obtained similar results to those presented by Althammer. Finally, P. Lesky [81], and F. W. Schafke and G. Wolf [156] worked on the case when higher order derivatives appear in the inner product. Lesky considered essentially the case of constants weights

$$\langle f, g \rangle_s = \sum_{k=0}^m \lambda_k \int_a^b f^{(k)}(x)g^{(k)}(x)dx.$$

and gave a Rodrigues' type formula for the corresponding Sobolev polynomials. On the other hand, F. W. Schafke and G. Wolf extended all earlier results considering a more general inner product, namely

$$\langle f, g \rangle_s = \sum_{j,k=0}^\infty \int_a^b f^{(j)}(x)g^{(k)}(x)v_{j,k}(x)w(x)dx, \quad (1.14)$$

where w is one of the classical weights and the functions $v_{j,k}$ are polynomials under some restrictions which were classified into eight different classes. However, the most general situation, where the integrals in (1.11) are considered with respect to arbitrary positive measures μ_k , see (1.9), or even measures defined by weights far from the classical ones, had to be postponed because of the lack of the integration by parts method in this new framework.

This first period (1962-1973) was characterized by watching to Sobolev orthogonal polynomials as an extension of the classical ones. Consequently, the main goal was to extend its algebraic and zero location properties as well as finding recursive and differential formulas that connect this two different kind of polynomials.

1.4.2 Discrete Sobolev Polynomials

For about fifteen years, the Sobolev orthogonal polynomials seemed to be a research area closed by the work of F. W. Schafke and G. Wolf [156]. It was at the end of the 80's when the Sobolev Polynomials waked up from this lethargy by means of the emergence of the so-called discrete case. These new Sobolev polynomials arose in the environment of the problem of finding orthogonal polynomial sequences $\{P_n\}_{n \geq 0}$ that solve linear differential equations with polynomial coefficients, whose degree does not depend on n .

In 1938 [72], H. L. Krall considered the linear differential equations in the form

$$\sum_{k=1}^r a_k(x)y_n^{(k)}(x) = \lambda_n y_n(x), \quad (1.15)$$

where every a_k has degree at most k . Note that the second differential equation of the classical polynomials (1.8) is the particular case $r = 2$. In this paper, Krall gave conditions equivalent to this equation has standard orthogonal polynomial sequences as solutions. By means of this characterization, he proved that the equations (1.15) of odd order ($r = 2k + 1$) are not solvable in this sense and gave an example of a fourth order differential equation that has a nonclassical polynomial solution. Two years later [73], he totally classified the standard orthogonal polynomials that solve the fourth order differential equations and showed that three of them, including the one founded in 1938, were nonclassical. However, its orthogonality measures are close to be one, since they are defined by classical weights plus a delta function at the end point(s) of the support interval:

1. Legendre type

$$d\mu(x) = \frac{\alpha}{2}dx + \frac{1}{2}(\delta(x-1) + \delta(x+1)) \quad \alpha > 0, \quad x \in [-1, 1].$$

2. Laguerre type

$$d\mu(x) = e^{-x}dx + \frac{1}{R}\delta(x), \quad R > 0, \quad x \in [0, \infty).$$

3. Jacobi type

$$d\mu(x) = (1-x)^\alpha dx + \frac{1}{M}\delta(x), \quad M > 0, \alpha > -1, \quad x \in [0, 1].$$

More than four decades later, with the rise of higher order self-adjoint boundary value problems these topics were resurrected by Allan M. Krall, the son of H. L. Krall. Allan studied exhaustively these three nonclassical families, naming them the Legendre type, Laguerre type and Jacobi type polynomials due to obvious reasons [74]. These results promoted the study of polynomials orthogonal with respect to an absolutely continuous measure perturbed by adding finitely many delta functions, which are currently known as Krall-type orthogonal polynomials. The references [128, Lem. 15], [85, 87, 71, 86, 23, 103, 79] and specially the more recent reviews [11] and [75, Chap. XV] are recommended for further studying of this kind of orthogonal polynomials.

Discrete-Sobolev orthogonal polynomials were studied by first time in 1989 by Bavinck and Meijer [14] as a natural extension of Krall-type polynomials. They considered the inner product

$$\begin{aligned} \langle f, g \rangle_s = & C_\alpha \int_{-1}^1 f(x)g(x)w(x)dx + M[f(1)g(1) + f(-1)g(-1)] \\ & + N[f'(1)g'(1) + f'(-1)g'(-1)], \end{aligned}$$

where w is the classical Gegenbauer weight $w(x) = (1-x^2)^\alpha$ (Jacobi weight with $\alpha = \beta$), $N, M \geq 0$ and C_α is a specific positive constant depending on $\alpha > -1$. In this paper, they proved that the corresponding Gegenbauer-Sobolev polynomials $\{S_n\}_{n \geq 0}$ satisfy the differential equation

$$p_{4,n}(x)(1-x^2)S_n''(x) + xq_{4,n}(x)S_n'(x) + r_{4,n}(x)S_n(x) = 0,$$

where $p_{4,n}$, $q_{4,n}$ and $r_{4,n}$ are appropriate polynomials of degree at most 4. It worth to remark that unlike (1.15), all the polynomial coefficients of this equation depend on n . In addition, they found that S_n can be written as

$$S_n(x) = \left[a_n x^2 \frac{d^2}{dx^2} + b_n x \frac{d}{dx} + c_n \right] R_n^{(\alpha)}(x),$$

where a_n , b_n and c_n are constants given by explicit formulas and $R_n^{(\alpha)}$ is the n th Gegenbauer polynomial. Furthermore, an Hypergeometric representation of the polynomials is provided. Subsequently, the authors [15] prove that the zeros of S_n are all real and simple and, for n large enough, exactly two of them lie outside of the interval support $[-1, 1]$. A five term recurrence relation for these polynomials is also provided.

Afterwards, many papers were published extending this work. R. Koekoek and H.G. Meijer [69, 70] dealt with the Laguerre case considering the inner product

$$\langle f, g \rangle_s = \frac{1}{\Gamma(\alpha+1)} \int_0^\infty f(x)g(x)x^\alpha e^{-x} dx + \sum_{v=0}^{\mathfrak{m}} M_v \cdot f^{(v)}(0)g^{(v)}(0),$$

where Γ denotes the gamma function, $\alpha > -1$ and $M_v \geq 0$ for $v = 0, 1, \dots, \mathfrak{m}$. They proved the following results similar to those given by Bavinck and Meijer, but with the notable improvement of considering an arbitrary derivation order.

1. Representation formula. S_n can be written as

$$S_n(x) = \sum_{k=0}^{\mathfrak{m}+1} a_{n,k} \frac{d^k}{dx^k} L_n^{(\alpha)}(x),$$

where $a_{n,k}$ are constants, explicitly calculated when $\mathfrak{m} = 1$.

2. Differential equation

$$x p_{2,n}(x) S_n''(x) - p_{1,n}(x) S_n'(x) + n p_{0,n}(x) S_n(x) = 0 \quad (1.16)$$

where $p_{2,n}$ and $p_{0,n}$ are polynomials of degree $\mathfrak{m} + 1$ and $p_{1,n}$ has degree $\mathfrak{m} + 2$.

3. S_n satisfies the $2\mathfrak{m} + 3$ term recurrence relation

$$x^{\mathfrak{m}+1} S_n(x) = \sum_{k=n-\mathfrak{m}-1}^{n+\mathfrak{m}+1} C_{k,n} S_k(x), \quad (1.17)$$

$C_{k,n}$ being real constants.

Finally, in [104], F. Marcellán and A. Ronveaux made a study of the inner product

$$\langle f, g \rangle_s = \int f(x)g(x)w(x)dx + M f^{(\mathfrak{m})}(c)g^{(\mathfrak{m})}(c),$$

where w is an arbitrary weight, $c \in \mathbb{R}$, $M \geq 0$ and $\mathfrak{m}$ is a nonnegative integer. They found the following equation, which links the Sobolev polynomials S_n with the standard polynomials P_n orthogonal with respect to $w(x)$:

$$(x - c)^{\mathfrak{m}+1} S_n(x) = A_{\mathfrak{m}+1}(x; n) P_n(x) + B_{\mathfrak{m}}(x; n) P_{n-1}(x)$$

where $A_{\mathfrak{m}+1}$ and $B_{\mathfrak{m}}$ are polynomials of degree at most $\mathfrak{m} + 1$ and $\mathfrak{m}$, respectively, depending on all parameters of the inner product and whose calculation formulas are given. A similar $2\mathfrak{m} + 3$ recurrence relation is provided and when the weight w is semiclassical they also found a second order differential equation like (1.16). A weight w is said to be semiclassical if, there exist polynomials ϕ and ψ such that

$$(\phi w)' = \psi w.$$

They conclude the paper applying these results to Hermite case ($w(x) = e^{-x^2}$) when $\mathfrak{m} = 1$ and $c = 0$.

The Discrete Sobolev inner products, also named Sobolev-type inner products can be defined, in general, as

$$\langle f, g \rangle_s = \int_a^b f^{(k)}(x) g^{(k)}(x) d\mu(x) + \sum_{j=1}^{\mathcal{N}} \sum_{k=0}^{\ell_j} M_{j,k} f^{(k)}(c_j) g^{(k)}(c_j),$$

Most of the properties of the polynomials orthogonal with respect to this kind of inner products are derived from the existence of a polynomial that kills the discrete part. It is easy to verify that the polynomial $\rho(x) = \prod_{i=1}^{\mathcal{N}} (x - c_i)^{\ell_i+1}$ satisfies

$$\langle \rho P, Q \rangle_s = \langle \rho P, Q \rangle_{\mu_0} = \langle P, \rho Q \rangle_s, \quad P, Q \in \mathbb{P}.$$

In a similar way as it was made for the standard case, see Theorem 1.2.3, using this property one obtains that S_n has at least $n - \deg(\rho)$ zeros in the interior of $\Delta = \mathbf{C}_h(\text{supp}(\mu))$. A comprehensive study of the location and interlacing properties of the zeros of the polynomials S_n , orthogonal with respect to inner products in the form

$$\langle f, g \rangle_s = \int f^{(k)}(x) g^{(k)}(x) d\mu(x) + \sum_{k=0}^{\mathfrak{m}} M_k f^{(k)}(c) g^{(k)}(c),$$

for arbitrary weights w , $\mathfrak{m} \in \mathbb{N}$, $c \in \mathbb{R}$ and $M_k \geq 0$ for $k = 0, 1, \dots, \mathfrak{m}$, can be found in [6, 99, 34, 117, 118, 119, 132, 3, 68]. The most general result on the location of the zeros of Sobolev polynomials is given in [3, Th. 2.2] where the authors prove that S_n has at least $n - \ell^*$ changes of sign in $\mathbf{C}_h(\text{supp}(\mu))^\circ$. In this manuscript this result is extended to inner products having more than one mass point, under some order restrictions in the discrete part of $\langle \cdot, \cdot \rangle_s$, see Theorem 3.1.2.

With the development of the theory and taking into account the $2\mathfrak{m} + 3$ term recurrence relation (1.17), the idea of generalize Theorem 1.2.2 for the discrete Sobolev polynomials arose in a natural way. It is not difficult to prove that the following conditions are equivalent for any monic polynomial ρ of degree ℓ and a polynomial sequence $\{p_n\}_{n \geq 0} \subset \mathbb{P}$ such that the first ℓ polynomials satisfy $\deg(p_n) = n$ and $\kappa(p_n) > 0$ for $n = 0, \dots, \ell - 1$.

1. There exist complex constants $\mathfrak{a}_{n,m}$ satisfying $\overline{\mathfrak{a}_{n,m}} = \mathfrak{a}_{m,n}$ and $\mathfrak{a}_{n,n+\ell} > 0$ for $n, m = 0, 1, 2, \dots$ such that the $2\ell + 1$ recurrence relation

$$\rho(x)p_n(x) = \sum_{k=n-\ell}^{n+\ell} \mathfrak{a}_{n,k} p_k(x), \quad n = 0, 1, 2, \dots \quad (1.18)$$

holds, where, $0 \equiv p_{-1} \equiv p_{-2} \equiv \dots \equiv p_{-\ell}$ as usual.

2. There exists $\langle \cdot, \cdot \rangle$ an inner product over $\mathbb{P}$ having the sequence p_n as its orthonormal polynomial sequence and such that the operator of multiplication by ρ is symmetric, i.e.

$$\langle \rho P, Q \rangle = \langle P, \rho Q \rangle \quad P, Q \in \mathbb{R}[x]. \quad (1.19)$$

Note that from 1.) we get $\mathfrak{a}_{n,n} \in \mathbb{R}$ and $\mathfrak{a}_{n,n \pm \ell} > 0$, but except for the case $\ell = 1$ (standard case) we cannot guarantee that the remaining coefficients $\mathfrak{a}_{n,m}$, and, consequently, the coefficients of p_n , are real, unless the trivial case $\mathbb{P} = \mathbb{R}[x]$.

In [46], A. J. Durán found a characterization for real inner products $\langle \cdot, \cdot \rangle$ (i.e. $\mathbb{P} = \mathbb{R}[x]$) to be written “discretely”. He proved that the following statements are equivalent for any polynomial $\rho = \prod_{i=1}^{\mathcal{N}} (x - c_i)^{\ell_i + 1}$ of degree $\ell := \mathcal{N} + \sum_{i=1}^{\mathcal{N}} \ell_i$ and having real zeros c_i , see [46, Lem. 3].

1. The inner product $\langle \cdot, \cdot \rangle$ satisfies
 - a) The operator multiplication by ρ is symmetric (see (1.19)).
 - b) The polynomial ρ commutes with $I(x) = x$

$$\langle \rho P(x), IQ \rangle = \langle IQ, \rho P \rangle \quad P, Q \in \mathbb{R}[x]$$

2. There exist a measure μ and real constants $M_{i,j,k,l}$ with $0 \leq i \leq \ell_k$, $0 \leq j \leq \ell_l$, $1 \leq k, l \leq \ell - 1$ and $M_{i,j,k,l} = M_{j,i,l,k}$ such that the inner product $\langle \cdot, \cdot \rangle$ takes the form

$$\langle P, Q \rangle = \int f(t)g(t)d\mu(t) + \sum_{k,l=1}^{\mathcal{N}} \sum_{i=0}^{\ell_k} \sum_{j=0}^{\ell_l} M_{i,j,k,l} f^{(i)}(c_k) g^{(j)}(c_l)$$

A second generalization of condition 1. in Theorem 1.2.2 was given in [48]. Here the authors proved that if the inner product is restricted to be a Sobolev inner product $\langle \cdot, \cdot \rangle_s$, i.e. defined by (1.9), then (1.19) and, consequently, also (1.18), are equivalent to the Sobolev inner product is discrete.

Once the principal algebraic results were obtained, the question about the behavior of the Sobolev polynomial sequence $\{S_n\}_{n \geq 0}$ when n tends to infinity quickly arised. The first paper dealing with theses asymptotic questions was published in 1993 by F. Marcellán and W. Van Assche [105] and generalized two years later by the authors together with G. López Lagomasino in [91]. They consider the measure μ belonging

to the Nevai class $\mathbf{M}(0, 1)$, this means that the coefficients in the three term recurrence relation (1.3) satisfy

$$\lim_{n \rightarrow \infty} a_n = \frac{1}{2}, \quad \lim_{n \rightarrow \infty} b_n = 0. \quad (1.20)$$

This class is broader than what it seems at first sight, E. A. Rakhmanov proved in [143] and [144] that $\mathbf{M}(0, 1)$ includes all absolutely continuous measures $d\mu = w(x)dx$, $-1 < x < 1$, such that $w(x) > 0$ almost everywhere on $[-1, 1]$. Simplified proofs may be found in [115] and [145]. In the case $c_j \in \mathbb{R} \setminus \text{supp}(\mu)$ for $j = 1, 2, \dots, \mathcal{N}$, they proved

$$\frac{\hat{S}_n(z)}{\hat{P}_n(z)} \Rightarrow \prod_{j=1}^{\mathcal{N}} \left(\frac{(\varphi(z) - \varphi(c_j))^2}{2\varphi(z)(z - c_j)} \right)^{\ell_j^*}, \quad K \subset \overline{\mathbb{C}} \setminus \text{supp}(\mu), \quad (1.21)$$

where $\{\hat{P}_n\}_{n \geq 0}$ is the sequence of orthogonal polynomials with respect to μ , ℓ_j^* is the number of coefficients $M_{j,i}$ different from zero ($i = 0, 1, \dots, \ell_j$) and recall that $\varphi(z) = z + \sqrt{z^2 - 1}$, with $\sqrt{z^2 - 1} > 0$ for $z > 1$, is the conformal function which maps the complement of $[-1, 1]$ onto the exterior of the unit circle. This is known as the outer relative asymptotics of the Sobolev-type polynomials. Thus, the (strong) asymptotic behavior of the polynomials $\hat{S}_n$ outside the support can be deduced directly from the one of $\{\hat{P}_n\}_{n \geq 0}$. When the measure $d\mu(x) = w(x)dx$ satisfies the Szegő condition, $\mu \in \mathcal{S}$, on $[-1, 1]$

$$\int_{-1}^1 \frac{\ln w(x)}{\sqrt{1-x^2}} dx > -\infty, \quad (1.22)$$

the strong asymptotic behavior of $\hat{P}_n$ is known, so we can obtain a general formula in this case. This was showed by A.F. Moreno and F. Marcellán [50, Cor. 5] for the more convenient normalization $S_n = \|\hat{S}_n\|_{\mu} \hat{S}_n$, where in such a paper they also found the strong inner asymptotics of S_n (for $z \in \text{supp}(\mu)$). See also [6] for a deeper study when $\mathcal{N} = 1$ and $\ell_1 = 1$. It is worth to be mentioned that all these works consider more general class of inner products not necessarily positive definite and the mas points c_j may be taken in $\mathbb{C} \setminus \text{supp}(\mu_0)$. Some other generalizations have been made when the support of μ is not located in the real line. When it is in the unit circle, a formula similar to (1.21) for the relative asymptotics of S_n can be found in [83] and [51] as well as in [21] when the measure is supported on a rectifiable Jordan curve in the complex plane.

The situation is quite different when the support of the measure μ is unbounded and there is not much that one can says. Results only appear essentially when the measure μ is the Laguerre measure and $\langle \cdot, \cdot \rangle_s$ takes the form

$$\langle f, g \rangle_s = \int_0^\infty f(x)g(x)x^\alpha e^{-x} dx + \sum_{k=0}^{\mathfrak{M}} M_k f^{(k)}(c)g^{(k)}(c).$$

See [12, Th. 1 and Th. 2] for the case $\mathfrak{M} = 1$ and $c = 0$ and also [45] for an alternative study when $M_0 = 0$. The case when an arbitrary order derivative $\mathfrak{M}$ appears in the inner

product was dealt in [7], where the authors also found some asymptotic properties for inner products in the above form but taking the measure $d\mu(x) = |x|^{2u}e^{-x^2}$ ($u > -\frac{1}{2}$) a generalized Hermite measure. Finally, the case $\mathfrak{M} = 1$ and $c < 0$ was studied in [107].

1.4.3 Continuous Sobolev Polynomials

The research on continuous Sobolev orthogonal polynomials is resumed again in 1991 when the notion of coherent pairs is introduced in [64]. In this paper, the authors rescued the classical Althammer's example (1.12) and considered the more general inner product

$$\langle f, g \rangle_s = \int_a^b f(x)g(x)d\mu_0(x) + \lambda \int_a^b f'(x)g'(x)d\mu_1(x), \quad \lambda > 0, \quad (1.23)$$

where $-\infty \leq a < b \leq \infty$ and $\vec{\mu} = (\mu_0, \mu_1)$ is a continuous vectorial measure. The definition of a coherent pair $\vec{\mu}$ given in [64] is equivalent to the existence of a nonzero real numbers sequence $\{a_n\}_{n \geq 1}$ satisfying

$$\hat{P}_{1,n}(z) = \frac{\hat{P}'_{0,n+1}(z)}{n+1} + a_n \frac{\hat{P}'_{0,n}(z)}{n}, \quad n \geq 1, \quad (1.24)$$

where $\hat{P}_{i,n}$ is the n th monic orthogonal polynomial of degree n with respect to μ_i , for $i = 0, 1$, see [64, Th. 3]. In addition, if $[a, b] = [-c, c]$ and the measures μ_i are both symmetric, then the vectorial measure is called symmetrically coherent if

$$\hat{P}_{1,n}(z) = \frac{\hat{P}'_{0,n+1}(z)}{n+1} + a_n \frac{\hat{P}'_{0,n-1}(z)}{n-1}, \quad n \geq 2.$$

The main idea that originated this definition is the search for a recurrence relation which, unlike the standard and discrete cases, does not hold in general. One of the first examples of each definition of coherent pairs, which are not included in the families considered by F. W. Schafke and G. Wolf (1.14), are

$$\text{Laguerre-Sobolev:} \quad d\mu_0 = d\mu_1 = x^\alpha e^{-x} dx, \quad x \geq 0, \quad \alpha > -1.$$

$$\text{Gegenbauer-Sobolev:} \quad d\mu_0 = d\mu_1 = (1-x^2)^{\alpha-\frac{1}{2}} dx, \quad -1 \leq x \leq 1, \quad \alpha > -\frac{1}{2}.$$

The algebraic and differential properties of the corresponding Sobolev orthogonal polynomials as well as its relation with the classical orthogonal polynomials was obtained in [101] and [100] respectively. In the case of Laguerre-Sobolev polynomials, the authors prove that their zeros are real and simple. Also, if $\alpha \geq 0$ all zeros are positive, otherwise if $-1 < \alpha < 0$, at most one of them is negative. Furthermore, they proved that these Sobolev polynomials satisfy a four-term recurrence relation, see [101, Cor. 4.4]. In the case of Gegenbauer-Sobolev polynomials some properties concerning the localization and separation of the zeros are also deduced, see [100, Th. 5.6].

Another interesting property on zero location of the Sobolev polynomials in the context of coherent pairs is obtained when values of λ large enough are considered. In 1993, H.G. Meijer [116] proved that if λ is sufficiently large, then the Sobolev polynomial S_n orthogonal with respect to the inner product (1.23) has all its zeros real, simple and interlace the zeros of $\hat{P}_{0,n-1}$ and $\hat{P}_{1,n-1}$. Meijer also proved that if (μ_0, μ_1) is a symmetrically coherent pair, the corresponding Sobolev polynomials may have a pair of complex conjugate zeros. Finally in 1995, Meijer [122] provided a complete classification of the coherent pairs, where he proved that if (μ_0, μ_1) is a coherent pair then necessarily one the measures must be classical, so this class is far from being a general case. For this reason, the study of coherent pairs has been expanded to more general definitions. For example, a vectorial measure (μ_0, μ_1) is called a generalized coherent pair if there exist real constants A_n, B_n and C_n with $C_n \leq 0$ such that

$$\hat{P}_{1,n}(z) = A_n \hat{P}'_{0,n+1}(z) + B_n \hat{P}'_{0,n}(z) + C_n \hat{P}'_{0,n-1}(z), \quad n \geq 1,$$

In [36, Th. 4.1], H. G. Meijer and M. G. de Bruin extended the above result on location of the zeros of S_n for λ sufficiently large, when the measures in the inner product (1.23) are a generalized coherent pair. In addition, they proved that, in this case, the Sobolev polynomials satisfy a 5-term recurrence relation, which is reduced to 4-term in the case of coherent pairs.

Finally, in 2002 a series of three papers was published by M. G. de Bruin, W. G. M. Groenevelt, and H. G. Meijer [37, 35, 61] where the zero location problem in the coherent pair context was totally solved. They proved that if (μ_0, μ_1) is a coherent pair of measures, then the zeros of S_n are real, simple, they are a shifted to the left with respect to the zeros of $P_{0,n}$ and are separated by the zeros of $P_{0,n-1}$ see [37, Th. 4.1]. For symmetrically coherent pairs they proved that S_n has at least $n - 2$ different real zeros and when the classical measure is the second one (μ_1), then all the zeros are real and simple, see [35, Th. 4.7 and Th. 5.7] and [61, Th. 5.1, Th. 5.3, Th. 5.5, Th. 5.6 and Th. 5.7]. In addition, if the classical measure is the first one (μ_0), certain conditions can be found such that S_n has a pair of conjugate complex zeros.

However, the situation is quite different outside of the coherent pair framework. In [120], Meijer considered the inner product

$$\langle f, g \rangle_s = \int_{-1}^3 f(x)g(x)dx + \lambda \int_{-1}^1 f'(x)g'(x)dx + \int_1^3 f'(x)g'(x)dx, \quad \lambda > 0$$

and proves that if λ is sufficiently large, then the corresponding Sobolev polynomials S_{2n} and S_{2n+1} have exactly two and one real zeros, respectively. One of the zeros of S_{2n} is contained in $(-3, -1)$ and the other one in $(1, 3)$, whereas the real zero of S_{2n+1} lies on $(1, 3)$.

An important paper which lights the way towards solving the problem on zero location of Sobolev Polynomials in the continuous case is [55]. Here, W. Gautschi and A. B. J. Kuijlaars proved that, under mild conditions over the measures μ_0 and μ_1 , the derivatives of the Sobolev polynomials orthogonal with respect to the inner product

$$\langle f, g \rangle_s = \int f g d\mu_0 + \int f' g' d\mu_1$$

have regular asymptotic zero distribution. This means

$$\lim_{n \rightarrow \infty} \int f d\sigma(S'_n) = \int f d\mu_{\Delta_0 \cup \Delta_1}$$

for every continuous function f on $\overline{\mathbb{C}}$, where $\mu_{\Delta_0 \cup \Delta_1}$ is the equilibrium measure of $\Delta_0 \cup \Delta_1$ (see [146, §3.3]) and $\sigma(S'_n)$ is the *normalized zero counting measure associated with S'_n* , defined by

$$\sigma(S'_n) = \frac{1}{n-1} \sum_{j=1}^{n-1} \delta_{z_j},$$

where $\{z_j\}_{j=1}^{n-1}$ are the $n-1$ zeros of S'_n . So, in this sense, the critical points of S_n have a behavior similar to the zeros of orthogonal polynomials in the standard case (see [162, Def. 3.1.3 and Th. 3.1.4]). Unfortunately, this does not imply that the critical points of S_n accumulate on $\Delta_0 \cup \Delta_1$, it does not even guarantee that the set of zeros of $\{S_n\}_{n \geq 0}$ is bounded. The authors also provided several numerical experiments where in all of them the critical points of S_n are contained in the convex hull of $\Delta_0 \cup \Delta_1$ (similar to what do the zeros in the standard case, see Theorem 1.2.3). This fact was proved by G. L. Lagomasino I. P. Izquierdo and H. Pijeira in [92, Th. 1.3.] when the interior of the intervals Δ_0 and Δ_1 are disjoint. In Chapter 2 we generalize this latter result to extremal polynomials, see Proposition 2.2.1, Theorem 2.1.1 and Corollary 2.3.0.1.

The problem of the boundedness of the zeros of Sobolev orthogonal polynomials, in the case of the support of the involved measures is bounded, has been solved for a wide class of vectorial measures, in several cases as a consequence of finding the relative asymptotics of the Sobolev polynomials. First, the study became by particular cases as the Gegenbauer-Sobolev polynomials (see [113, Th. 1]) and extended later to the coherent pairs framework, see [112, Th. 1] for coherent pairs and [94, Th. 7] for symmetrically coherent pairs. In [111, Th. 2] a first step out of this framework was made by proving that the same relative asymptotics holds considering the inner product

$$\langle f, g \rangle_s = \int f(x)g(x)d\mu(x) + \lambda \int_{-1}^1 f'(x)g'(x)(1-x)^\alpha(1+x)^\beta dx$$

where $\alpha, \beta > 0$ and μ is a measure on $[-1, 1]$ satisfying a certain condition more general than forming a coherent pair with $d\mu_1 = (1-x)^\alpha(1+x)^\beta dx$. Finally, A. Martínez Finkelshtein generalized all these previous results in [109], proving that for every continuous vectorial measure (μ_0, μ_1) such that $\mu_0, \mu_1 \in \mathcal{S}$ (Szegő condition on $[-1, 1]$ see (1.22)) the following relative asymptotics holds

$$\frac{S_n(z)}{P_{1,n}(z)} \xrightarrow[n]{\Rightarrow} \frac{2}{\varphi'(z)}, \quad K \subset \overline{\mathbb{C}} \setminus [-1, 1].$$

This series of papers concerning the relative asymptotics of the continuous Sobolev polynomials finished with the extension of the above result to the case when higher order derivatives appear in the inner product (i.e. $\mathfrak{M}$ arbitrary). In [114], A. Martínez

Finkelshtein and H. Pijera proved that if the vectorial measure $(\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ satisfies that every measure $\mu_k \in \mathcal{S}$ and

$$\text{supp}(\mu_k) \subset \text{supp}(\mu_{\mathfrak{m}}) = [-1, 1], \quad k = 0, 1, \dots, \mathfrak{m} - 1 \quad (1.25)$$

then

$$\frac{S_n(z)}{P_{\mathfrak{m},n}(z)} \xrightarrow[n]{} \frac{2^{\mathfrak{m}}}{[\varphi'(z)]^{\mathfrak{m}}}, \quad K \subset \overline{\mathbb{C}} \setminus [-1, 1]. \quad (1.26)$$

In all of these papers, the authors proved that the zeros of S_n accumulate on $\text{supp}(\mu_{\mathfrak{m}})$ as a consequence of the relative asymptotics. Note also that, as it was mentioned in the above section for the discrete case, the strong asymptotics of Sobolev polynomials is a direct consequence of the relative one. The surveys [108] and [110] can be recommended for an in depth study of this subject.

In a very different way, H. Pijera and G. L. Lagomasino proved in [90] that if the multiplication operator is bounded, i.e.

$$\|xP(x)\|_s \leq C \|P(x)\|_s, \quad P \in \mathbb{P},$$

for some constant C independent on P , then the zeros of S_n are contained in the set $\{z \in \mathbb{C} : |z| \leq 2C\}$. In [92] the proof of this result was reduced to the following very simple one

Let z_0 be a zero of S_n , then $S_n(z) = (z - z_0)q_{n-1}(z)$ with $\deg(q_{n-1}) \leq n - 1$, then

$$\begin{aligned} |z_0|^2 \|q_{n-1}\|_s^2 &= \|z_0 q_{n-1}\|_s^2 \leq \|S_n\|_s^2 + \|z_0 q_{n-1}\|_s^2 = \langle S_n + z_0 q_{n-1}, S_n + z_0 q_{n-1} \rangle_s \\ &= \|z q_{n-1}(z)\|_s^2 \leq C^2 \|q_{n-1}\|_s^2. \end{aligned}$$

So, simplifying $\|q_{n-1}\|_s^2$ and taking square root we conclude that the zeros of S_n are actually in $\{z \in \mathbb{C} : |z| \leq C\}$. $\square$

In contrast with the inclusion conditions (1.25), required for the relative asymptotics, the authors in [90] introduced the following class of vectorial measures for which the multiplication operator is bounded, whereas the supports of the involved measures are also bounded. A vectorial measure $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ is called sequentially dominated if satisfies

- $\Delta_0 \supset \Delta_1 \supset \dots \supset \Delta_{\mathfrak{m}}$.
- There exist functions $f_i \in L^\infty(\mu_{k-1})$, for $k = 1, 2, \dots, \mathfrak{m}$ such that

$$d\mu_k = f_k f_{k-1} \dots f_1 d\mu_0, \quad k = 1, 2, \dots, \mathfrak{m}.$$

Consequently, if this condition is satisfied the zeros of $\{S_n\}_{n \geq 0}$ are bounded.

In [148], J. M. Rodríguez proved that the boundedness of the multiplication operator over $\mathbb{P}$ implies that the Sobolev inner product is essentially sequentially dominated. This means that there exists a Sobolev norm sequentially dominated over $\mathbb{P}$ that is topologically equivalent to the induced norm of the Sobolev inner product. On

the other hand, in [26] M. Castro and A. J. Durán proved that the boundedness of the multiplication operator implies the compactness of the support of every measure involved in the inner product. So, putting it all together one obtains that the multiplication operator is bounded with respect to the norm induced by $\langle \cdot, \cdot \rangle_s$ if and only if there exists an inner product sequentially dominated with bounded support whose induced norm is topologically equivalent to that of $\langle \cdot, \cdot \rangle_s$.

In [47, Th. 1.1] A. J. Durán and E. B. Saff proved that the condition of being sequentially dominated is also sufficient to prove the location of the zeros in the unbounded case. Of course, here location means that the zeros are at a finite distance from the support. Namely, they proved that if z_0 is a zero of S_n orthogonal with respect to a sequentially dominated inner product, then

$$d(z_0, \Delta_0) := \inf_{x \in \Delta_0} |z_0 - x| \leq \frac{1}{2} \sqrt{\sum_{k=1}^{\mathfrak{M}} k^2 \|f_k\|_{\mathbf{L}^\infty(\mu_{k-1})}}.$$

This proof was later simplified in [43, Th. 1.3]. The authors also proved that this bound is sharp by means of an example.

In the bounded case, besides the strong asymptotics, which can be derived from (1.26), and the asymptotic zero distribution obtained by W. Gautschi and A. B. J. Kuijlaars in [55] for $\mathfrak{M} = 1$ and for general $\mathfrak{M}$ in [90], G. L. Lagomasino and H. Pijeira also found the n th root asymptotic behavior of $S_n^{(\mathfrak{M})}$ in the complement of a compact set containing the zeros of S_n , see [90, Th. 4], whenever the vectorial measure $\vec{\mu}$ is essentially sequentially dominated. This result was extended to the case of extremal Sobolev polynomials (i.e. polynomials satisfying condition (1.2) with the norm $\|\cdot\|_{s,p}$ defined in (1.10)) by the same authors together with I. P. Izquierdo in [93, Th. 6]. In Chapter 2 we found the n -root asymptotic behavior of the extremal polynomials but when the interior of the intervals Δ_0 and Δ_1 are disjoint, which is certainly not sequentially dominated. In addition, the convergence is obtained in the larger set $\mathbb{C} \setminus \mathbf{C}_h(\Delta_0 \cup \Delta_1)$, see Theorem 2.4.1.

In the continuous case with unbounded support the first asymptotic result was founded in 1976 by J. Brenner for the inner product (1.13) in the earlier paper [22]. Despite of this, it was not until 1997 when F. Marcellán et al. [95] took back this problem in consideration and extended the results of Brenner to the more general inner product

$$\langle f, g \rangle_s := \int_0^\infty f(x)g(x)x^\alpha e^{-x} dx + \lambda \int_0^\infty f'(x)g'(x)x^\alpha e^{-x} dx, \quad \lambda > 0, \alpha > -1.$$

Namely, they proved

$$\frac{S_n(x)}{L_n^{(\alpha-1)}(x)} \xrightarrow{n} \frac{2}{\sqrt{\lambda^2 + 4\lambda} - \lambda}, \quad k \in \mathbb{C} \setminus [0, \infty),$$

where, for $\alpha \in (-1, 0]$, $L_n^{(\alpha-1)}$ is interpreted in the generalized sense (1.6). As in the bounded case, the next goal was to extend this result in the framework of coherent pairs. This was achieved addressing separately each family of coherent pairs with

unbounded support, see [97, Th. 2 and Th. 4], [123, Th. 4.11], [8, Th. 2.3] and [125, Th. 2].

A first step taken out of the coherent pairs framework was given in [24]. Here the authors found the asymptotic properties of the polynomials orthogonal with respect to the inner product

$$\langle f, g \rangle_s = \int f(x)g(x)e^{-x^4} dx + \lambda \int_{\mathbb{R}} f'(x)g'(x)e^{-x^4} dx, \quad \lambda > 0.$$

They proved that

$$\begin{aligned} \frac{S_n(x)}{P_n(x)} &\underset{n}{\Rightarrow} \frac{3}{2}, \quad K \subset \mathbb{C} \setminus \mathbb{R} \quad \text{and} \\ \frac{S_n(\sqrt[4]{nx})}{P_n(\sqrt[4]{nx})} &\underset{n}{\Rightarrow} \frac{3\varphi^2(\sqrt[4]{\frac{3}{4}}x)}{3\varphi^2(\sqrt[4]{\frac{3}{4}}x) + 1}, \quad K \subset \mathbb{C} \setminus \left[-\sqrt[4]{3/4}, \sqrt[4]{3/4}\right], \end{aligned}$$

where P_n is orthogonal with respect to $e^{-x^4} dx$, see [24, Th. 2 and Th. 3]. Finally, a more general result was obtained by J. S. Geronimo, D. S. Lubinsky and F. Marcellán in [56]. He proved a close asymptotic relationship between the Sobolev polynomial S_n orthogonal with respect to the general inner product

$$\langle f, g \rangle_s = \int_{\mathbb{R}} fg\psi^2 e^{-2Q(x)} dx + \lambda \int_{\mathbb{R}} f'g' e^{-2Q(x)} dx$$

and the polynomial orthogonal with respect to $e^{-2Q(x)} dx$, where Q is a convex and continuous real function which satisfies some extra mild conditions and $\psi \in \mathbf{L}^\infty(\mathbb{R})$. In addition, the conditions on Q allows to take $Q(x) = |x|^\alpha$, $\alpha > 1$, and in such a case the strong asymptotics of S_n can be derived, see [56, Th. 1.1]. The reader is referred to the survey [97], where this topic is addressed in detail.

Concerning the weak asymptotic behaviors in the unbounded case, the n th root asymptotics and the zero distribution was found in [89, Cor. 3.1 and Cor. 3.2] for inner products of the general form

$$\langle f, g \rangle_s = \sum_{k=0}^m \int f^{(k)}(x)g^{(k)}(x)w_k(x)dx$$

where every weight function w_k satisfies the following relaxed condition as compared to [56]

$$\lim_{|x| \rightarrow \infty} \frac{-\ln(w_k(x))}{\tau|x|^\alpha} = 1$$

for some constants $\alpha, \tau > 0$. These results were extended in [43, Th. 1.2 and Cor. 1.1] and [44, Th. 1.2 and Th 1.3] for a wider class of weight functions. The reader is referred to the papers [89, 43, 44] to avoid the introduction of new terminologies.

1.4.4 Discrete-Continuous Sobolev polynomials

The first example in the literature involving discrete-continuous Sobolev polynomials appeared in the first period of the development of the Sobolev orthogonal polynomials (see Section 1.4.1) when Cohen [32] considered the case

$$\langle f, g \rangle_s = f(-1)g(-1) + \int_{-1}^1 f'(x)g'(x)dx.$$

Cohen, however, did not deal with the corresponding orthogonal polynomials. Indeed, this inner product does not appear explicitly in his paper, only its induced norm does it. He was just interested on the problem of finding the polynomial P_n , of degree at most n , that is the best approximation to a function f with respect to the norm induced by this inner product. Namely, he proved that this polynomial P_n is the limit when λ goes to infinite of its counterpart polynomial $P_{n,\lambda}$ with respect to the norm induced by the inner product (1.12) introduced by Althammer.

The interest in the study of discrete-continuous Sobolev polynomials grew up significantly in the second half of the 1990s with the problem of finding inner products on $\mathbb{P}$ having the generalized classical polynomials (1.6) and (1.7) as its orthogonal polynomial sequence. The most interesting case is when α or β take negative integer values, because in this case a_n in the three-term recurrence relation (1.3) vanishes for some values of n and Favard's theorem (see [27, Th. 4.4]) cannot be applied. For Laguerre case $\{L_n^{(-k)}\}_{n \geq 0}$ with $k \in \mathbb{N}$ this problem was solved by K. H. Kwon and L. L. Littlejohn [77] finding that this sequence is orthogonal with respect to the discrete-continuous inner product

$$\langle f, g \rangle_s = F(0)L(k)G(0)^t + \int_0^\infty f^{(k)}(x)g^{(k)}(x)e^{-x}dx$$

with $L(k)$ a given positive definite $k \times k$ real matrix (see [133, Lem. 3.1 and Eq. (4.1)]), $F(0) = (f(0), f'(0), \dots, f^{(k-1)}(0))$ and $G(0) = (g(0), g'(0), \dots, g^{(k-1)}(0))$, see also the previous paper of the same authors [78] for the case $k = 1$.

Although some other particular cases of discrete-continuous inner products were studied previously, see [132] and [67], the first unified approach was made by M. Alfaro, T. E. Pérez, M. A. Piñar, and M. L. Rezola [9] in 1999 considering the discrete-continuous Sobolev inner product

$$\langle f, g \rangle_s = F(c)AG(c)^t + \int_0^\infty f^{(\mathfrak{m})}(x)g^{(\mathfrak{m})}(x)d\mu(x), \quad (1.27)$$

where A is a positive-definite symmetric $\mathfrak{m} \times \mathfrak{m}$ real matrix, $c \in \mathbb{R}$, $F(c) = (f(c), f'(c), \dots, f^{(\mathfrak{m}-1)}(c))$ and $G(c) = (g(c), g'(c), \dots, g^{(\mathfrak{m}-1)}(c))$. Actually, the authors considered the orthogonality in a more general framework, but we do not present it here to avoid introducing new terminologies. They proved in [9, Th. 2.1 and Th. 2.2] that a monic polynomial sequence $\{\hat{S}_n\}_{n \geq 0}$ is orthogonal with respect to a discrete-continuous Sobolev inner product of the form (1.27) if and only if

$$1. \text{ deg}(\hat{S}_n) = n, \quad n \geq 0.$$

$$2. \hat{S}_n^{(k)}(c) = 0, \quad 0 \leq k \leq \mathfrak{m} - 1, n \geq \mathfrak{m}.$$

$$3. \hat{S}_n^{(\mathfrak{m})} = \frac{n!}{(n - \mathfrak{m})!} \hat{P}_{n - \mathfrak{m}}, \quad n \geq \mathfrak{m}.$$

where $\hat{P}_n$ denotes the n th monic orthogonal polynomial with respect to μ . In addition, in the affirmative case we have $A = Q^{-1}D(Q^{-1})^t$, where Q is the non-singular and lower triangular matrix

$$Q = \begin{pmatrix} \hat{S}_0(c) & \hat{S}'_0(c) & \cdots & \hat{S}_0^{(\mathfrak{m}-1)}(c) \\ \hat{S}_1(c) & \hat{S}'_1(c) & \cdots & \hat{S}_1^{(\mathfrak{m}-1)}(c) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{S}_{\mathfrak{m}-1}(c) & \hat{S}'_{\mathfrak{m}-1}(c) & \cdots & \hat{S}_{\mathfrak{m}-1}^{(\mathfrak{m}-1)}(c) \end{pmatrix}$$

and D is the diagonal matrix with positive entries

$$D = \begin{pmatrix} \langle \hat{S}_0, \hat{S}_0 \rangle_s & 0 & \cdots & 0 \\ 0 & \langle \hat{S}_1, \hat{S}_1 \rangle_s & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \langle \hat{S}_{\mathfrak{m}-1}, \hat{S}_{\mathfrak{m}-1} \rangle_s \end{pmatrix}.$$

Notice that the positive diagonal entries of the matrix D can be arbitrary chosen and for $m = 0, 1, \dots, \mathfrak{m} - 1$ and $n \in \mathbb{Z}_+$

$$\langle \hat{S}_n, \hat{S}_m \rangle_s = \left(\hat{S}_n(c), \hat{S}'_n(c), \dots, \hat{S}_n^{(\mathfrak{m}-1)}(c) \right) A \begin{pmatrix} \hat{S}_m(c) \\ \hat{S}'_m(c) \\ \vdots \\ \hat{S}_m^{(\mathfrak{m}-1)}(c) \end{pmatrix} = \begin{cases} 0, & \text{if } n \neq m, \\ \langle \hat{S}_n, \hat{S}_n \rangle_s, & \text{if } n = m \end{cases}$$

holds.

When A is the identity matrix and the derivatives in the discrete part of (1.27) are considered in different points we obtain a similar result in Theorem 4.5.1. Using this result it is easy to find the corresponding discrete-continuous Sobolev inner product of the generalized Jacobi polynomials (1.7) when one of its parameters (α or β) is the negative integer $-\mathfrak{m}$ and the other one is greater than $-\mathfrak{m} - 1$, see [9, §3.2]. For the remainder set of negative integer parameters, orthogonality relations in a discrete-continuous sense have been founded, but this orthogonality is again out of the inner products framework. We refer the reader to [2, Th. 3.1] to avoid introducing new terminologies. See also [39, Prop. 3] for the case of generalized Gegenbauer polynomials (generalized Jacobi polynomials with $\alpha = \beta$) $\{C_n^{(-k+1/2)}\}_{n \geq 0}$ for $k \in \mathbb{N}$.

On the other hand, Alfaro et al. also proved that the only monic polynomials orthogonal with respect to an inner product of the form (1.27) and satisfying the classical differential equation (1.8) or/and a three-term recurrence relation like (1.5), but allowing a_n to vanish for some values of n , are

Polynomials	value of c	measure μ
$\hat{L}_n^{(-\mathfrak{m})}$, $\mathfrak{m} \in \mathbb{N}$	$c = 0$	$d\mu = e^{-x}dx$
$\hat{P}_n^{(-\mathfrak{m}, \beta)}$, $\mathfrak{m} \in \mathbb{N}$, $\beta + \mathfrak{m} > -1$	$c = 1$	$d\mu = (1+x)^{\beta+\mathfrak{m}}dx$
$\hat{P}_n^{(\alpha, -\mathfrak{m})}$, $\mathfrak{m} \in \mathbb{N}$, $\alpha + \mathfrak{m} > -1$	$c = -1$	$d\mu = (1-x)^{\alpha+\mathfrak{m}}dx$

Table 1.2: Classical discrete-continuous Sobolev polynomials.

In [53] the inner product

$$\langle f, g \rangle_s = (f(c_0), f(c_1), \dots, f(c_{\mathfrak{m}-1})) A \begin{pmatrix} g(c_0) \\ g(c_1) \\ \vdots \\ g(c_{\mathfrak{m}-1}) \end{pmatrix} + \int f^{(\mathfrak{m})}(x) g^{(\mathfrak{m})}(x) d\mu(x)$$

is considered, where A is a positive-definite and symmetric $\mathfrak{m} \times \mathfrak{m}$ real matrix and $c_1, c_2, \dots, c_{\mathfrak{m}-1}$ are different real numbers. In this paper, some relations are found between the corresponding discrete-continuous Sobolev polynomials and a problem of simultaneous polynomial interpolation and approximation, see [53, Th. 5.2].

About the asymptotic and zero location properties of this kind of Sobolev polynomials, as far as we known, the only available results have been published in [16], where the authors consider the Gegenbauer discrete-continuous Sobolev inner product

$$\langle f, g \rangle_s = \eta f(c)g(c) + \int_{-1}^1 f'(x)g'(x)(1-x^2)^{\lambda-\frac{1}{2}}dx$$

with $c \in \mathbb{C}$, $\eta > 0$ and $\lambda > -\frac{1}{2}$. In this paper, the fundamental formula

$$\hat{S}_n^\lambda(x) = \tilde{C}_n^\lambda(x) - \tilde{C}_n^\lambda(c), \quad n \geq 3,$$

plays a main role, where $\tilde{C}_n^\lambda = \hat{C}_n^\lambda(z) - \frac{n(n-1)}{4(n+\lambda-1)(n+\lambda-2)}\hat{C}_{n-2}^\lambda(z)$ and $\{\hat{C}_n^\lambda\}_{n \geq 0}$ are the monic classical Gegenbauer polynomials orthogonal with respect to $w(x) = (1-x^2)^{\lambda-1/2}$ i.e. Jacobi weight with $\alpha = \beta = \lambda - 1/2$. Among other results, they proved the following ones, see [16, Th. 2.1, Th. 2.2 and Th. 2.3]

1. The zeros of S_n are located on the lemniscate

$$\mathbf{E}_n(c) := \left\{ z \in \mathbb{C} : \prod_{k=1}^n |z - x_{n,k}| = \prod_{k=1}^n |c - x_{n,k}| \right\},$$

where $x_{n,1}, x_{n,2}, \dots, x_{n,n}$ are the zeros of $\tilde{C}_n^\lambda$.

2. All the zeros of the polynomials $\{S_n\}_{n \geq 0}$ are in the open ball centered at 0 and radius $r = \max_{x \in [-1,1]} |c - x| + \inf_{x \in [-1,1]} |c - x|$.
3. The multiplicity of the zeros of S_n is at most two and an example with a double zero is provided.

4. Every multiple zero of S_n lies on $[-1, 1]$ and when $\inf_{x \in [-1, 1]} |c - x| > 1$ all the zeros are simple.
5. If $c \in \mathbb{C} \setminus [-1, 1]$ and $\inf_{x \in [-1, 1]} |c - x| > 1$ the accumulation points of the zeros of $\{S_n\}_{n \geq 0}$ are located on the ellipse

$$\mathbf{E}(c) := \left\{ z = x + iy \in \mathbb{C} : \frac{x^2}{(|\varphi(c)|^2 + 1)^2} + \frac{y^2}{(|\varphi(c)|^2 - 1)^2} = \frac{1}{4|\varphi(c)|^2} \right\}.$$

In addition, they found the non-consecutive three-term recurrence relation

$$\hat{S}_{n+2}(z) = z\hat{S}_{n+1}(z) + a_n\hat{S}_n(z) + b_n\hat{S}_1(z), \quad n \geq 2,$$

where $\hat{S}_0 \equiv 1$, $\hat{S}_1(z) = z - c$,

$$a_n = -\frac{(n+1)(n+2\lambda-2)}{4(n+\lambda)(n+\lambda-1)} \quad \text{and} \quad b_n = \tilde{C}_{n+1}^\lambda(c).$$

Finally, in [16, Th. 5.1] the authors also proved the following relative asymptotic

$$\frac{\hat{S}_n(z)}{\hat{C}_n^\lambda(z)} \rightrightarrows_n \frac{2}{\varphi'(z)}, \quad K \subset \left\{ z : |z| \geq \sup_{x \in [-1, 1]} |c - x| + 1 \right\}.$$

Chapter 4 of this thesis dissertation deals with a discrete-continuous inner product whose vectorial measure $\vec{\mu}$ has the classical Jacobi measure as its continuous component measure (i.e. $d\mu_{\mathfrak{m}} = (1-x)^\alpha(1+x)^\beta dx$, $\alpha, \beta > -1$). The greatest space of functions f for which $\|f\|_{s,p}$ has a sense is studied and the convergence of the Fourier series in the norm $\|\cdot\|_{s,p}$ is totally described.

Sobolev Extremal Polynomials with respect to singular measures

2.1 Introduction

The study of zeros, and critical points of extremal polynomials is of great interest because they can be interpreted in various ways from the standpoint of physics function theory, and numerical analysis.

Let μ_0 be a standard measure (see Definition 1.3.1) and $1 < p < \infty$. It is well known that $\|\cdot\|_{\mathbf{L}^p(\mu_0)}$ is a *strictly convex norm* over the space of polynomials, i.e. the unit ball is a strictly convex set. Then, there exists a unique monic polynomial $P_n \in \mathbb{P}_n^*$ such that

$$\|P_n\|_{\mathbf{L}^p(\mu_0)} = \min_{Q \in \mathbb{P}_n^*} \|Q\|_{\mathbf{L}^p(\mu_0)}, \quad (2.1)$$

It is also known that $\|\cdot\|_{\mathbf{L}^p(\mu_0)}$ defines a *Fejér or monotonic norm* when restricted to the space $\mathbb{P}$ of polynomials (i.e., for distinct $f, g \in \mathbb{P}$ the condition $|f(z)| \leq |g(z)|$ for all $z \in \text{supp}(\mu_0)$, with equality only if $g(z) = 0$, implies $\|f\|_{\mathbf{L}^p(\mu_0)} < \|g\|_{\mathbf{L}^p(\mu_0)}$). Hence, from *Fejér's convex hull theorem* [33, Th. 10.2.2] we get that the zeros of P_n lie in $\Delta_0 = \mathbf{C}_h(\text{supp}(\mu_0))$.

Let us mention a characterization of the solution of the extremal problem (2.1) (cf. [20, §2.2 and Ex. 7-h]). A polynomial $P_n \in \mathbb{P}_n^*$ is the n th monic extremal polynomial in $\mathbf{L}^p(\mu_0)$ if and only if for all $Q \in \mathbb{P}_{n-1}$

$$\int Q \text{sgn}(P_n) |P_n|^{p-1} d\mu_0 = 0, \text{ where } \text{sgn}(y) = \begin{cases} y/|y|, & \text{if } y \neq 0, \\ 0, & \text{if } y = 0. \end{cases} \quad (2.2)$$

Hence, if P_n has a zero of multiplicity at least two at x^* then, $\frac{P_n(x)}{(x-x^*)^2}$ is a polynomial of degree $(n-2)$ and we have the contradiction

$$0 < \int_{\Delta_0} \frac{|P_n(x)|^p}{(x-x^*)^2} d\mu_0(x) = \int_{\Delta_0} \frac{P_n(x)}{(x-x^*)^2} \text{sgn}(P_n(x)) |P_n(x)|^{p-1} d\mu_0(x) = 0.$$

Consequently, *all the zeros of P_n are simple*.

Let $\vec{\mu} = (\mu_0, \mu_1)$ be a standard vectorial measure. For $1 < p < \infty$, we consider on the space $\mathbb{P}$ of polynomials, the corresponding p -Sobolev norm

$$\|f\|_{s,p} = \left(\|f\|_{\mathbf{L}^p(\mu_0)}^p + \|f'\|_{\mathbf{L}^p(\mu_1)}^p \right)^{\frac{1}{p}} = \left(\int |f|^p d\mu_0 + \int |f'|^p d\mu_1 \right)^{\frac{1}{p}} \quad (2.3)$$

It is not difficult to prove that (2.3) is a strictly convex norm (See [93, Lem. 1]) and, therefore, for each $n \in \mathbb{Z}_+$ there exists a unique monic polynomial $L_n \in \mathbb{P}_n^*$ such that

$$\|L_n\|_{s,p} = \min_{Q \in \mathbb{P}_n^*} \|Q\|_{s,p}. \quad (2.4)$$

The polynomial L_n is called the *n*th *monic extremal polynomial* relative to $\|\cdot\|_{s,p}$. In Proposition 2.2.1 we provide an alternative direct proof of the uniqueness of L_n . Obviously, $\|\cdot\|_{s,p}$ is not a Fejér norm because we can construct (piecewise continuously differentiable) functions such that $|f(z)| < |g(z)|, x \in \Delta_0 \cup \Delta_1$, with $|f'|$ much larger than $|g'|$, μ_1 a.e. on Δ_1 , so that $\|f\|_{s,p} > \|g\|_{s,p}$. Specific examples are easy to produce.

Example 2.1.1

Take $0 < a < c < 1$ and assume that μ_0 and μ_1 are probability measures satisfying $\text{supp}(\mu_0) \cup \text{supp}(\mu_1) \subset [-a, a]$. If $f(x) = x$ and $g(x) \equiv c$, then $|f(x)| < |g(x)|, x \in \Delta_0 \cup \Delta_1$, whereas

$$\|g\|_{s,p}^p = c^p < 1 < \int |x|^p d\mu_0(x) + 1 = \|f\|_{s,p}^p.$$

As it was mentioned in Section 1.4, P. Althammer shows in an early example (cf. [10], where $p = 2$) that in the Sobolev case the zeros of the orthogonal polynomials may lie outside the interior of the convex hull of $\text{supp}(\mu_0) \cup \text{supp}(\mu_1)$. Other examples of the previous fact can be found in [55, §2, where $p = 2$].

However, in the numerical experiments carried out in [55, §2] (for $p = 2$), the authors found that in all the cases considered, the critical points of L_n were real numbers. Their experiments conclude with two conjectures about the location of zeros and critical points of the Sobolev-type orthogonal polynomials (see [55, Conj. 1 and 2]). In the following theorem, we solve the problem derived from these conjectures for extremal polynomials when μ_0 and μ_1 in (2.3) are supported on mutually disjoint intervals.

Theorem 2.1.1 (Location of critical points)

Let (μ_0, μ_1) be a standard vectorial measure such that $\Delta_0^\circ \cap \Delta_1 = \emptyset$ (recall that $\Delta_i := \text{supp}(\mu_i)$) and $1 < p < \infty$. Then

2.1.1.1. the zeros of L_n in Δ_0° are simple for all $n \geq 1$ and

$$n - 1 \leq \mathbf{N}_s(L_n; \Delta_0^\circ) + \mathbf{N}_s(L'_n; \Delta_1^\circ) \leq n. \quad (2.5)$$

2.1.1.2. for $n \geq 2$, the critical points of the extremal polynomial L_n are simple and contained in $\mathbf{C}_h(\Delta_0 \cup \Delta_1)^\circ$.

2.1.1.3. the number of zeros (or critical points) of the extremal polynomial L_n lying in $\mathbf{C}_h(\Delta_0 \cup \Delta_1)^\circ \setminus (\Delta_0^\circ \cup \Delta_1^\circ)$ is at most one.

2.1.1.4. the zeros of L'_n in Δ_0° interlace the zeros of L_n on that set.

The following example shows that, in general, the lower bound in inequality (2.5) cannot be improved.

Example 2.1.2

Set $d\mu_0(x) = dx$, $x \in \Delta_0 = [-11, -2]$, $d\mu_1(x) = dx$, $x \in \Delta_1 = [22.5, 36]$ and

$$\|f\|_{s,2}^2 = \int_{-11}^{-2} |f(x)|^2 dx + \int_{22.5}^{36} |f'(x)|^2 dx.$$

The extremal polynomial of degree $n = 2$ is $L_2(z) = z^2 - 49$ with zeros $z_1 = -7$, $z_2 = 7$ and critical point $z_3 = 0$. Then $n - 1 = 1 = \mathbf{N}_s(L_2; \Delta_0^\circ) + \mathbf{N}_s(L_2'; \Delta_1^\circ)$.

The next result provides a natural and intrinsic characterization of the extremal polynomials defined by (2.4), and an extension of (2.2) for the Sobolev case.

Theorem 2.1.2

Let (μ_0, μ_1) be a vectorial measure and $1 < p < \infty$. L_n is the n th monic extremal polynomial with respect to $\|\cdot\|_{s,p}$ if and only if

$$\begin{aligned} \langle Q, L_n \rangle_{s,p} &:= \int Q(x) \operatorname{sgn}(L_n(x)) |L_n(x)|^{p-1} d\mu_0(x) \\ &+ \int Q'(x) \operatorname{sgn}(L_n'(x)) |L_n'(x)|^{p-1} d\mu_1(x) = 0, \end{aligned}$$

for every polynomial $Q \in \mathbb{P}_{n-1}$.

Note that unless $p = 2$, $\langle \cdot, \cdot \rangle_{s,p}$ does not define an inner product.

Theorem 2.1.2 is a particular case of Theorem 2.2.1 below, where (2.3) is replaced by a Sobolev norm with derivatives of higher order (1.10).

According to (2.4), $L_n \in \mathbb{P}_n^*$ (the n th monic extremal polynomial with respect to (1.10)) is a monic polynomial that verifies

$$\|L_n\|_{s,p} = \min_{Q \in \mathbb{P}_n^*} \|Q\|_{s,p}. \quad (2.6)$$

When $p = 2$, and the norm (1.10) is given by the inner product (1.9), the corresponding Sobolev extremal polynomials are the monic Sobolev orthogonal polynomials, which, as we have seen in Section 1.4, have been extensively studied. However, for $p \neq 2$ ($1 < p < \infty$) not much has been attained and the basic references are [93] for the so-called “*sequentially dominated norms*” and [89] for measures with unbounded support on the real line.

Section 2.2 is devoted to the study of the existence and uniqueness of the extremal polynomial with respect to the norm (1.10). Theorem 2.2.1, which is of independent interest, is the main tool for locating zeros and critical points. In Section 2.3, we prove Theorem 2.1.1 and Corollary 2.3.0.1 on the location and algebraic properties of the zeros and critical points of the extremal polynomials L_n . The last section is devoted to the study of the asymptotic distribution of the zeros and critical points of the Sobolev extremal polynomials. Next, we state the main result in this direction after introducing some needed terminologies.

For any complex polynomial $Q_n(z) = c \prod_{k=1}^n (z - z_k)$, with $c, z_1, \dots, z_n \in \mathbb{C}$, we denote by $\sigma(Q_n)$ the so-called *normalized zero counting measure associated with Q_n* , as

$$\sigma(Q_n) = \frac{1}{n} \sum_{j=1}^n \delta_{z_j}.$$

Following the usual terminology, if $\{\mu_n\}_{n \geq 1}$ is a sequence of measures on a compact set $K \subset \mathbb{C}$, we say that a measure μ is the limit of $\{\mu_n\}$ in the weak star topology of measures, if

$$\lim_{n \rightarrow \infty} \int f d\mu_n = \int f d\mu,$$

for every continuous function f on K . In this case we write $\text{w-lim}_{n \rightarrow \infty} \mu_n = \mu$.

Let μ be a finite Borel measure whose compact support $\text{supp}(\mu) \subset \mathbb{C}$ has positive logarithmic capacity $\text{cap}(\text{supp}(\mu))$ and let P_n be the associated monic orthogonal polynomial with respect to μ of degree n . We say that μ is regular and write $\mu \in \mathbf{Reg}$ if

$$\lim_{n \rightarrow \infty} \|P_n\|_{\mu}^{1/n} = \text{cap}(\text{supp}(\mu)),$$

Theorem 3.1.1 in [162] contains several equivalent forms of defining regular measures (see also [162, Th. 3.2.1 and Th. 3.2.3]). Recall that for any compact set $K \subset \mathbb{C}$ with $\text{cap}(K) > 0$ there exists a unique probability measure $\mu_K, \text{supp}(\mu_K) \subset K$, called the *equilibrium measure* of K , which is characterized by

$$\int \log \frac{1}{|z - x|} d\mu_K(x) \begin{cases} = \gamma, & \text{if } z \in K \setminus A, \\ \leq \gamma, & \text{if } z \in \mathbb{C}. \end{cases}$$

where A is a Borel set whose logarithmic capacity is zero ($\text{cap}(A) = 0$), and γ is some uniquely determined constant (actually $e^{-\gamma} = \text{cap}(K)$).

Theorem 2.1.3 (Asymptotic zero distribution of the zeros of SL_n)

Let (μ_0, μ_1) be a standard vectorial measure such that $\mu_0, \mu_1 \in \mathbf{Reg}$ and its respective supports $\text{supp}(\mu_0)$ and $\text{supp}(\mu_1)$ are compact intervals of $\mathbb{R}$, so we get $\text{supp}(\mu_i) = \Delta_i$, $i = 0, 1$. Denote by $\{L_n\}_{n \geq 0}$ the sequence of monic extremal polynomials relative to the corresponding Sobolev norm (2.3) with $p \in (1, \infty)$. Then, for each integer $j \geq 1$

$$\lim_{n \rightarrow \infty} \|L_n^{(j)}\|_{\Delta}^{1/n} = \text{cap}(\Delta), \quad (2.7)$$

$$\text{w-lim}_{n \rightarrow \infty} \sigma(L_n^{(j)}) = \mu_{\Delta}, \quad (2.8)$$

where $\Delta = \Delta_0 \cup \Delta_1$ and μ_{Δ} is the equilibrium measure on Δ .

Notice that (2.8) holds for $j \geq 1$. For $j = 0$ the zeros of the polynomials L_n can leave Δ and their asymptotic zero distribution is governed by the balayage of μ_{Δ} onto a certain region which we describe later (for details, see Theorem 2.4.2 below).

The content of this chapter is included in [40].

2.2 The Characterization Theorem

Throughout this section we consider the standard vectorial measure $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ and the more general Sobolev norm (1.10) with $1 < p < \infty$, and L_n verifies (2.6). Since $\mathbb{P}_{n-1}$ is a finite dimensional linear space, the existence of $L_n \in \mathbb{P}_n^*$ is obvious. In addition, L_n has real coefficients. Otherwise L_n could be rewritten as $L_n = P + iQ$ where Q and P are polynomials with real coefficients, $Q \not\equiv 0$ and P is a monic polynomial of degree n satisfying

$$\begin{aligned} \|L_n\|_{s,p}^p &= \sum_{k=0}^{\mathfrak{m}} \int |P^{(k)} + iQ^{(k)}|^p d\mu_k = \sum_{k=0}^{\mathfrak{m}} \int \left((P^{(k)})^2 + (Q^{(k)})^2 \right)^{\frac{p}{2}} d\mu_k \\ &> \sum_{k=0}^{\mathfrak{m}} \int |P^{(k)}|^p d\mu_k = \|P\|_{s,p}^p \end{aligned}$$

The next proposition contains a direct proof of the uniqueness of the Sobolev extremal polynomial L_n and shows that (1.10) is a strictly convex norm.

Proposition 2.2.1 (Uniqueness)

Let $\vec{\mu}$ be a k -standard vectorial measure and let $\|\cdot\|_{s,p}$ be the Sobolev type norm defined by (1.10). Then, there exists a unique monic polynomial L_n with $\mathbf{deg}(L_n) = n$ such that

$$\|L_n\|_{s,p} = \inf_{Q_n \in \mathbb{P}_n^*} \|Q_n\|_{s,p}.$$

In addition, such a polynomial L_n has real coefficients.

Proof. It was already proved the existence of the monic extremal polynomial L_n with degree n and real coefficients, so it only remains to prove the uniqueness. If L_n and $\tilde{L}_n$ are two different monic extremal polynomials of degree n , from the extremality and the triangular inequality it is obvious that $\frac{1}{2}(L_n + \tilde{L}_n)$ is also a monic extremal polynomial. Hence

$$\|L_n + \tilde{L}_n\|_{s,p} = \|L_n\|_{s,p} + \|\tilde{L}_n\|_{s,p}. \quad (2.9)$$

From the Minkowski inequality (applied twice) we obtain

$$\begin{aligned} \|L_n + \tilde{L}_n\|_{s,p} &= \left(\sum_{j=0}^{\mathfrak{m}} \left\| L_n^{(j)} + \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)}^p \right)^{1/p} \\ &\leq \left(\sum_{j=0}^{\mathfrak{m}} \left(\left\| L_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} + \left\| \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} \right)^p \right)^{1/p} \\ &\leq \|L_n\|_{s,p} + \|\tilde{L}_n\|_{s,p}. \end{aligned}$$

Therefore, by (2.9), the first inequality just shown is in fact the equality

$$\sum_{j=0}^{\mathfrak{m}} \left\| L_n^{(j)} + \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)}^p = \sum_{j=0}^{\mathfrak{m}} \left(\left\| L_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} + \left\| \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} \right)^p.$$

However

$$\left\| L_n^{(j)} + \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)}^p \leq \left(\left\| L_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} + \left\| \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} \right)^p, \quad j = 0, 1, \dots, \mathfrak{m}.$$

Therefore,

$$\left\| L_n^{(j)} + \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} = \left\| L_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)} + \left\| \tilde{L}_n^{(j)} \right\|_{\mathbf{L}^p(\mu_j)}, \quad j = 0, 1, \dots, \mathfrak{m}.$$

In particular, from the Minkowski inequality for $\mathbf{L}^p(\mu_j)$, there exists a constant $\alpha_j \geq 0$ such that $L_n^{(j)} = \alpha_j \tilde{L}_n^{(j)}$ almost everywhere with respect to μ_j , for $j = 0, 1, \dots, \mathfrak{m}$.

On the other hand, the second inequality also holds in the equality, so there exists a constant α such that

$$\left\| L_n^{(j)} \right\|_{s,p} = \alpha \left\| \tilde{L}_n^{(j)} \right\|_{s,p}, \quad j = 0, 1, \dots, \mathfrak{m}.$$

So we get, $\alpha_0 = \alpha_1 = \dots = \alpha_{\mathfrak{m}} = \alpha$. But, L_n and $\tilde{L}_n$ are monic polynomials and $\text{supp}(\mu_k)$ is an infinite set, hence $\alpha = 1$ and $L_n^{(k)} \equiv \tilde{L}_n^{(k)}$. Since μ_j , $j = 0, 1, \dots, k-1$ are no null measures, there exist constants $c_j \in \text{supp}(\mu_j)$ such that

$$L_n^{(j)}(c_j) = \tilde{L}_n^{(j)}(c_j), \quad j = 0, 1, \dots, k-1,$$

and, finally, we obtain $L_n \equiv \tilde{L}_n$. □

Theorem 2.2.1 (Characterization)

Let $\tilde{\mu}$ be a k -standard vectorial measure and let $\|\cdot\|_{s,p}$ ($1 < p < \infty$) be the Sobolev type norm defined in (1.10). Then, the monic polynomial L_n is the n th monic extremal polynomial relative to $\|\cdot\|_{s,p}$ if and only if

$$\langle Q, L_n \rangle_{s,p} := \sum_{k=0}^{\mathfrak{m}} \int Q^{(k)} \text{sgn}(L_n^{(k)}) \left| L_n^{(k)} \right|^{p-1} d\mu_k = 0, \quad (2.10)$$

for every polynomial $Q \in \mathbb{P}_{n-1}$.

Proof. Assume that L_n is the n th monic extremal polynomial relative to the norm $\|\cdot\|_{s,p}$ and let $Q \in \mathbb{P}_{n-1}$. Then

$$\|L_n + \alpha Q\|_{s,p} \leq \|L_n\|_{s,p}, \quad \text{for all } \alpha \in \mathbb{R}. \quad (2.11)$$

Let $F(\alpha)$ be the auxiliary function defined for all $\alpha \in \mathbb{R}$ by the expression

$$F(\alpha) = \|L_n + \alpha Q\|_{s,p}^p = \sum_{k=0}^{\mathfrak{m}} \int \left| L_n^{(k)} + \alpha Q^{(k)} \right|^p d\mu_k.$$

From Proposition 2.2.1 and (2.11), $\alpha = 0$ is the unique minimum point of F , thus

$$0 = F'(0) = p \sum_{k=0}^{\mathfrak{m}} \int Q^{(k)} \text{sgn}(L_n^{(k)}) \left| L_n^{(k)} \right|^{p-1} d\mu_k = p \langle Q, L_n \rangle_{s,p}$$

and we get (2.10). Now, assume that (2.10) takes place for every polynomial $Q \in \mathbb{P}_{n-1}$. Obviously, each monic polynomial P of degree n can be written as the sum $P = L_n + Q$ where $Q \in \mathbb{P}_{n-1}$.

Let q be the conjugate exponent of p , i.e. $q = \frac{p}{p-1}$. For $k = 0, 1, \dots, \mathfrak{m}$ we have

$$G_k = \operatorname{sgn}\left(L_n^{(k)}\right) |L_n^{(k)}|^{p-1} \in \mathbf{L}^q(\mu_k), \quad \int |G_k|^q d\mu_k = \|L_n\|_{\mathbf{L}^p(\mu_k)}^p.$$

Thus

$$\|L_n\|_{s,p}^p = \sum_{k=0}^{\mathfrak{m}} \left\| L_n^{(k)} \right\|_{\mathbf{L}^p(\mu_k)}^p = \sum_{k=0}^{\mathfrak{m}} \int |G_k|^q d\mu_k = \sum_{k=0}^{\mathfrak{m}} \|G_k\|_{\mathbf{L}^q(\mu_k)}^q. \quad (2.12)$$

Let $\alpha, \beta \geq 0$, $p > 1$ and $q = \frac{p}{p-1}$. It is well known that

$$\sqrt[p]{\alpha} \cdot \sqrt[q]{\beta} \leq \frac{\alpha}{p} + \frac{\beta}{q}, \quad (2.13)$$

with equality if and only if $\alpha = \beta$ (cf. [124, §2.1.1 and Th. 2]). From (2.10), Hölder's inequality, (2.13) and (2.12), we get

$$\begin{aligned} \|L_n\|_{s,p}^p &= \langle L_n, L_n \rangle_{s,p} = \langle L_n + Q, L_n \rangle_{s,p} = \langle P, L_n \rangle_{s,p}, \\ &= \sum_{k=0}^{\mathfrak{m}} \int P^{(k)} G_k d\mu_k \leq \sum_{k=0}^{\mathfrak{m}} \|P^{(k)}\|_{\mathbf{L}^p(\mu_k)} \cdot \|G_k\|_{\mathbf{L}^q(\mu_k)}, \\ &\leq \sum_{k=0}^{\mathfrak{m}} \left(\frac{\|P^{(k)}\|_{\mathbf{L}^p(\mu_k)}^p}{p} + \frac{\|G_k\|_{\mathbf{L}^q(\mu_k)}^q}{q} \right) = \frac{\|P\|_{s,p}^p}{p} + \frac{\|L_n\|_{s,p}^p}{q}. \end{aligned}$$

Thus $\|L_n\|_{s,p} \leq \|P\|_{s,p}$, which completes the proof. $\square$

Corollary 2.2.1.1

Let $\vec{\mu}$ be a standard vectorial measure, if $n \geq 1$ then L_n has at least one zero of odd multiplicity $x_0 \in \mathbf{C}_h(\operatorname{supp}(\mu_0))^\circ$.

Proof. This is an immediate consequence of

$$\int \operatorname{sgn}(L_n) |L_n|^{p-1} d\mu_0 = \langle 1, L_n \rangle_{s,p} = 0.$$

$\square$

Corollary 2.2.1.2

Let $\vec{\mu}$ be a standard vectorial measure, if $n \geq 2$ then L'_n has at least one zero of odd multiplicity in $\mathbf{C}_h(\operatorname{supp}(\mu_0) \cup \operatorname{supp}(\mu_1))^\circ$.

Proof. If L'_n has no zeros of odd multiplicity on $\mathbf{C}_h(\text{supp}(\mu_0) \cup \text{supp}(\mu_1))^\circ$, then L_n is monotone on $\mathbf{C}_h(\text{supp}(\mu_0) \cup \text{supp}(\mu_1))^\circ$. From Corollary 2.2.1.1, L_n has exactly one zero x_0 of odd multiplicity on $\mathbf{C}_h(\text{supp}(\mu_0))^\circ$, so

$$\text{sgn}((x - x_0)L_n(x)) = \text{sgn}(L'_n(x)) = c$$

is constant for all $x \in \mathbf{C}_h(\text{supp}(\mu_0))^\circ$ with $c = \pm 1$. Hence, by Theorem 2.2.1 we have

$$\begin{aligned} 0 &= \langle c(x - x_0), L_n \rangle_{s,p} \\ &= \int c(x - x_0) \text{sgn}(L_n) |L_n|^{p-1} d\mu_0 + \int c \cdot \text{sgn}(L'_n) |L'_n|^{p-1} d\mu_1 > 0 \end{aligned}$$

which is a contradiction. $\square$

2.3 Two disjoint intervals. Proof of Theorem 2.1.1

Here, we prove the results on the location of zeros and critical points announced previously. The first lemma in this section is a consequence of Biernacki's theorem [141, Th. 4.5.2], which in turn is a converse of the Gauss-Lucas theorem.

Lemma 2.3.1 (Biernacki [141, Th. 4.5.2])

Let K be the convex hull of the critical points of a polynomial f , and let $f(z_0) = 0$. Then, the zeros of f lie in the union of all the closed disks centered at the vertices of K and radius equal to the distance from the vertex to z_0 .

Let $I \subset \mathbb{R}$ be an interval and $Q \in \mathbb{P}$. Additionally, for all $n \geq 1$ we write $\ell_n := \mathbf{N}_s(L_n; \Delta_0^\circ) + \mathbf{N}_s(L'_n; \Delta_1^\circ)$. The key to the proof of Corollary 2.3.0.1 below is the following trivial consequence of Rolle's theorem.

Lemma 2.3.2

[92, Lem. 2.1] *Let I be an interval of the real line and Q a non-constant polynomial of degree n with real coefficients, then $\mathbf{N}_z(Q; I) + \mathbf{N}_z(Q'; \mathbb{C} \setminus I) \leq n$.*

Proof of Theorem 2.1.1. For $n = 1, 2$, the statements of the Lemma are immediate consequences of Corollaries 2.2.1.1 and 2.2.1.2. So, in the sequel we assume that $n \geq 3$. From Lemma 2.3.2 we have $\ell_n \leq n$. The simplicity of the zeros of L_n in Δ_0° follows directly from the inequality $\ell_n \geq n - 1$ and Lemma 2.3.2. Therefore, to complete the proof of the statement 2.0.1.1 it suffices to show that

$$\ell_n \geq n - 1. \quad (2.14)$$

Without loss of generality, we can assume that $\Delta_0^\circ = (a, b)$, $\Delta_1^\circ = (c, d)$ and $-\infty \leq a < b \leq c \leq d \leq \infty$, the case $d \leq a$ is solved similarly.

Fix $n \geq 3$ and let x_0 be the point in (a, b) closest to $[c, d]$ where L_n changes sign. This point exists due to Corollary 2.2.1.1. There are two possible cases, either

$$\text{sgn}(L'_n(x_0 + \varepsilon) \cdot L'_n(c + \varepsilon)) = 1 \quad (\text{I})$$

for all sufficiently small $\varepsilon > 0$, or

$$\operatorname{sgn}(L'_n(x_0 + \varepsilon) \cdot L'_n(c + \varepsilon)) = -1 \quad (\text{II})$$

for all sufficiently small $\varepsilon > 0$. Let us consider each case separately. In case I we can prove more than (2.14); namely, $\ell_n = n$.

To the contrary, suppose that $\ell_n \leq n - 1$ in case I or $\ell_n \leq n - 2$ in case II. We shall see that we can find a polynomial $Q \in \mathbb{P}_{n-1}$ such that

$$Q(x)L_n(x) \geq 0, \quad x \in [a, b] \quad \text{and} \quad Q'(x)L'_n(x) \geq 0, \quad x \in [c, d]. \quad (2.15)$$

Suppose that (2.15) holds, using Theorem 2.1.2 we get

$$0 = \langle Q, L_n \rangle_{s,p} = \int_a^b Q \operatorname{sgn}(L_n) |L_n|^{p-1} d\mu_0 + \int_c^d Q' \operatorname{sgn}(L'_n) |L'_n|^{p-1} d\mu_1 > 0,$$

which is a contradiction and the proof of the statement 2.1.1.1. would be complete. Therefore, it is sufficient to find such a polynomial Q .

Case I

Suppose that $\ell_n \leq n - 1$ and take Q to be a polynomial of degree $\leq \ell_n$ with real coefficients, not identically equal to zero, which has a zero at each point of (a, b) where L_n changes sign and whose derivative has a zero at each point of (c, d) where L'_n changes sign. The existence of Q reduces to solving a homogeneous linear system of ℓ_n equations on $\ell_n + 1$ unknowns (the coefficients of Q); thus, a non trivial solution always exists. Notice that

$$\ell_n \leq \mathbf{N}_z(Q; (a, b)) + \mathbf{N}_z(Q'; (c, d))$$

with strict inequality if either Q (resp. Q') has on (a, b) (resp. (c, d)) zeros of multiplicity greater than one or distinct from those assigned by construction. On the other hand, because of Corollary 2.2.1.1 the degree of Q is at least 1; therefore, using Lemma 2.3.2, we have that

$$\ell_n \leq \mathbf{N}_z(Q; (a, b)) + \mathbf{N}_z(Q'; (c, d)) \leq \deg(Q) \leq \ell_n.$$

Thus

$$\ell_n = \mathbf{N}_z(Q; (a, b)) + \mathbf{N}_z(Q'; (c, d)) = \deg(Q). \quad (2.16)$$

Hence Q (resp. Q') has on (a, b) (resp. (c, d)) simple zeros and has no other zero different from those given by construction. So, QL_n and $Q'L'_n$ have constant sign on $[a, b]$ and $[c, d]$, respectively. We can choose Q in such a way that $QL_n \geq 0$ on $[a, b]$ (if this was not, then replace Q by $-Q$). In order to prove (2.15) it remains to check that $\operatorname{sgn}(Q'(c + \varepsilon)L'_n(x_0 + \varepsilon)) = 1$ for all ε sufficiently small. From Rolle's Theorem and (2.16) we have

$$\ell_n - 1 = \mathbf{N}_s(Q; (a, x_0)) + \mathbf{N}_s(Q'; (c, d)) \leq \mathbf{N}_s(Q'; (a, x_0)) + \mathbf{N}_s(Q'; (c, d)) \leq \ell_n - 1$$

Hence $\mathbf{N}_s(Q'; (a, x_0)) + \mathbf{N}_s(Q'; (c, d)) = \deg(Q')$ and all the zeros of Q' are contained in $(a, x_0) \cup (c, d)$. So, for all ε sufficiently small, we have

$$\operatorname{sgn}(Q'(x_0 + \varepsilon) \cdot Q'(c + \varepsilon)) = 1.$$

Now, from this expression and (I), we obtain

$$\begin{aligned}
 \operatorname{sgn}(Q'(c + \varepsilon)L'_n(c + \varepsilon)) &= \operatorname{sgn}(Q'(c + \varepsilon)) \operatorname{sgn}(L'_n(c + \varepsilon)) \\
 &= \operatorname{sgn}(Q'(x_0 + \varepsilon)) \operatorname{sgn}(L'_n(x_0 + \varepsilon)) \\
 &= \operatorname{sgn}(Q(x_0 + \varepsilon)) \operatorname{sgn}(L_n(x_0 + \varepsilon)) \\
 &= \operatorname{sgn}(Q(x_0 + \varepsilon)L_n(x_0 + \varepsilon)) = 1.
 \end{aligned}$$

Therefore, we get (2.15) and hence $\ell_n = n$.

In order to prove the remaining statements for this case, notice that

$$\begin{aligned}
 n - 1 &= l_n - 1 = \mathbf{N}_s(L_n; (a, x_0)) + \mathbf{N}_s(L'_n; (c, d)) \\
 &\leq \mathbf{N}_s(L'_n; (a, x_0)) + \mathbf{N}_s(L'_n; (c, d)) + \mathbf{N}_s(L'_n; (x_0, c]) \leq n - 1.
 \end{aligned}$$

Therefore, $\mathbf{N}_s(L'_n; (a, d)) = n - 1$, $\mathbf{N}_s(L'_n; (x_0, c]) = 0$, and

$$\mathbf{N}_s(L_n; (a, x_0)) = \mathbf{N}_s(L'_n; (a, x_0))$$

which proves 2.1.1.2, 2.1.1.3 and 2.1.1.4.

Case II

Suppose that $\ell_n \leq n - 2$. The difference consists in that the polynomial L'_n has different signs to the right of x_0 and c . Here, we construct Q of degree $\leq \ell_n + 1 \leq n - 1$ with real coefficients, not identically equal to zero, with the same interpolation conditions as above plus $Q'(c) = 0$. Following the same line of reasoning, we have

$$\ell_n + 1 = \mathbf{N}_z(Q; (a, b)) + \mathbf{N}_z(Q'; [c, d)) = \mathbf{deg}(Q).$$

Hence Q (resp. Q') has on (a, b) (resp. $[c, d)$) simple zeros and no other zero except those given by construction. So QL_n and $Q'L'_n$ have constant sign on $[a, b]$ and $[c, d]$, respectively. As in the previous case, $\mathbf{N}_s(Q'; (a, x_0)) + \mathbf{N}_s(Q'; [c, d)) = \mathbf{deg}(Q')$ and all the zeros of Q' are contained in $(a, x_0) \cup [c, d)$. Now, using that Q changes sign at c and (II) we obtain

$$\begin{aligned}
 \operatorname{sgn}(Q'(c + \varepsilon)L'_n(c + \varepsilon)) &= \operatorname{sgn}(Q'(c + \varepsilon)) \operatorname{sgn}(L'_n(c + \varepsilon)) \\
 &= -\operatorname{sgn}(Q'(x_0 + \varepsilon)) (-\operatorname{sgn}(L'_n(x_0 + \varepsilon))) \\
 &= \operatorname{sgn}(Q(x_0 + \varepsilon)) \operatorname{sgn}(L_n(x_0 + \varepsilon)) \\
 &= \operatorname{sgn}(Q(x_0 + \varepsilon)L_n(x_0 + \varepsilon)) = 1
 \end{aligned}$$

which proves that Q satisfies (2.15) and hence (2.14) is true.

Now, notice that (II) and the intermediate value theorem imply that L'_n has at least an odd zero on the interval $(x_0, c]$, thus

$$\begin{aligned}
 n - 1 &\leq (\ell_n - 1) + 1 = \mathbf{N}_s(L_n; (a, x_0)) + \mathbf{N}_s(L'_n; (c, d)) + 1 \\
 &\leq \mathbf{N}_s(L'_n; (a, x_0)) + \mathbf{N}_s(L'_n; (c, d)) + \mathbf{N}_s(L'_n; (x_0, c]) \leq n - 1.
 \end{aligned}$$

Therefore, $\mathbf{N}_s(L'_n; (a, d)) = n - 1$, $\mathbf{N}_s(L'_n; (x_0, c]) = 1$, and

$$\mathbf{N}_s(L_n; (a, x_0)) = \mathbf{N}_s(L'_n; (a, x_0))$$

from which 2.1.1.2, 2.1.1.3, and 2.1.1.4 follow. $\square$

Corollary 2.3.0.1

Let $p \in (1, \infty)$ and let (μ_0, μ_1) be a standard vectorial measure such that $\Delta_0^\circ \cap \Delta_1 = \emptyset$. Assume also that $\text{supp}(\mu_0)$ and $\text{supp}(\mu_1)$ are compact sets. If we take $[a, b] = \Delta_0$ and $[c, d] = \Delta_1$, where $a, b, c, d \in \mathbb{R}$, then the zeros of L_n lie in

$$D_\Delta = \begin{cases} \mathbf{B}(d, d-a) \cup \mathbf{B}(a, b-a), & \text{if } b \leq c, \\ \mathbf{B}(c, b-c) \cup \mathbf{B}(b, b-a), & \text{if } d \leq a, \end{cases}$$

where $\mathbf{B}(a, r) = \{z \in \mathbb{C} : |z - a| < r\}$.

Proof. For $n = 1$ the statement is certainly true because of Corollary 2.2.1.1. For $n \geq 2$ the result follows directly from statement 2.1.1.2 in Theorem 2.1.1 and the Biernacki Lemma 2.3.1. $\square$

In order to show how the bounding area D_Δ looks like, we conclude this section with a simple example for which its corresponding bounding area D_Δ has been plotted in Figure 2.1.

Example 2.3.1

Consider a standard vectorial measure (μ_0, μ_1) , where the supports are $\text{supp}(\mu_0) = [-2, -1]$ and $\text{supp}(\mu_1) = [1, 2]$, then

$$D_\Delta = \mathbf{B}(2, 4) \cup \mathbf{B}(-2, 1)$$

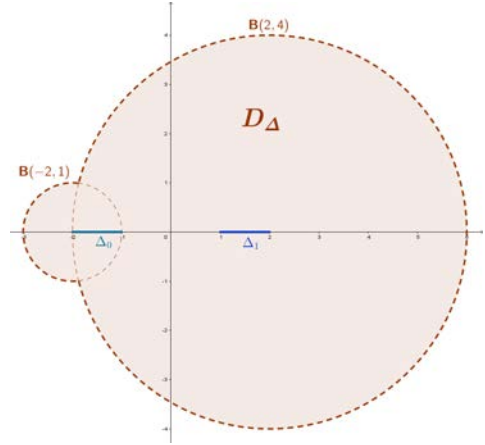


Figure 2.1: Bounding area of zeros of L_n

2.4 Regular asymptotic distribution of critical points

A compact set K of the complex plane is said to be *regular* if the Green's function with singularity at ∞ relative to the unbounded connected component of $\mathbb{C} \setminus K$ can be extended continuously to the boundary. We refer the reader to [80, 146] and for short [162, Appendix] for this and other notions related with logarithmic potential theory. For example, the union of a finite number of compact intervals in the real line is a regular compact set.

Suppose that μ is a finite positive Borel measure such that $\text{supp}(\mu)$ is a regular compact set and $1 \leq p < \infty$. It is well known (see [162, Th. 3.4.3]) that $\mu \in \mathbf{Reg}$ if and only if

$$\lim_{n \rightarrow \infty} \left(\frac{\|Q_n\|_{\text{supp}(\mu)}}{\|Q_n\|_{\mathbf{L}^p(\mu)}} \right)^{1/n} = 1, \quad (2.17)$$

where $\{Q_n\}_{n \geq 0}$, is any sequence of polynomials such that $\deg(Q_n) = n$. From the Cauchy's integral theorem for the derivative of a holomorphic function, with the same hypothesis on $\text{supp}(\mu)$ it is easy to show (see [90, Lem. 3]) that for all $j \in \mathbb{Z}_+$

$$\limsup_{n \rightarrow \infty} \left(\frac{\|Q_n^{(j)}\|_{\text{supp}(\mu)}}{\|Q_n\|_{\text{supp}(\mu)}} \right)^{1/n} \leq 1. \quad (2.18)$$

One last result that will be used in this chapter is contained in [18, Th. 2.1]. Let K be a compact subset of the real line with $\text{cap}(K) > 0$ and let $\{Q_n\}$ be a sequence of monic polynomials, $\deg(Q_n) = n$. Then

$$\limsup_{n \rightarrow \infty} \|Q_n\|_K^{1/n} = \text{cap}(K) \quad \Rightarrow \quad \text{w-lim}_{n \rightarrow \infty} \sigma(Q_n) = \mu_K. \quad (2.19)$$

Proof of Theorem 2.1.3. From (2.19), (2.8) follows from (2.7). Therefore, let us prove (2.7).

Let T_n denote the n th monic Chebyshev polynomial with respect to Δ ; that is, $\|T_n\|_\Delta \leq \|Q_n\|_\Delta$ for any monic polynomial Q_n of degree n . In particular,

$$\begin{aligned} \liminf_{n \rightarrow \infty} \|L_n^{(j)}\|_\Delta^{1/n} &\geq \liminf_{n \rightarrow \infty} \left(\frac{n!}{(n-j)!} \|T_{n-j}\|_\Delta \right)^{1/n} \\ &\geq \liminf_{n \rightarrow \infty} \|T_{n-j}\|_\Delta^{1/n} = \text{cap}(\Delta), \end{aligned}$$

since it is well known that $\lim_{n \rightarrow \infty} \|T_n\|_\Delta^{1/n} = \text{cap}(\Delta)$ (see, for example, [146, Cor. 5.5.5]). Hence, we only need to prove that

$$\limsup_{n \rightarrow \infty} \|L_n^{(j)}\|_\Delta^{1/n} \leq \text{cap}(\Delta). \quad (2.20)$$

Since $\mu_0, \mu_1 \in \mathbf{Reg}$ and Δ_0, Δ_1 are regular compact sets, from (2.17) and (2.18) we obtain

$$\begin{aligned} \limsup_{n \rightarrow \infty} \|L_n^{(j)}\|_{\Delta_0}^{p/n} &\leq \limsup_{n \rightarrow \infty} \|L_n\|_{\Delta_0}^{p/n} \leq \limsup_{n \rightarrow \infty} \|L_n\|_{\mathbf{L}^p(\mu_0)}^{p/n} \\ &\leq \limsup_{n \rightarrow \infty} \|L_n\|_{s,p}^{p/n}, \end{aligned} \quad (2.21)$$

and

$$\begin{aligned} \limsup_{n \rightarrow \infty} \|L_n^{(j)}\|_{\Delta_1}^{p/n} &\leq \limsup_{n \rightarrow \infty} \|L_n\|_{\Delta_1}^{p/n} \leq \limsup_{n \rightarrow \infty} \|L_n\|_{\mathbf{L}^p(\mu_1)}^{p/n} \\ &\leq \limsup_{n \rightarrow \infty} \|L_n\|_{s,p}^{p/n}. \end{aligned}$$

These relations yield

$$\limsup_{n \rightarrow \infty} \|L_n^{(j)}\|_\Delta^{1/n} \leq \limsup_{n \rightarrow \infty} \|L_n\|_{s,p}^{1/n}, \quad j \geq 1. \quad (2.22)$$

Now, from the extremality of L_n in the Sobolev norm, we have

$$\begin{aligned} \|L_n\|_{s,p}^p &\leq \|T_n\|_{s,p}^p = \|T_n\|_{\mathbf{I}^p(\mu_0)}^p + \|T_n'\|_{\mathbf{I}^p(\mu_1)}^p \\ &\leq \mu_0(\Delta_0) \|T_n\|_{\Delta_0}^p + \mu_1(\Delta_1) \|T_n'\|_{\Delta_1}^p \\ &\leq \mu_0(\Delta_0) \|T_n\|_{\Delta}^p + \mu_1(\Delta_1) \|T_n'\|_{\Delta}^p. \end{aligned}$$

On the other hand, using again (2.17) and (2.18) it follows that

$$\limsup_{n \rightarrow \infty} \left(\frac{\|L_n\|_{s,p}}{\|T_n\|_{\Delta}} \right)^{1/n} \leq \limsup_{n \rightarrow \infty} \left(\mu_0(\Delta_0) + \mu_1(\Delta_1) \frac{\|T_n'\|_{\Delta}^p}{\|T_n\|_{\Delta}^p} \right)^{1/pn} \leq 1,$$

whence

$$\limsup_{n \rightarrow \infty} \|L_n\|_{s,p}^{1/n} \leq \mathbf{cap}(\Delta). \quad (2.23)$$

Now, (2.22) and (2.23) give (2.20) and our statement follows. $\square$

Remark 2.4.1

If $\Delta_0 = [a_0, b_0]$ and $\Delta_1 = [a_1, b_1]$ are bounded and non trivial intervals of $\mathbb{R}$, for the equilibrium measure μ_{Δ} in Theorem 2.1.3, we have two possibilities:

- If $\Delta_0 \cap \Delta_1 \neq \emptyset$, then $\Delta = [a, b]$ (an interval) and μ_{Δ} is the arcsine measure

$$d\mu_{\Delta} = \frac{dx}{\pi \sqrt{|(x-a)(x-b)|}}, \quad x \in (a, b). \quad (2.24)$$

- If $\Delta_0 \cap \Delta_1 = \emptyset$ then there exists $x_* \in \Delta_g = \mathbf{C}_h(\Delta_0 \cup \Delta_1) \setminus (\Delta_0 \cup \Delta_1)$ such that

$$\int_{\Delta_g} \frac{(x-x_*) dx}{\sqrt{|(x-a_0)(x-b_0)(x-b_1)(x-a_1)|}} = 0,$$

and (see [162, Lem. 4.4.1])

$$d\mu_{\Delta} = \frac{|x-x_*| dx}{\pi \sqrt{|(x-a_0)(x-b_0)(x-a_1)(x-b_1)|}}, \quad x \in (a_0, b_0) \cup (a_1, b_1). \quad (2.25)$$

Combining Theorems 2.1.1 and 2.1.3 we obtain

Theorem 2.4.1 ((uniform) n th root asymptotic behavior)

Let (μ_0, μ_1) be a standard vectorial measure whose component measures belong to class **Reg**. Assume also that $\mathbf{supp}(\mu_0)$ and $\mathbf{supp}(\mu_1)$ are bounded and non trivial intervals of $\mathbb{R}$ such that $\Delta_0^{\circ} \cap \Delta_1 = \emptyset$. If $\{L_n\}_{n \geq 0}$ denotes the sequence of monic extremal polynomials relative to the corresponding Sobolev norm (2.3) with $p \in (1, \infty)$, then, for all $j \geq 1$

$$\limsup_{n \rightarrow \infty} \left| L_n^{(j)}(z) \right|^{1/n} = \mathbf{cap}(\Delta) e^{g_{\Omega}(z; \infty)}, \quad z \in \mathbb{C}, \quad (2.26)$$

except on a set of zero capacity, where $g_\Omega(z; \infty)$ is the Green's function for $\Omega = \overline{\mathbb{C}} \setminus \Delta$ with singularity at infinity (cf. [162, Appendix V]).

Moreover, uniformly on compact subsets of $\widehat{\Omega} = \mathbb{C} \setminus \mathbf{C}_h(\Delta_0 \cup \Delta_1)$

$$\lim_{n \rightarrow \infty} \left| L_n^{(j)}(z) \right|^{1/n} = \mathbf{cap}(\Delta) e^{g_\Omega(z; \infty)}, \quad (2.27)$$

and

$$\lim_{n \rightarrow \infty} \frac{L_n^{(j+1)}(z)}{n L_n^{(j)}(z)} = \int \frac{d\mu_\Delta(x)}{z - x}, \quad (2.28)$$

where $d\mu_\Delta$ is given by (2.24) when Δ_0 and Δ_1 have a common boundary point and by (2.25) if they are disjoint.

Remark 2.4.2

To give an explicit expression for the function on the right side of (2.26)-(2.27), we assume without loss of generality that $\Delta_0 = [a_0, b_0]$ and $\Delta_1 = [a_1, b_1]$ with $-\infty < a_0 < b_0 \leq a_1 < b_1 < \infty$. As we show below, there are closed formulas for the Green's function $g_\Omega(z; \infty)$ and the logarithmic capacity of Δ . When both intervals have the same length, $g_\Omega(z; \infty)$ and $\mathbf{cap}(\Delta)$ are given by elementary formulas (see (2.31) below). In general, (formulas (2.29)-(2.30)), they can be expressed in terms of theta-functions (see [154, 160]) defined by

$$\theta(u, \tau, r, s) = \sum_{k \in \mathbb{Z}} e^{2\pi i \left((k+r)^2 \frac{\tau}{2} + (k+r)(u+s) \right)}$$

where $u, \tau \in \mathbb{C}$, $\text{Im}(\tau) > 0$ and $r, s \in \mathbb{R}$. Here, we will be particularly interested in the functions

$$\vartheta(u, \tau) = \theta\left(u, \tau, \frac{1}{2}, \frac{1}{2}\right) \quad \text{and} \quad \vartheta_0(\tau) = \theta\left(0, \tau, 0, \frac{1}{2}\right).$$

Following the procedure for computing the logarithmic capacity of two segments given in [49, §1.3.3] and [154, Chap. 2], let $\Psi(z)$ be the function

$$\Psi(z) = \sqrt{\frac{(z - a_0)(b_1 - b_0)}{(z - b_0)(b_1 - a_0)}},$$

where $\sqrt{z} > 0$ for $z > 0$, and $\Upsilon(w)$ is the elliptic integral

$$\Upsilon(z) = \int_0^z \frac{dx}{\sqrt{(1-x^2)(1-v^2x^2)}}, \quad \text{where } v = \Psi(a_1)^{-1}.$$

Putting $\Phi(z) = \Upsilon(\Psi(z))$ and $\tau = \frac{\Phi(a_1)}{\Phi(b_1)}$, we obtain

$$g_\Omega(z; \infty) = -\log \left| \frac{\vartheta((2\Phi(b_1))^{-1}(\Phi(z) - \Phi(\infty)), \tau)}{\vartheta((2\Phi(b_1))^{-1}(\Phi(z) + \overline{\Phi(\infty)}), \tau)} \right|, \quad (2.29)$$

$$\mathbf{cap}(\Delta) = \left| \frac{\vartheta_0(\tau) \sqrt[4]{(a_1 - a_0)(a_1 - b_0)(b_1 - a_0)(b_1 - b_0)}}{2 \vartheta(\Phi^{-1}(b_1)\text{Re}(\Phi(\infty)), \tau)} \right|. \quad (2.30)$$

We recall that $a_0 < b_0 < a_1 < b_1$. Hence if $-a_0 = b_1$ and $-b_0 = a_1$ for the Green's function and the logarithmic capacity, we obtain

$$g_\Omega(z; \infty) = \frac{1}{2} \log^+ \left| \frac{\sqrt{z^2 - a_1^2} + \sqrt{z^2 - b_1^2}}{\sqrt{z^2 - a_1^2} - \sqrt{z^2 - b_1^2}} \right| \quad \text{and} \quad \mathbf{cap}(\Delta) = \frac{1}{2} \sqrt{b_1^2 - a_1^2}, \quad (2.31)$$

where

$$\log^+ |z| = \begin{cases} \log |z|, & \text{if } |z| > 1, \\ 0, & \text{if } |z| \leq 1. \end{cases}$$

Proof of Theorem 2.4.1. From Theorem 2.1.1, we have that for all $n \geq 2$, the critical points of L_n are simple and contained in $\mathbf{C}_h(\Delta_0 \cup \Delta_1)^\circ$. Now, Rolle's theorem implies that the zeros of all derivatives of higher order of L_n lie in the convex hull of the set of its critical points. Therefore, for all $n \geq 2$ and $j \geq 1$, the $(n-j)$ zeros $\{x_k^{(j)}\}$ of $L_n^{(j)}$ lie on $\mathbf{C}_h(\Delta_0 \cup \Delta_1)^\circ$. Thus, for each fixed $j \geq 1$, the support of the measure $\sigma(L_n^{(j)})$ is contained in $\mathbf{C}_h(\Delta_0 \cup \Delta_1)$. From the lower envelope theorem (cf. [162, Appendix III]) and (2.8), we get

$$\liminf_{n \rightarrow \infty} \int \log \frac{1}{|z-x|} d\sigma(L_n^{(j)})(x) = \int \log \frac{1}{|z-x|} d\mu_\Delta(x), \quad (2.32)$$

for all $z \in \mathbb{C}$ except on a set of zero capacity. But, from [162, Chap.1 and Eq. (2.3)]

$$\int \log \frac{1}{|z-x|} d\mu_\Delta(x) = \log \frac{1}{\mathbf{cap}(\Delta)} - g_\Omega(z; \infty),$$

hence (2.32) is equivalent to (2.26).

In order to prove (2.27), notice that for each fixed $j \geq 1$, the family of functions

$$\left\{ \int \log \frac{1}{|z-x|} d\sigma(L_n^{(j)})(x) \right\}, \quad n \in \mathbb{Z}_+,$$

is harmonic and uniformly bounded on compact subsets of $\widehat{\Omega}$. From (2.26), any subsequence which converges uniformly on compact subsets of $\widehat{\Omega}$ must tend to $\int \log |z-x|^{-1} d\mu_\Delta(x)$ (independent of the convergent subsequence chosen). Therefore, due to the uniform boundedness of the sequence of functions, the whole sequence converges uniformly on compact subsets of $\widehat{\Omega}$ to this function. This is equivalent to (2.27).

Finally, expanding the rational function in the right side of (2.28) in partial fractions, we get

$$\frac{L_n^{(j+1)}(z)}{nL_n^{(j)}(z)} = \frac{1}{n} \sum_{k=1}^{n-j} \frac{1}{z-x_k^{(j)}} = \frac{n-j}{n} \int \frac{d\sigma(L_n^{(j)})(x)}{z-x}.$$

Hence, it is straightforward that, for each fixed $j \geq 1$, the sequence of rational functions $\{L_n^{(j+1)}(z)/nL_n^{(j)}(z)\}_{n \geq j}$ is uniformly bounded on each compact subset of $\widehat{\Omega}$.

Therefore, from (2.8), we find that any subsequence of $\{L_n^{(j+1)}(z)/nL_n^{(j)}(z)\}_{n \geq j}$ which converges uniformly on compact subsets of $\hat{\Omega}$ converges pointwise to

$$\int \frac{d\mu_{\Delta}(x)}{z-x}.$$

Thus, the whole sequence converges uniformly on compact subsets of $\hat{\Omega}$ to this function as stated in (2.28). $\square$

To discuss the zeros of Sobolev extremal polynomials L_n , we need to introduce some notation. Recall that $\Omega = \mathbb{C} \setminus \Delta$ and that in the case we are now considering $g_{\Omega}(z, \infty)$ is given by (2.29). For $\rho > 0$, consider the set $\{z \in \mathbb{C} : g_{\Omega}(z, \infty) < \rho\}$. Depending on ρ this set has either one connected component (when ρ is large enough) or two connected components one of which contains Δ_0 and the other one Δ_1 . Let G_{ρ} be the connected components of $\{z \in \mathbb{C} : g_{\Omega}(z, \infty) < \rho\}$ which is disjoint from Δ_0 (See Figure 2.2).

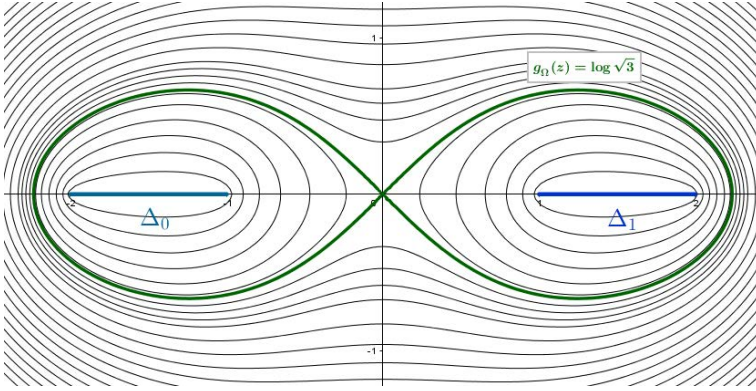


Figure 2.2: Level curves of the Green's function $g_{\Omega}(\cdot, \infty)$ of Example 2.3.1

If the intervals are not disjoint, obviously, $G_{\rho} = \emptyset$, for all $\rho > 0$, otherwise there exists $\rho_0 > 0$ such that $G_{\rho} = \emptyset$ for all $\rho > \rho_0 > 0$. Set $G = \bigcup_{\rho > 0} G_{\rho}$ and

$$\Gamma = \partial G \cup \Delta_0,$$

where ∂G is the boundary of G . In the sequel, $[\mu]_{\Gamma}$ denotes the balayage of a measure μ onto Γ . See [162, Appendix A:VII] for a brief introduction to the notion of balayage of a measure and for more details we refer to [80, Chap. IV].

Theorem 2.4.2 (Asymptotic zero distribution of the zeros of L_n)

Under the hypotheses of Theorem 2.1.3, let σ be a limit of a subsequence of $\{\sigma(L_n)\}_{n \geq 0}$ in the sense of the weak star topology of measures. Then $S(\sigma) \subset \Delta \cup \overline{G}$ and $[\sigma]_{\Gamma} = [\mu_{\Delta}]_{\Gamma}$. In addition, when Δ_0 and Δ_1 overlap, the full sequence converge to $\sigma = \mu_{\Delta}$.

Proof. The proof of this result is similar to that of [55, Th. 2 and Cor. 3] with $p = 2$. The main tool in the proof of [55, Th. 2] is [55, Th. 5] on the distribution of zeros of certain family of weighted polynomials. For the application of [55, Th. 5] it is necessary to prove that if $z \in \mathbb{C} \setminus G$ then

$$\limsup_{n \rightarrow \infty} |L_n(z)|^{1/n} \leq \mathbf{cap}(\Delta) e^{g_\Omega(z, \infty)}. \quad (2.33)$$

Replacing in the proof of [55, Lem. 8] the expressions [55, Eq. (4.1)-(4.2)] by (2.7), (2.21) and (2.23), it is straightforward to deduce (2.33). $\square$

To finish, we illustrate the results of the chapter for the Example 2.3.1. From (2.31) we get

$$g_\Omega(z; \infty) = \frac{1}{2} \log^+ \left| \frac{\sqrt{z^2 - 1} + \sqrt{z^2 - 4}}{\sqrt{z^2 - 1} - \sqrt{z^2 - 4}} \right| \quad \text{and} \quad \mathbf{cap}(\Delta) = \frac{\sqrt{3}}{2}.$$

By straightforward calculation we obtain $x_* = 0$ and

$$d\mu_\Delta = \frac{1}{\pi} \frac{|x|dx}{\sqrt{(4-x^2)(x^2-1)}}, \quad x \in (-2, -1) \cup (1, 2).$$

Thus, from Theorem 2.4.1 it follows that, for $j \geq 1$, we obtain

$$\limsup_{n \rightarrow \infty} |L_n^{(j)}(z)|^{1/n} = \frac{1}{2} \max \left\{ \sqrt{|2z^2 + 2\sqrt{z^2 - 1}\sqrt{z^2 - 4} - 5|}, 1 \right\}, \quad z \in \mathbb{C},$$

except on a set of zero capacity. Moreover

$$\begin{aligned} |L_n^{(j)}(z)|^{1/n} &\rightrightarrows \max \left\{ \sqrt{|2z^2 + 2\sqrt{z^2 - 1}\sqrt{z^2 - 4} - 5|}, 1 \right\}, & K \subset \mathbb{C} \setminus [-2, 2], \\ \frac{L_n^{(j+1)}(z)}{nL_n^{(j)}(z)} &\rightrightarrows \frac{1}{\pi} \int_1^2 \frac{2xzdx}{(z^2 - x^2)\sqrt{(4-x^2)(x^2-1)}}, & K \subset \mathbb{C} \setminus [-2, 2]. \end{aligned}$$

$$\begin{aligned} \Gamma &= \{z \in \mathbb{C} : |2z^2 + 2\sqrt{z^2 - 1}\sqrt{z^2 - 4} - 5| = 9, \Re z \geq 0\} \cup [-2, -1] \\ \Delta \cup \overline{G} &= [-2, -1] \cup \{z \in \mathbb{C} : |2z^2 + 2\sqrt{z^2 - 1}\sqrt{z^2 - 4} - 5| \leq 9, \Re z \geq 0\} \end{aligned}$$

and every limit σ of a subsequence of $\{\sigma(L_n)\}_{n \geq 0}$ (in weak star topology of measures) satisfies $\mathbf{supp}(\sigma) \subset \Delta \cup \overline{G}$ and $[\sigma]_\Gamma = [\mu_\Delta]_\Gamma$. It is worth to remark that this does not imply that the zeros of L_n lie on $\Delta \cup \overline{G}$, in this sense we only know that they are inside of the region $\mathbf{B}(-2, 1) \cup \mathbf{B}(2, 4)$, see Figure 2.3. However, it is expected (and numerical experiments seem to indicate) that the accumulation points of the zeros of the polynomials L_n draw Γ .

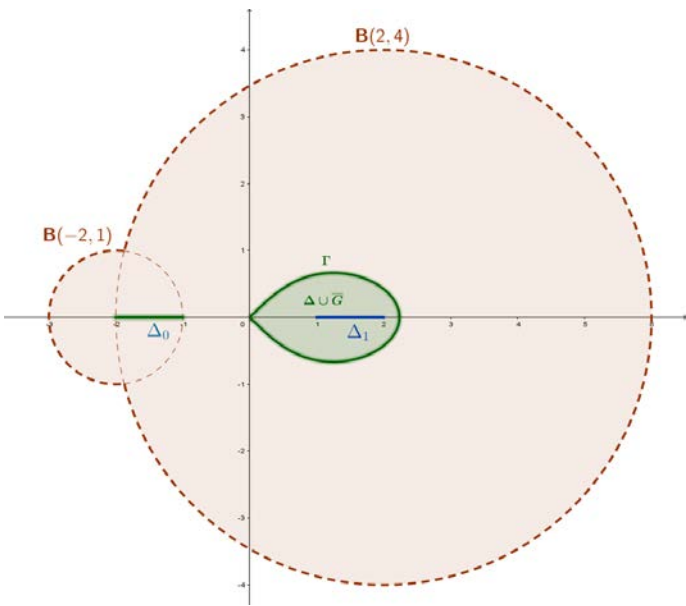


Figure 2.3: Scheme of results of Theorem 2.4.2 for Example 2.3.1

Sobolev-type Orthogonality and Rational Approximation

3.1 Introduction

As it was argued in Section 1.4, most of the arguments for the standard theory of orthogonal polynomials fail in the Sobolev case and the location of its zeros is one of them. In contrast to Theorem 1.2.3, M. Alfaro et al. [4, Th. 8] proved that there exist discrete Sobolev inner products in the form

$$\langle f, g \rangle_s = \int_{-a}^a f g d\mu + M f^{(\mathfrak{m})}(0) g^{(\mathfrak{m})}(0), \quad (3.1)$$

such that S_n has two complex conjugate zeros for infinitely many values of n , where $a > 0$, $\mathfrak{m} \geq 2$, $M > 0$ and μ is a symmetric measure. M. Alfaro et al. also found inner products in the above form, such that at least one of its orthogonal polynomials has a double zero at $z = 0$.

In this chapter, we will consider discrete Sobolev inner products, also named Sobolev-type inner products, which are usually defined by the expression

$$\langle f, g \rangle_s = \int f(x)g(x)d\mu(x) + \sum_{j=1}^{\mathcal{N}} \sum_{k=0}^{\ell_j} M_{j,k} f^{(k)}(c_j) g^{(k)}(c_j), \quad (3.2)$$

where μ is a standard measure, $M_{j,k} \geq 0$ and $M_{j,\ell_j} > 0$. It is easy to see that this formula agrees with the Sobolev inner products (1.9) whose component measures of $\vec{\mu}$ involving derivatives are trivial measures. Namely, $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ with $\mathfrak{m} = \max_{1 \leq j \leq \mathcal{N}} \{\ell_j\}$,

$$\mu_0 = \mu + \sum_{j=1}^{\mathcal{N}} M_{j,0} \delta_{c_j}, \quad \mu_k = \sum_{j=1}^{\mathcal{N}} M_{j,k} \delta_{c_j}, \quad k = 1, 2, \dots, \mathfrak{m},$$

and $M_{j,k} = 0$ for $\ell_j < k \leq \mathfrak{m}$, $j = 1, 2, \dots, \mathcal{N}$. Thus (1.9) becomes

$$\langle f, g \rangle_s = \int f(x)g(x)d\mu_0(x) + \sum_{j=1}^{\mathcal{N}} \sum_{k=1}^{\ell_j} M_{j,k} f^{(k)}(c_j) g^{(k)}(c_j). \quad (3.3)$$

We recall that $\Delta, \Delta_0, \dots, \Delta_{\mathfrak{m}}$ denote the respective convex hulls of the support of the measures $\mu, \mu_0, \dots, \mu_{\mathfrak{m}}$ and S_n denotes the n th polynomial orthogonal with respect to $\langle \cdot, \cdot \rangle_s$. Frequently, the points c_j will be called mass points.

Notice that, unlike (3.3), the representation (3.2) of $\langle \cdot, \cdot \rangle_s$ is not unique, it depends on how many Dirac deltas measures, of the discrete part of μ_0 , are included in the

measure μ . Thus, we assume without loss of generality that $M_{j,0} = 0$ for all j such that $c_j \in \Delta^\circ$. In general, the representation (3.2) is unique once the measure μ is fixed, so this dependence will be omitted for brevity.

Given μ a standard measure and $\langle \cdot, \cdot \rangle_s$ an inner product defined by (3.2) and satisfying $c_j \notin \Delta^\circ = (a, b)$ for $j = 1, 2, \dots, \mathcal{N}$, we consider the standard inner product

$$\langle f, g \rangle_{\mu_\rho} = \int_a^b f(x)g(x)d\mu_\rho(x), \quad (3.4)$$

where

$$\rho(z) = \prod_{c_j \leq a} (z - c_j)^{\ell_j+1} \prod_{c_j \geq b} (c_j - z)^{\ell_j+1} \quad \text{and} \quad d\mu_\rho(x) = \rho(x)d\mu(x).$$

Note that ρ is a polynomial of degree $\ell := \mathcal{N} + \sum_{j=1}^{\mathcal{N}} \ell_j$ and positive on (a, b) .

If $n > \ell$, then S_n satisfies the following quasi orthogonality relations with respect to μ_ρ

$$\langle S_n, P \rangle_{\mu_\rho} = \langle S_n, P\rho \rangle_\mu = \int S_n \cdot P\rho d\mu = \langle S_n, P\rho \rangle_s = 0, \quad (3.5)$$

for all $P \in \mathbb{P}_{n-\ell-1}$. Hence, the polynomial S_n is quasi-orthogonal of order ℓ with respect to μ_ρ , we thus obtain the following result.

Proposition 3.1.1 (Zero location in the discrete Sobolev case)

Let S_n be the n th orthogonal polynomial with respect to (3.2) which satisfies $c_j \notin \Delta^\circ$ for $j = 1, 2, \dots, \mathcal{N}$, and $n > \ell$, then S_n has at least $(n - \ell)$ changes of sign on Δ° .

This theorem can be seen as a generalization of the standard zero location Theorem 1.2.3. However, a result proved by M. G. Bruin, see [34, Th. 4.1], in 1993 seems to suggest that the number of zeros of S_n in Δ° does not depend on the higher order derivatives ℓ_j in (3.2) but on the number of terms in the discrete part

$$\ell^* := \left| \{M_{j,k} > 0 : j = 1, 2, \dots, \mathcal{N}, k = 0, 1, \dots, \ell_j\} \right|,$$

where $|A|$ denotes the number of elements of a set A . This suspicion became even stronger when the relative asymptotic of discrete Sobolev polynomials (1.21) was found. Finally, in 1996, G. López Lagomasino et al. proved that when (3.2) has only one mass point ($\mathcal{N} = 1$) this suspicion was correct.

Theorem 3.1.1 ([3, Th. 2.2])

Let μ be a standard measure such that $c_1 \in \mathbb{R} \setminus \Delta^\circ$. If S_n denotes the n th Sobolev polynomial with respect to

$$\langle f, g \rangle_s = \int f(x)g(x)d\mu(x) + \sum_{k=0}^{\mathfrak{m}} M_{1,k} f^{(k)}(c_1) g^{(k)}(c_1).$$

Then S_n has at least $n - \ell^*$ changes of sign in Δ° .

The next examples show that this theorem is not longer true if we consider arbitrary mass point configurations with more than one point (i.e. $\mathcal{N} \geq 2$ in (3.2)), at least not for every value of n .

Example 3.1.1

Set $\langle f, g \rangle_s = \int_{-1}^1 f(x)g(x)dx + 8f'(4)g'(4) + 6f''(2)g''(2)$, then

$$S_4(x) = k_4 \left(x^4 - \frac{2595}{803}x^3 - \frac{5232}{539}x^2 - \frac{837735}{39347}x + \frac{8181}{2695} \right),$$

whose zeros are approximately $\xi_1 \approx 0.13$, $\xi_2 \approx -5.62$, $\xi_3 \approx -1.26 + 1.56i$ and $\xi_4 \approx -1.26 - 1.56i$.

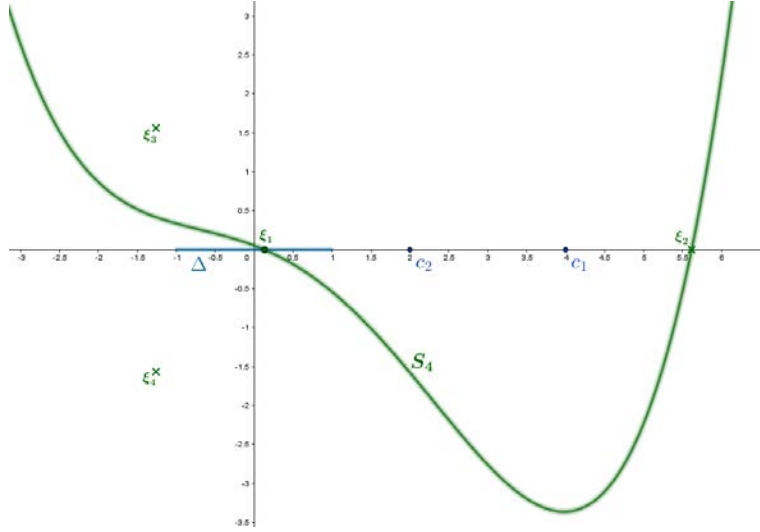


Figure 3.1: Sobolev polynomial of Example 3.1.1 for $k_4 = \frac{1}{56}$.

Example 3.1.2

Set $\langle f, g \rangle_s = \int_0^\infty f(x)g(x)e^{-x}dx + 3f'(-4)g'(-4) + 8f''(0)g''(0)$, then

$$S_4(x) = k_4 \left(x^4 - \frac{128}{97}x^3 - \frac{2536}{97}x^2 + \frac{8800}{97}x - \frac{5288}{97} \right),$$

whose zeros are approximately $\xi_1 \approx 0.79$, $\xi_2 \approx -5.93$, $\xi_3 \approx 3.24 + 1.16i$ and $\xi_4 \approx 3.24 - 1.16i$.

Note that, in both cases, three zeros of S_4 are out of Δ° and two of them are non-real.

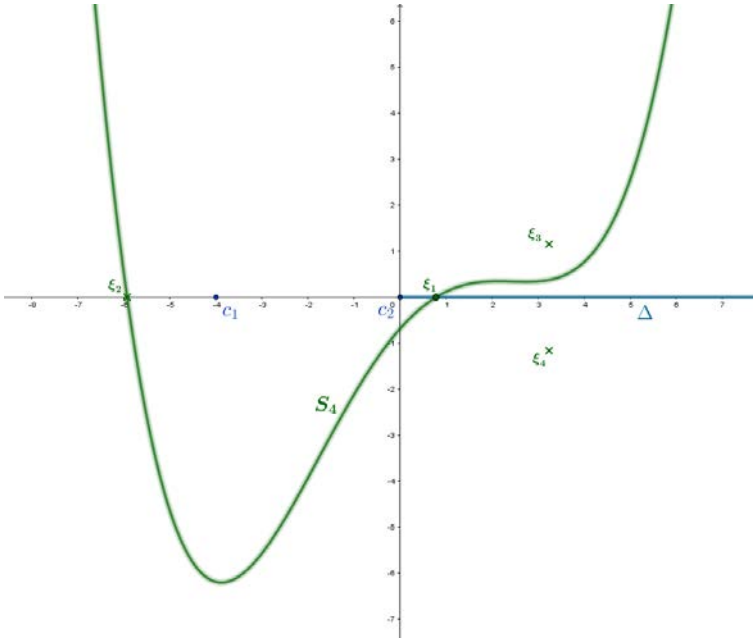


Figure 3.2: Sobolev polynomial of Example 3.1.2 for $k_4 = \frac{1}{80}$

In our first result of this chapter we generalize Theorem 3.1.1 to an arbitrary number of mass points c_j under some order restrictions. To do this, we first need to introduce the way in which they should be ordered.

Definition 3.1.1 (Sequentially-ordered Sobolev inner product)

We say that a Sobolev inner product $\langle \cdot, \cdot \rangle_s$ defined by (3.2), or its equivalent reformulation (3.3), is sequentially ordered if the conditions

$$\Delta_k \cap \mathbf{C}_h \left(\bigcup_{i=0}^{k-1} \Delta_i \right)^\circ = \emptyset, \quad k = 1, 2, \dots, \mathfrak{m},$$

hold. We recall that $\Delta_k = \mathbf{C}_h(\text{supp}(\mu_k))$, so they can be rewritten as

$$\Delta_k = \begin{cases} \mathbf{C}_h(\Delta \cup \{c_j : M_{j,0} > 0\}), & \text{if } k = 0, \\ \mathbf{C}_h(\{c_j : M_{j,k} > 0\}), & \text{if } 1 \leq k \leq \mathfrak{m}. \end{cases}$$

Example 3.1.3

The following inner products are sequentially ordered

$$\begin{aligned}\langle f, g \rangle_s &= \int_{-1}^1 f(x)g(x)d\mu(x) + f'(-1)g'(-1) + f'(-3)g'(-3) \\ &\quad + f''(2)g''(2) + f^{(5)}(-3)g^{(5)}(-3), \\ \langle f, g \rangle_s &= \int_{-1}^1 f(x)g(x)d\mu(x) + \sum_{k=0}^{\ell_1} M_{1,k}f^{(k)}(-1)g^{(k)}(-1) + \sum_{k=0}^{\ell_2} M_{2,k}f^{(k)}(1)g^{(k)}(1).\end{aligned}$$

where $M_{1,k}M_{2,k} = 0$ for $k = 0, 1, \dots, \min\{\ell_1, \ell_2\}$.

Theorem 3.1.2

If $\langle \cdot, \cdot \rangle_s$, defined by (3.2), is a sequentially-ordered discrete Sobolev inner product, then S_n has at least $n - \ell^*$ changes of sign on Δ° .

It is worth to remark that, although the theorem is enunciated depending on which representation (3.2) is considered, the definition of sequentially ordered inner product only depends on (3.3). In addition, notice that the condition $c_j \notin \Delta^\circ$ remains present in this theorem as a consequence of the assumption $M_{j,0} = 0$ whenever $c_j \in \Delta^\circ$ and the sequential order conditions.

On the other hand, if what we are looking for is to have located the largest number of zeros as possible, in the theorem, we should calculate ℓ^* considering the representation (3.3), rather than (3.2), but in this case we would have the zeros located in the bigger set $\Delta_0 \supset \Delta$. Because of the previous assumption on $M_{j,0}$, this inclusion is strict except for the trivial case when (3.3) and (3.2) are the same ($\mu \equiv \mu_0$).

Unlike what Corollary 2.2.1.1 ($p = 2$) would make us think, if $n \leq \ell^*$, S_n may does not change sign on Δ° . For example, in the case

$$\langle f, g \rangle_s = \int_{-1}^1 f(x)g(x)dx + f(6)g(6) + f'(10)g'(10),$$

$S_1(z) = z - 2$, which is negative on $\Delta^\circ = (-1, 1)$. This occurs because is due to we cannot confuse μ in (3.2) with μ_0 in (3.3), so there is no contradiction with Corollary 2.2.1.1.

Notice that the inner product (3.1) considered by M. Alfaro et al. as well as those inner products given in Example 3.1.1 and Example 3.1.2 are not sequentially-ordered. Thus, this order restriction in the discrete part seems to be optimal for having the most number of zeros simple and located on Δ° , at least for every value of n .

One major advantage of Theorem 3.1.2 comes from put it together with the relative asymptotic behavior (1.21). From this result, Theorem 1.2.3, and Rouché's Theorem we can ensure that every mass point attracts exactly ℓ_j^* zeros of S_n as n goes to infinity, whenever the measure μ is supported in $[-1, 1]$, belongs to the Nevai class $\mathbf{M}(0, 1)$, and $c_j \notin [-1, 1]$ for $j = 1, \dots, \mathcal{N}$. Since $\ell^* = \sum_{j=1}^{\mathcal{N}} \ell_j^*$, Theorem 3.1.2 states that, for n large enough, S_n has exactly $n - \ell^*$ simple zeros on $(-1, 1)$. This will be seen thoroughly in Lemma 3.3.3.

Let $\{P_n\}_{n \geq 0}$ be a sequence of orthogonal polynomials with respect to the standard measure μ . Let us denote by $P_n^{[1]}$ the usually called the n th polynomial associated to

P_n , defined by the expression

$$P_n^{[1]}(z) = \int_{-1}^1 \frac{P_{n+1}(z) - P_{n+1}(x)}{z - x} d\mu(x).$$

Note that $P_n^{[1]}$ is a polynomial of degree n with leading coefficient $\kappa(P_{n+1})\mu([-1, 1])$. From the standard three-term recurrence relation (1.5) satisfied by the monic orthogonal polynomial sequence $\{\hat{P}_n\}$ we have the following three-term recurrence relation for $\hat{P}_n^{[1]}$

$$\begin{aligned} \hat{P}_{n+1}^{[1]}(z) &= (z - b_{n+1})\hat{P}_n^{[1]}(z) - a_{n+1}^2\hat{P}_{n-1}^{[1]}(z), \quad n \geq 0, \\ \hat{P}_{-1}^{[1]}(z) &= 0 \quad \text{and} \quad \hat{P}_0^{[1]}(z) = \mu([-1, 1]). \end{aligned} \quad (3.6)$$

As it is known, some particular families of orthogonal polynomials were studied in detail before a general theory was stated. One of the starting points of this theory is closely related to the study of the convergence of certain sequences of rational functions, as it can be seen in the first monographs on the subject [161, Chap. I and §4] and [163, §3.5]. The analysis of the convergence of these sequences entails essential difficulties. One of the first, and most remarkable, general results in this sense is the following theorem established by A.A. Markov in 1895.

Markov's Theorem ([130, Th. 6.1])

Let μ be a standard measure supported in $[-1, 1]$. Then

$$\frac{P_n^{[1]}(z)}{P_{n+1}(z)} \xrightarrow[n]{} \hat{\mu}(z), \quad K \subset \Omega_\infty = \overline{\mathbb{C}} \setminus [-1, 1],$$

where $\hat{\mu}(z) = \int_{-1}^1 \frac{d\mu(x)}{z - x}$ is known as the Markov's function of μ .

Note that $\hat{\mu}(z)$ is well defined and holomorphic in Ω_∞ ($\hat{\mu} \in \mathcal{H}(\Omega_\infty)$ for short). Some examples can be seen in [130, p. 64]. This classical theorem admits several generalizations, some of which are discussed in [38, 58, 59, 88] and the references therein.

Let $\{\hat{Q}_n\}_{n \geq 0}$ be the sequence of monic orthogonal polynomials with respect to (3.4). We associate to the sequence $\{S_n\}_{n \geq 0}$ orthogonal with respect to (3.2), the next sequences of polynomials

$$S_n^{[k]}(z) = \int_{-1}^1 \frac{S_{n+k}(z) - S_{n+k}(x)}{z - x} \hat{Q}_{k-1}(x) d\mu_p(x), \quad (3.7)$$

for $k \in \mathbb{N}$ and $n \geq 0$. Additionally, we adopt the convention $S_n^{[0]} = S_n$. We call $\{S_n^{[k]}\}_{n \geq 0}$ the sequence of k th polynomials associated to $\{S_n\}_{n \geq 0}$.

As far as we know, the only extension of Markov's Theorem for Sobolev orthogonal polynomials appears in [96, Th. 5.5], when the inner product (3.2) is such that $\mathcal{N} = 1$, $\ell_1 = 1$, $c_1 = 0$, $M_{1,0} = 0$, $M_{1,1} > 0$ and μ is supported on $[\alpha, \beta]$ with $\beta < 0$. The main goal of the present chapter is to prove the following theorem, which provides a natural extension of the Markov's Theorem for the Sobolev case.

Theorem 3.1.3 (Extended Markov's Theorem)

Let $\Omega_\infty^* = \Omega_\infty \setminus \{c_1, c_2, \dots, c_N\}$ and let (3.2) be a sequentially-ordered discrete Sobolev inner product with $\mu \in \mathbf{M}(0, 1)$, $\text{supp}(\mu) = [-1, 1]$, and $c_j \notin [-1, 1]$ for $j = 1, \dots, N$. Then, for $k \in \mathbb{N}$,

$$R_n^{[k]}(z) = \frac{S_n^{[k]}(z)}{S_{n+k}(z)} \xrightarrow{n} \hat{\mu}_k(z) := \int_{-1}^1 \frac{\hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x), \quad K \subset \Omega_\infty^*. \quad (3.8)$$

$\hat{\mu}_k$ is said to be the k th Markov-type function associated with μ_ρ .

Also, in Corollary 3.5.0.1, we give the following estimate for the degree of convergence of the sequence of rational functions $\{R_n^{[k]}\}_{n \geq 0}$ to the corresponding Markov-type function $\hat{\mu}_k$.

$$\limsup_n \left\| \hat{\mu}_k - R_n^{[k]} \right\|_K^{1/2n} \leq \|\varphi\|_K^{-1} < 1.$$

The content of this chapter is based on [42].

3.2 Recurrence relations

Unlike the rest of the chapter, the inner product (3.2) does not necessarily have to be sequentially-ordered in this section. It will be enough that $c_j \notin \Delta^\circ$ for $j = 1, 2, \dots, N$.

Proposition 3.2.1

Let $S_n^{[k]}$ be the k th associated polynomial defined by (3.7). Then $S_n^{[k]}$ is a polynomial of degree n and leading coefficient equal to $\kappa(S_{n+k}) \|\hat{Q}_{k-1}\|_{\mu_\rho}^2$.

Proof. Let $S_{n+k}(x) = \sum_{i=0}^{n+k} \theta_i x^i$, then

$$\begin{aligned} S_n^{[k]}(z) &= \int_{-1}^1 \frac{S_{n+k}(z) - S_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) \\ &= \sum_{i=1}^{n+k} \theta_i \int_{-1}^1 \frac{z^i - x^i}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) \\ &= \sum_{i=1}^{n+k} \theta_i \int_{-1}^1 \left(\sum_{j=0}^{i-1} z^{i-1-j} x^j \right) \hat{Q}_{k-1}(x) d\mu_\rho(x) \\ &= \sum_{i=1}^{n+k} \theta_i \sum_{j=0}^{i-1} z^{i-1-j} \left(\int_{-1}^1 x^j \hat{Q}_{k-1}(x) d\mu_\rho(x) \right) \\ &= \sum_{i=1}^{n+k} \theta_i \sum_{j=k-1}^{i-1} \langle x^j, \hat{Q}_{k-1} \rangle_{\mu_\rho} z^{i-1-j} \\ &= \theta_{n+k} \sum_{j=k-1}^{n+k-1} \langle x^j, \hat{Q}_{k-1} \rangle_{\mu_\rho} z^{n+k-1-j} + \sum_{i=1}^{n+k-1} \theta_i \sum_{j=k-1}^{i-1} \langle x^j, \hat{Q}_{k-1} \rangle_{\mu_\rho} z^{i-1-j} \end{aligned}$$

$$\begin{aligned}
&= \theta_{n+k} \langle x^{k-1}, \hat{Q}_{k-1} \rangle_{\mu_\rho} z^n + f_{n-1}(z) \\
&= \theta_{n+k} \|\hat{Q}_{k-1}\|_{\mu_\rho}^2 z^n + f_{n-1}(z),
\end{aligned}$$

where f_{n-1} is a polynomial of degree at most $n-1$. $\square$

In the standard case, where the polynomials $\{\hat{P}_n\}_{n \geq 0}$ satisfy the three terms recurrence relation (1.5), the sequence of associated polynomials $\{P_n^{[1]}\}_{n \geq 0}$ can be generated by the recurrence relation (3.6). The following proposition is an analogous result for the sequence of associated polynomials $\{S_n^{[k]}\}_{n \geq 0}$.

Proposition 3.2.2 (Recurrence relation of $s_n^{[k]}$)

For $n \geq 2\ell - 1$, the Sobolev polynomials sequences of orthonormal polynomials $\{s_n^{[k]}\}$ satisfy the following $2\ell + 1$ term recurrence relation

$$\rho(z)s_n^{[k]}(z) = \sum_{j=n-\ell}^{n+\ell} a_{n+k,j+k} s_j^{[k]}(z), \quad (3.9)$$

where $a_{n,j} = \langle s_n, \rho s_j \rangle_s$ and $a_{n,n \pm \ell} > 0$.

Proof. For $k = 0$, (3.9) agrees with the higher order term recurrence relation (1.18) and it can be obtained directly from (3.5), i.e.,

$$\rho(z)s_n(z) = \sum_{j=n-\ell}^{n+\ell} a_{n,j} s_j(z), \quad \text{where } a_{n,j} = \langle s_n, \rho s_j \rangle_s.$$

Hence, if $k > 0$

$$\frac{\rho(z)s_{n+k}(z) - \rho(x)s_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) = \sum_{j=n-\ell}^{n+\ell} a_{n+k,j+k} \frac{s_{j+k}(z) - s_{j+k}(x)}{z-x} \hat{Q}_{k-1}(x),$$

so that

$$\int \frac{\rho(z)s_{n+k}(z) - \rho(x)s_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) = \sum_{j=n-\ell}^{n+\ell} a_{n+k,j+k} s_j^{[k]}(z).$$

As $n \geq 2\ell - 1$, from (3.5), we get $\int s_{n+k}(x) \frac{\rho(z) - \rho(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) = 0$. Hence,

$$\begin{aligned}
\rho(z)s_n^{[k]}(z) &= \int \frac{\rho(z)(s_{n+k}(z) - s_{n+k}(x))}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) \\
&\quad + \int s_{n+k}(x) \frac{\rho(z) - \rho(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) \\
&= \int \frac{\rho(z)s_{n+k}(z) - \rho(x)s_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x),
\end{aligned}$$

and we get (3.9). $\square$

Remember that $\{\hat{Q}_n\}_{n \geq 0}$ is the sequence of monic orthogonal polynomials with respect to μ_ρ , which was defined in (3.4). As it is was seen in Theorem 1.2.2, this sequence satisfies the three-term recurrence relation

$$\hat{Q}_{n+1}(z) = (z - \beta_n)\hat{Q}_n(z) - \alpha_n^2\hat{Q}_{n-1}(z), \quad n \geq 0,$$

where $\hat{Q}_{-1} = 0$, $\hat{Q}_0 = 1$, $\beta_n = \langle \hat{Q}_n, x\hat{Q}_n \rangle_{\mu_\rho} / \|\hat{Q}_n\|_{\mu_\rho}^2$, $\alpha_n = \|\hat{Q}_n\|_{\mu_\rho} / \|\hat{Q}_{n-1}\|_{\mu_\rho}$ and $\alpha_0^2 = \int_{-1}^1 d\mu_\rho(x)$.

Following [165], we define its k th sequence of associated polynomials ($k \in \mathbb{Z}_+$) as

$$Q_n^{[k]}(z) = \int \frac{Q_{n+k}(z) - Q_{n+k}(x)}{z - x} \hat{Q}_{k-1}(x) d\mu_\rho(x),$$

where $Q_n^{[0]} = \hat{Q}_n$. Note that $Q_n^{[k]}$ is a polynomial in z of degree n . From [165, Eq. (1.3) and Eq. (2.13)]

$$Q_{n+1}^{[k]}(x) = (x - \beta_{n+k})Q_n^{[k]}(x) - \alpha_{n+k}^2 Q_{n-1}^{[k]}(x).$$

The next proposition is the analogous of [165, Eq. (2.5)] for the Sobolev case.

Proposition 3.2.3

For $n \geq \ell - 1$ and $k \geq 2$, the relation

$$S_n^{[k]}(z) = (z - \beta_{k-2})S_{n+1}^{[k-1]}(z) - \alpha_{k-2}^2 S_{n+2}^{[k-2]}(z) \quad (3.10)$$

holds.

Proof. We deal with the case $k = 2$ and $k \geq 3$ separately. For $k = 2$, we have

$$\begin{aligned} S_n^{[2]}(z) &= \int_{-1}^1 \frac{S_{n+2}(z) - S_{n+2}(x)}{z - x} ((x - \beta_0)\hat{Q}_0(x) - \alpha_0^2\hat{Q}_{-1}(x)) d\mu_\rho(x) \\ &= \int_{-1}^1 \frac{S_{n+2}(z) - S_{n+2}(x)}{z - x} x d\mu_\rho(x) - \beta_0 S_{n+1}^{[1]}(z). \end{aligned}$$

Otherwise, if $k \geq 3$, then

$$\begin{aligned} S_n^{[k]}(z) &= \int \frac{S_{n+k}(z) - S_{n+k}(x)}{z - x} ((x - \beta_{k-2})\hat{Q}_{k-2}(x) - \alpha_{k-2}^2\hat{Q}_{k-3}(x)) d\mu_\rho(x) \\ &= \int \frac{S_{n+k}(z) - S_{n+k}(x)}{z - x} x \hat{Q}_{k-2}(x) d\mu_\rho(x) - \beta_{k-2} S_{n+1}^{[k-1]}(z) - \alpha_{k-2}^2 S_{n+2}^{[k-2]}(z). \end{aligned}$$

From orthogonality property of the sequence $\{\hat{Q}_n\}_{n \geq 0}$ and quasi-orthogonality relations (3.5),

$$\int \frac{S_{n+k}(z) - S_{n+k}(x)}{z - x} (z - x) \hat{Q}_{k-2}(x) d\mu_\rho(x) = \begin{cases} \alpha_0^2 S_{n+2}(z), & \text{if } k = 2, \\ 0, & \text{if } k \geq 3. \end{cases}$$

Therefore,

$$\begin{aligned} \int \frac{S_{n+k}(z) - S_{n+k}(x)}{z-x} x \hat{Q}_{k-2}(x) d\mu_\rho(x) \\ = \begin{cases} zS_{n+1}^{[1]}(z) - \alpha_0^2 S_{n+2}(z), & \text{if } k=2, \\ zS_{n+1}^{[k-1]}(z), & \text{if } k \geq 3. \end{cases} \end{aligned} \quad (3.11)$$

Substituting (3.11) into the previous expressions for $S_n^{[k]}$, we get (3.10). $\square$

3.3 Proof of Theorem 3.1.2

The next lemma is an extension of [92, Lem. 2.1].

Lemma 3.3.1

Let $\{I_i\}_{i=0}^m$ be a set of intervals on the real line with $m \in \mathbb{Z}_+$ and let P be a polynomial with real coefficients of degree $\geq m$. If

$$I_k \cap \mathbf{C}_h \left(\bigcup_{i=0}^{k-1} I_i \right)^\circ = \emptyset, \quad k = 1, 2, \dots, m, \quad (3.12)$$

then

$$\begin{aligned} \mathbf{N}_z(P; J) + \mathbf{N}_o(P; I_0 \setminus J) + \sum_{i=1}^m \mathbf{N}_o(P^{(i)}; I_i) \leq \mathbf{N}_z(P^{(m)}; J) + \\ \mathbf{N}_o(P^{(m)}; \mathbf{C}_h(\bigcup_{i=0}^m I_i) \setminus J) + m, \end{aligned} \quad (3.13)$$

for every closed subinterval J of I_0° (both empty set and unitary sets are assumed to be intervals).

Proof. First, we are going to point out the following consequence of Rolle's Theorem. If I is a real interval and J is a closed subinterval of I° , then

$$\mathbf{N}_z(P; J) + \mathbf{N}_o(P; I \setminus J) \leq \mathbf{N}_z(P'; J) + \mathbf{N}_o(P'; I^\circ \setminus J) + 1. \quad (3.14)$$

For $m = 0$ (3.13) trivially holds. We now proceed by induction on m . Suppose that we have $m+1$ intervals $\{I_i\}_{i=0}^m$ satisfying (3.12), and that (3.13) is true for the first m intervals $\{I_k\}_{k=0}^{m-1}$. From (3.14) we obtain

$$\begin{aligned} \mathbf{N}_z(P; J) + \mathbf{N}_o(P; I_0 \setminus J) + \sum_{i=1}^m \mathbf{N}_o(P^{(i)}; I_i) \\ \leq \mathbf{N}_z(P^{(m-1)}; J) + \mathbf{N}_o(P^{(m-1)}; \mathbf{C}_h(\bigcup_{i=0}^{m-1} I_i) \setminus J) + m - 1 + \mathbf{N}_o(P^{(m)}; I_m) \\ \leq \mathbf{N}_z(P^{(m)}; J) + \mathbf{N}_o(P^{(m)}; \mathbf{C}_h(\bigcup_{i=0}^{m-1} I_i)^\circ \setminus J) + m + \mathbf{N}_o(P^{(m)}; I_m) \\ \leq \mathbf{N}_z(P^{(m)}; J) + \mathbf{N}_o(P^{(m)}; \mathbf{C}_h(\bigcup_{i=0}^m I_i) \setminus J) + m. \end{aligned}$$

$\square$

Corollary 3.3.0.1

Under the hypotheses of the above lemma we have

$$\mathbf{N}_z(P; J) + \mathbf{N}_o(P; I_0 \setminus J) + \sum_{i=1}^m \mathbf{N}_o(P^{(i)}; I_i) \leq \mathbf{deg}(P) \quad (3.15)$$

for every J closed subinterval of I_0° . In particular for $J = \emptyset$ we get

$$\sum_{i=0}^m \mathbf{N}_o(P^{(i)}; I_i) \leq \mathbf{deg}(P). \quad (3.16)$$

Definition 3.3.1

We say that a sequence of ordered pairs $\{(r_i, v_i)\}_{i=1}^M \subset \mathbb{R} \times \mathbb{Z}_+$ is sequentially-ordered, if the following conditions hold

1. $v_1 \leq v_2 \leq \dots \leq v_M$.
2. The set of intervals

$$I_k = \mathbf{C}_h(\{r_i : v_i = k\}), \quad k = 0, 1, \dots, v_M,$$

satisfy conditions (3.12).

Lemma 3.3.2

Let $\{(r_i, v_i)\}_{i=1}^M \subset \mathbb{R} \times \mathbb{Z}_+$ be a sequence of M ordered pairs, then there exists a unique monic polynomial U_M of minimal degree ($\leq M$), such that

$$U_M^{(v_i)}(r_i) = 0, \quad i = 1, 2, \dots, M. \quad (3.17)$$

Furthermore, if $\{(r_i, v_i)\}_{i=1}^M$ is sequentially-ordered, then the degree of U_M is $u_M = \min \mathfrak{I}_M - 1$, where

$$\mathfrak{I}_M = \{i : 1 \leq i \leq M \text{ and } v_i \geq i\} \cup \{M+1\}.$$

Proof. The existence of a not identically zero polynomial with degree $\leq M$ satisfying (3.17) reduces to solving a homogeneous linear system of M equations with $M+1$ unknowns (its coefficients). Thus, a non trivial solution always exists. In addition, if we suppose that there exist two different minimal monic polynomials U_M and $\tilde{U}_M$, then the polynomial $\hat{U}_M = U_M - \tilde{U}_M$ is not identically zero, it satisfies (3.17), and $\mathbf{deg}(\hat{U}_M) < \mathbf{deg}(U_M)$. Then, if we divide $\hat{U}_M$ by its leading coefficient, we reach a contradiction.

The rest of the proof runs by induction on the number of points M . For $M = 1$, the result follows taking

$$U_1(x) = \begin{cases} x - r_1, & \text{if } v_1 = 0, \\ 1, & \text{if } v_1 \geq 1. \end{cases}$$

Suppose that, for each sequentially-ordered sequence of M ordered pairs, the corresponding minimal polynomial U_M has degree u_M .

Let $\{(r_i, v_i)\}_{i=1}^M$ be a sequentially-ordered sequence of M ordered pairs. Obviously, $\{(r_i, v_i)\}_{i=1}^{M-1}$ is a sequence of $M-1$ ordered pairs which is sequentially-ordered, $\deg(U_M) \geq \deg(U_{M-1})$, and from the induction hypothesis $\deg(U_{M-1}) = u_{M-1}$. Now, we shall split the proof in two cases:

1. If $u_M = M$, then for all $1 \leq i \leq M$ we have $v_i < i$, which yields

$$\deg(U_M) \geq \deg(U_{M-1}) = u_{M-1} = M-1 \geq v_M.$$

Since $\{(r_i, v_i)\}_{i=1}^M$ is sequentially-ordered, from (3.16) we get

$$M \leq \sum_{i=0}^{v_M} N_o(U_M^{(i)}; I_i) \leq \deg(U_M),$$

which implies that $\deg(U_M) = M = u_M$.

2. If $u_M \leq M-1$, then there exists a minimal j ($1 \leq j \leq M$), such that $v_j \geq j$, and $v_i < i$ for all $1 \leq i \leq j-1$. Therefore, $u_M = j-1 = u_{M-1}$. From the induction hypothesis

$$\deg(U_{M-1}) = u_{M-1} = j-1 \leq v_j - 1 \leq v_M - 1,$$

which gives $U_{M-1}^{(v_M)} \equiv 0$. Hence, $U_M \equiv U_{M-1}$ and, consequently, we get

$$\deg(U_M) = \deg(U_{M-1}) = u_{M-1} = u_M.$$

□

Note that, in Lemma 3.3.2, the assumption of $\{(r_i, v_i)\}_{i=1}^M$ being sequentially ordered is necessary for asserting that the polynomial U_M has degree u_M . In fact, if we consider $\{(-1, 0), (1, 0), (0, 1)\}$, which is not sequentially ordered, we obtain $U_3 = x^2 - 1$ and $u_3 = 3 \neq \deg(U_3)$.

Proof of Theorem 3.1.2. Let $\xi_1 < \xi_2 < \dots < \xi_\eta$ be the points on Δ° where S_n changes sign and suppose that $\eta < n - \ell^*$. Since $\langle \cdot, \cdot \rangle_s$ is sequentially-ordered, the sequence of $\ell^* + \eta$ ordered pairs

$$\{(r_i, v_i)\}_{i=1}^{\ell^* + \eta} = \{(\xi_i, 0)\}_{i=1}^\eta \cup \{(c_j, i) : M_{j,i} > 0, j = 1, 2, \dots, \mathcal{N}, i = 0, 1, \dots, \ell_j\}$$

is sequentially ordered (we can assume without loss of generality that $v_1 \leq v_2 \leq \dots \leq v_{\ell^* + \eta}$). Consequently, from Lemma 3.3.2, there exists a unique monic polynomial $U_{\ell^* + \eta}$ of minimal degree, such that

$$\begin{aligned} U_{\ell^* + \eta}(\xi_i) &= 0; & \text{for } i = 1, \dots, \eta, \\ U_{\ell^* + \eta}^{(i)}(c_j) &= 0; & \text{for } (j, i) : M_{j,i} > 0, \end{aligned} \quad (3.18)$$

and

$$\deg(U_{\ell^* + \eta}) = \min \mathcal{J}_{\ell^* + \eta} - 1 \leq \ell^* + \eta,$$

where

$$\mathfrak{I}_{\ell^*+\eta} = \{i : 1 \leq i \leq \ell^* + \eta \text{ and } v_i \geq i\} \cup \{\ell^* + \eta + 1\}. \quad (3.19)$$

Now, we must consider the following two cases.

1. If $\deg(U_{\ell^*+\eta}) = \ell^* + \eta$, from (3.19), we get $\deg(U_{\ell^*+\eta}) = \ell^* + \eta \geq v_{\eta+\ell^*} + 1$. Thus, taking $I_i = \Delta_i$, $i = 0, 1, \dots, \mathfrak{m}$ and the closed interval $J = [\xi_1, \xi_\eta] \subset \Delta^\circ \subset \Delta_0^\circ$ in (3.15), we get

$$\begin{aligned} \ell^* + \eta &\leq \sum_{k=0}^{v_{\ell^*+\eta}} \mathbf{N}_o(U_{\ell^*+\eta}^{(k)}; \Delta_k) \leq \mathbf{N}_z(U_{\ell^*+\eta}; J) + \mathbf{N}_o(U_{\ell^*+\eta}; \Delta_0 \setminus J) \\ &\quad + \sum_{k=1}^{v_{\ell^*+\eta}} \mathbf{N}_o(U_{\ell^*+\eta}^{(k)}; \Delta_k) \leq \deg(U_{\ell^*+\eta}) = \ell^* + \eta. \end{aligned}$$

2. If $\deg(U_{\ell^*+\eta}) < \ell^* + \eta$, from (3.19), there exists $1 \leq j \leq \ell^* + \eta$ such that $\deg(U_{\ell^*+\eta}) = j - 1$, $v_j \geq j$ and $v_i \leq i - 1$ for $i = 1, 2, \dots, j - 1$. Hence,

$$v_{j-1} + 1 \leq j - 1 = \deg(U_{\ell^*+\eta})$$

and, again, from (3.15) we have

$$\begin{aligned} j - 1 &\leq \sum_{k=0}^{v_{j-1}} \mathbf{N}_o(U_{\ell^*+\eta}^{(k)}; \Delta_k) \leq \mathbf{N}_z(U_{\ell^*+\eta}; J) + \mathbf{N}_o(U_{\ell^*+\eta}; \Delta_0 \setminus J) \\ &\quad + \sum_{k=1}^{v_{j-1}} \mathbf{N}_o(U_{\ell^*+\eta}^{(k)}; \Delta_k) \leq \deg(U_{\ell^*+\eta}) = j - 1. \end{aligned}$$

In both cases, we obtain that $U_{\ell^*+\eta}$ has no other zeros than those given by construction and from $\mathbf{N}_o(U_{\ell^*+\eta}; J) = \mathbf{N}_z(U_{\ell^*+\eta}; J)$ all the zeros on Δ° are simple. Thus, in addition to (3.18), we have that $S_n U_{\ell^*+\eta}$ does not change sign on Δ° . Now, since $\deg(U_{\ell^*+\eta}) \leq \ell^* + \eta < n$, we arrive to the contradiction

$$\begin{aligned} 0 &= \langle S_n, U_{\ell^*+\eta} \rangle_s = \int S_n(x) U_{\ell^*+\eta}(x) d\mu(x) + \sum_{j=1}^{\mathcal{N}} \sum_{k=0}^{\ell_j} M_{j,k} S_n^{(k)}(c_j) U_{\ell^*+\eta}^{(k)}(c_j) \\ &= \int_{\Delta} S_n(x) U_{\ell^*+\eta}(x) d\mu(x) \neq 0. \end{aligned}$$

□

Now, combining Theorem 3.1.2 and (1.21), we obtain the following useful lemma.

Lemma 3.3.3

Let (3.2) be a sequentially-ordered Sobolev inner product such that $\text{supp}(\mu) = [-1, 1]$, $\mu \in \mathbf{M}(0, 1)$ and $c_j \notin [-1, 1]$ for $j = 1, 2, \dots, \mathcal{N}$. Then

1. For all n sufficiently large, each sufficiently small neighborhood of c_j ; $j = 1, \dots, N$; contains exactly ℓ_j^* zeros of S_n , and the remaining $n - \ell^*$ zeros are simple and lie on $(-1, 1)$.
2. The set of zeros of $\{S_n\}_{n \geq 1}$ is uniformly bounded.

Proof. The first part of assertion 1. of the lemma is a direct consequence (1.21) and Rouché's Theorem (see [62, Th. 9.2.3]) and the second one is just Theorem 3.1.2. The second assertion is a direct consequence of the first one. $\square$

3.4 Auxiliary lemmas

Let $\hat{S}_n$ be the n th monic orthogonal polynomial with respect to the sequentially ordered inner product (3.2) under the hypothesis of Lemma 3.3.3. Taking into consideration Lemma 3.3.3, let $\{\xi_{n,i}\}_{i=1}^{n-\ell^*}$ be the $n - \ell^*$ simple zeros of $\hat{S}_n$ on $(-1, 1)$ for all sufficiently large n and let $\{\xi_{n,n-\ell^*+i}\}_{i=1}^{\ell^*}$ be the remaining ℓ^* zeros of $\hat{S}_n$. Obviously, $\hat{S}_n$ admits the representation

$$\hat{S}_n(x) = S_{n,1}(x)S_{n,2}(x),$$

where

$$S_{n,1}(x) = \prod_{i=1}^{n-\ell^*} (x - \xi_{n,i}) \quad \text{and} \quad S_{n,2}(x) = \prod_{i=n-\ell^*+1}^n (x - \xi_{n,i}).$$

From Proposition 2.2.1 ($p = 2$) $\hat{S}_n$ has real coefficients, so $S_{n,2}(x)$ also has them. From Lemma 3.3.3, for all sufficiently large n , the last ℓ^* zeros of $\hat{S}_n$ are outside of $[-1, 1]$, so the sign of $S_{n,2}$ is constant on $[-1, 1]$ and does not depend on n . Indeed, the sign is equal to $(-1)^{\mathfrak{N}}$, where $\mathfrak{N} = \sum_{c_j > 1} \ell_j^*$. Thus, the polynomial $S_{n,2}^+(x) = (-1)^{\mathfrak{N}} S_{n,2}(x)$ is positive on $[-1, 1]$.

The following Lemma is an analogous of the Gauss-Jacobi quadrature formula for the sequentially-ordered Sobolev inner product (3.2), when n is large enough.

Lemma 3.4.1

Let S_n and $\{\xi_{n,i}\}_{i=1}^{n-\ell^*}$ as above. If n is large enough, then for every polynomial T with $\deg(T) \leq 2n - \ell - \ell^* - 1$,

$$\int_{-1}^1 T(x) S_{n,2}^+(x) d\mu_p(x) = \sum_{i=1}^{n-\ell^*} \lambda_{n,i} S_{n,2}^+(\xi_{n,i}) T(\xi_{n,i}), \quad (3.20)$$

$$\text{where } \lambda_{n,i} = \int_{-1}^1 \frac{S_n(x)}{S'_n(\xi_{n,i})(x - \xi_{n,i})} d\mu_p(x).$$

Moreover, the number of positive coefficients $\lambda_{n,i}$ is greater than or equal to $\left(n - \frac{\ell + \ell^*}{2}\right)$.

We call Christoffel-type coefficients to the numbers $\{\lambda_{n,i}\}_{i=1}^{n-\ell^*}$.

Proof. Let T be an arbitrary polynomial of degree at most $2n - \ell - \ell^* - 1$ and denote by L the Lagrange polynomial interpolating the polynomial T at the points $\xi_{n,1}, \dots, \xi_{n,n-\ell^*}$ ($\deg(L) < n - \ell^*$), i.e.

$$L(z) = \sum_{i=1}^{n-\ell^*} T(\xi_{n,i}) \frac{S_{n,1}(z)}{S'_{n,1}(\xi_{n,i})(z - \xi_{n,i})}.$$

Then, $T - L = fS_{n,1}$ where $\deg(f) \leq n - \ell - 1$. From (3.5)

$$\int_{-1}^1 (T - L)(x) S_{n,2}^+(x) d\mu_\rho(x) = (-1)^{\mathfrak{N}} \int_{-1}^1 f(x) \hat{S}_n(x) d\mu_\rho(x) = 0.$$

Hence,

$$\begin{aligned} \int_{-1}^1 T(x) S_{n,2}^+(x) d\mu_\rho(x) &= \int_{-1}^1 L(x) S_{n,2}^+(x) d\mu_\rho(x), \\ &= \int_{-1}^1 \left(\sum_{i=1}^{n-\ell^*} T(\xi_{n,i}) \frac{S_{n,1}(x)}{S'_{n,1}(\xi_{n,i})(x - \xi_{n,i})} \right) S_{n,2}^+(x) d\mu_\rho(x), \\ &= \sum_{i=1}^{n-\ell^*} \left(\int_{-1}^1 \frac{\hat{S}_n(x)}{S'_{n,1}(\xi_{n,i})(x - \xi_{n,i})} d\mu_\rho(x) \right) (-1)^{\mathfrak{N}} T(\xi_{n,i}), \\ &= \sum_{i=1}^{n-\ell^*} \left(\int_{-1}^1 \frac{\hat{S}_n(x)}{\hat{S}'_n(\xi_{n,i})(x - \xi_{n,i})} d\mu_\rho(x) \right) S_{n,2}^+(\xi_{n,i}) T(\xi_{n,i}), \\ &= \sum_{i=1}^{n-\ell^*} \left(\int_{-1}^1 \frac{S_n(x)}{S'_n(\xi_{n,i})(x - \xi_{n,i})} d\mu_\rho(x) \right) S_{n,2}^+(\xi_{n,i}) T(\xi_{n,i}), \end{aligned}$$

which establishes (3.20). Assume that n is fixed, let $I_+ = \{1 \leq i \leq n - \ell^* : \lambda_{n,i} > 0\}$ and $\Lambda_+^2(x) = \prod_{i \in I_+} (x - \xi_{n,i})^2$. If $\deg(\Lambda_+^2) < 2n - \ell - \ell^*$, from (3.20),

$$0 < \int_{-1}^1 \Lambda_+^2(x) S_{n,2}^+(x) d\mu_\rho(x) = \sum_{\substack{i=1 \\ i \notin I_+}}^{n-\ell^*} \lambda_{n,i} \Lambda_+^2(\xi_{n,i}) S_{n,2}^+(\xi_{n,i}) \leq 0,$$

which is a contradiction and the second assertion is established. $\square$

Let us denote for $k \in \mathbb{N}$

$$R_{n,1}^{[k]}(z) = \frac{S_{n,1}^{[k]}(z)}{S_{n+k,1}(z)},$$

where

$$S_{n,1}^{[k]}(z) = \int_{-1}^1 \frac{S_{n+k,1}(z) - S_{n+k,1}(x)}{z - x} \hat{Q}_{k-1}(x) d\mu_{\rho,n}(x)$$

and

$$d\mu_{\rho,n}(x) = S_{n,2}^+(x)\rho(x)d\mu(x).$$

From Lemma 3.3.3, it is straightforward to see that:

1. If n is sufficiently large, $S_{n,2}^+(x)\rho(x) > 0$ for all $x \in [-1, 1]$.
2. There exists a constant $\mathfrak{M}_\rho > 0$ such that for all $n \in \mathbb{Z}_+$

$$\mu_{\rho,n}([-1, 1]) = \int_{-1}^1 S_{n,2}^+(x)\rho(x)d\mu(x) \leq \mathfrak{M}_\rho. \quad (3.21)$$

Lemma 3.4.2 (Principal Lemma)

Let $\{\hat{S}_n\}_{n \geq 0}$ be the monic orthogonal polynomial sequence with respect to a sequentially-ordered Sobolev inner product (3.2) with $c_j \notin [-1, 1]$, for $j = 1, 2, \dots, \mathcal{N}$. Then, for n sufficiently large

$$R_{n-k,1}^{[k]}(z) = \sum_{j=1}^{n-\ell^*} \frac{\lambda_{n,j} \hat{Q}_{k-1}(\xi_{n,j}) S_{n,2}^+(\xi_{n,j})}{(z - \xi_{n,j})}. \quad (3.22)$$

Furthermore, $\{R_{n,1}^{[k]}\}$ is uniformly bounded on each compact subset $K \subset \mathbb{C} \setminus [-1, 1]$.

Proof. From Lemma 3.3.3, for n sufficiently large, we have that S_n has exactly $n - \ell^*$ zeros on $(-1, 1)$ and all of them are simple. For simplicity of notation, we write ξ_j instead of $\xi_{n,j}$. Then, $\{\xi_j\}_{j=1}^{n-\ell^*}$ is the set of zeros of S_n on $(-1, 1)$.

Thus, $S'_n(\xi_j) \neq 0$ for $j = 1, \dots, n - \ell^*$, and

$$R_{n-k,1}^{[k]}(z) = \sum_{j=1}^{n-\ell^*} \frac{b_j}{z - \xi_j},$$

where

$$\begin{aligned} b_j &= \lim_{z \rightarrow \xi_j} (z - \xi_j) R_{n-k,1}^{[k]}(z) = \lim_{z \rightarrow \xi_j} \frac{(z - \xi_j)}{S_{n,1}(z)} \lim_{z \rightarrow \xi_j} S_{n-k,1}^{[k]}(z) \\ &= \lim_{z \rightarrow \xi_j} S_{n,2}(z) \frac{(z - \xi_j)}{\hat{S}_n(z)} \lim_{z \rightarrow \xi_j} \int_{-1}^1 \frac{S_{n,1}(z) - S_{n,1}(x)}{z - x} \hat{Q}_{k-1}(x) d\mu_{\rho,n}(x) \\ &= \frac{S_{n,2}(\xi_j)}{\hat{S}'_n(\xi_j)} \int_{-1}^1 \frac{(-1)^{\mathfrak{N}} \hat{S}_n(x) \hat{Q}_{k-1}(x) d\mu_{\rho}(x)}{(x - \xi_j)} \\ &= S_{n,2}^+(\xi_j) \hat{Q}_{k-1}(\xi_j) \int_{-1}^1 \frac{S_n(x) d\mu_{\rho}(x)}{S'_n(\xi_j)(x - \xi_j)} \\ &= \lambda_{n,j} \hat{Q}_{k-1}(\xi_j) S_{n,2}^+(\xi_j), \end{aligned}$$

and we get (3.22).

The second part of the proof, as [142, Lem. 1], is based on the second proof of Chebyshev-Markov-Stieltjes's Separation Theorem in [163, §3.41]. Through the proof, we use the following notations:

$$d\vartheta_n(x) = \sum_{j=1}^{n-\ell^*} \lambda_{n,j} S_{n,2}^+(\xi_j) \delta_{\xi_j}(x), \quad \delta_{\xi_j}(x) = \begin{cases} 1, & \text{if } x = \xi_j, \\ 0, & \text{if } x \neq \xi_j, \end{cases}$$

$$\vartheta_n(x) = \int_{-1}^x d\vartheta_n(t), \quad u_{\rho,n}(x) = \int_{-1}^x d\mu_{\rho,n}(t) \quad \text{and} \quad \omega_n(x) = u_{\rho,n}(x) - \vartheta_n(x).$$

Let us recall that the function $u_{\rho,n}$ is monotone non-decreasing on $[-1, 1]$. Set $\xi_0 = -1$ and $\xi_{n-\ell^*+1} = 1$. Then, ϑ_n is a step-function, which is constant on each interval (ξ_j, ξ_{j+1}) for $j = 0, 1, \dots, n - \ell^*$. Hence, ω_n is monotone non-decreasing on each of these open intervals.

With these notations, we can rewrite (3.20) as

$$\int_{-1}^1 T(x) d\omega_n(x) = 0, \quad (3.23)$$

for any polynomial T of degree at most $(2n - \ell - \ell^* - 1)$. Since

$$\omega_n(-1) = u_{\rho,n}(-1) - \vartheta_n(-1) = 0$$

and

$$\omega_n(1) = u_{\rho,n}(1) - \vartheta_n(1) = \mu_{\rho,n}([-1, 1]) - \mu_{\rho,n}([-1, 1]) = 0,$$

by integrating (3.23) by parts we get

$$\int_{-1}^1 \omega_n(x) T'(x) dx = 0. \quad (3.24)$$

Obviously, in (3.24), the polynomial T' can be replaced by any other polynomial of degree at most $2n - \ell - \ell^* - 2$ and, consequently, we can assert that

$$\mathbf{N}_s(\omega_n; (-1, 1)) \geq 2n - \ell - \ell^* - 1.$$

Note that $\mathbf{N}_s(\omega_n; (\xi_0, \xi_1)) = \mathbf{N}_s(\omega_n; (\xi_{n-\ell^*}, \xi_{n-\ell^*+1})) = 0$. Take into account that ω_n is monotone non-decreasing on each interval (ξ_j, ξ_{j+1}) , $j = 1, \dots, n - \ell^* - 1$. Hence, it has at most one sign change on each of them. Therefore, we can conclude that the total number of sign changes of ω_n on $\bigcup_{j=1}^{n-\ell^*-1} (\xi_j, \xi_{j+1})$ is not greater than $n - \ell^* - 1$. On the other hand, ω_n could change sign at each of the $n - \ell^*$ points ξ_j . In conclusion,

$$2(n - \ell^*) - (\ell - \ell^*) - 1 \leq \mathbf{N}_s(\omega_n; (-1, 1)) \leq 2(n - \ell^*) - 1.$$

It thus follows that the number of intervals (ξ_j, ξ_{j+1}) where ω_n does not change sign is at most $(\ell - \ell^*)$.

We say that $\xi_j \in E_1$ if the function ω_n changes sign in each of the consecutive intervals (ξ_{j-1}, ξ_j) and (ξ_j, ξ_{j+1}) . In any other case, we say that $\xi_j \in E_2$.

Observe that if ω_n does not change sign on (ξ_j, ξ_{j+1}) , then $\xi_j, \xi_{j+1} \in E_2$. From the previous considerations, the number of intervals, where ω_n does not change sign is at most $(\ell - \ell^*)$. Therefore, E_2 cannot contain more than $2(\ell - \ell^*)$ elements.

Suppose that $\lambda_{n,j} \leq 0$. If $\xi_j \in E_1$, we know that ω_n changes sign in each of the consecutive intervals (ξ_{j-1}, ξ_j) and (ξ_j, ξ_{j+1}) . Let $x_1 \in (\xi_{j-1}, \xi_j)$ such that $\omega_n(x_1) > 0$ and let $x_2 \in (\xi_j, \xi_{j+1})$ such that $\omega_n(x_2) < 0$. As $u_{\rho,n}(x)$ is monotone non-decreasing on $(-1, 1)$, we get

$$0 < \omega_n(x_1) - \omega_n(x_2) = (u_{\rho,n}(x_1) - u_{\rho,n}(x_2)) + \lambda_{n,j} S_{n,2}^+(\xi_j) \leq 0.$$

This contradiction proves that $\xi_j \in E_1$ implies that $\lambda_{n,j} > 0$ (i.e., the Christoffel coefficients corresponding to the zeros $\xi_j \in E_1$ are positive).

Now, let $\xi_j \in E_1$, $x_1 \in (\xi_{j-1}, \xi_j)$ such that $\omega_n(x_1) \leq 0$ and $x_2 \in (\xi_j, \xi_{j+1})$ such that $\omega_n(x_2) \geq 0$. Recalling again that $u_{\rho,n}(x)$ is monotone non-decreasing on $(-1, 1)$, then

$$0 \geq \omega_n(x_1) - \omega_n(x_2) = (u_{\rho,n}(x_1) - u_{\rho,n}(x_2)) + \lambda_{n,j} S_{n,2}^+(\xi_j)$$

and

$$\lambda_{n,j} S_{n,2}^+(\xi_j) \leq u_{\rho,n}(x_2) - u_{\rho,n}(x_1) \leq \mu_{\rho,n}(\xi_{j+1}) - \mu_{\rho,n}(\xi_{j-1}).$$

From the last inequality, we get

$$\sum_{\xi_j \in E_1} |\lambda_{n,j} S_{n,2}^+(\xi_j)| = \sum_{\xi_j \in E_1} \lambda_{n,j} S_{n,2}^+(\xi_j) \leq 2\mu_{\rho,n}([-1, 1]).$$

Set $K \subset \mathbb{C} \setminus [-1, 1]$ a compact set, $\mathfrak{m} = \min_{\substack{x \in [-1, 1] \\ z \in K}} |x - z| > 0$ and $\mathfrak{M}_k = \max_{-1 \leq x \leq 1} |\hat{Q}_{k-1}(x)|$,

then

$$\sum_{\xi_j \in E_1} \left| \frac{\lambda_{n,j} \hat{Q}_{k-1}(\xi_{n,j}) S_{n,2}^+(\xi_{n,j})}{(z - \xi_{n,j})} \right| \leq \frac{2\mathfrak{M}_k \mu_{\rho,n}([-1, 1])}{\mathfrak{m}} \leq \frac{2\mathfrak{M}_k \mathfrak{M}_\rho}{\mathfrak{m}}, \quad (3.25)$$

where $\mathfrak{M}_\rho$ was defined in (3.21).

The aim of the last step of the proof is to show that the sum

$$G_2(z) = \sum_{\xi_j \in E_2} \frac{\lambda_{n,j} \hat{Q}_{k-1}(\xi_{n,j}) S_{n,2}^+(\xi_{n,j})}{(z - \xi_{n,j})}$$

is uniformly bounded on K . We renumber the zeros of $S_{n,1}$ in such a way that $E_2 = \{\xi_1, \dots, \xi_m\}$ and $E_1 = \{\xi_{m+1}, \dots, \xi_{n-\ell^*}\}$. From the previous result, $m \leq 2(\ell - \ell^*)$.

First, we introduce several notations. Let σ_η be the η -th elementary symmetric polynomials evaluated in $(\xi_1, \dots, \xi_m)$ (see [141, Eq. (1.2.4)]), i.e.

$$\sigma_0 = \sigma_0(\xi_1, \dots, \xi_m) = 1,$$

$$\sigma_\eta = \sigma_\eta(\xi_1, \dots, \xi_m) = \sum_{1 \leq \nu_1 < \dots < \nu_\eta \leq m} \prod_{l=1}^{\eta} \xi_{\nu_l}, \quad \text{for } \eta = 1, \dots, m.$$

The symbol $\sigma_{\eta,j} = \sigma_{\eta}(\xi_1, \dots, \xi_{j-1}, \xi_{j+1}, \dots, \xi_m)$ denotes the η -th elementary symmetric polynomial evaluated in $(\xi_1, \dots, \xi_{j-1}, \xi_{j+1}, \dots, \xi_m)$. It is straightforward to see that $\sigma_{\eta,j} = \sigma_{\eta} - \xi_j \sigma_{\eta-1,j}$ for $\eta = 1, \dots, m-1$, and iteratively applying this equality η times, we have

$$\sigma_{\eta,j} = \sum_{l=0}^{\eta} (-\xi_j)^l \sigma_{\eta-l}.$$

For simplicity of notation, we write $\varrho_{n,j} = \lambda_{n,j} \hat{Q}_{k-1}(\xi_j) S_{n,2}^+(\xi_j)$. Hence, for $i = 1, \dots, m$,

$$\sum_{j=1}^m \varrho_{n,j} \sigma_{i,j} = \sum_{j=1}^m \varrho_{n,j} \left(\sum_{l=0}^i (-\xi_j)^l \sigma_{i-l} \right) = \sum_{l=0}^i \sigma_{i-l} \left(\sum_{j=1}^m \varrho_{n,j} (-\xi_j)^l \right).$$

From Lemma 3.4.1 we have for $l \leq m \leq 2(\ell - \ell^*)$

$$\begin{aligned} \int_{-1}^1 (-x)^l \hat{Q}_{k-1}(x) d\mu_{\rho,n}(x) &= \sum_{j=1}^{n-\ell^*} \varrho_{n,j} (-\xi_j)^l \\ &= \sum_{j=1}^m \varrho_{n,j} (-\xi_j)^l + \sum_{j=m+1}^{n-\ell^*} \varrho_{n,j} (-\xi_j)^l. \end{aligned}$$

Thus, from (3.25)

$$\begin{aligned} \left| \sum_{j=1}^m \varrho_{n,j} (-\xi_j)^l \right| &\leq \left| \sum_{j=m+1}^{n-\ell^*} \varrho_{n,j} (-\xi_j)^l \right| + \left| \int_{-1}^1 (-x)^l \hat{Q}_{k-1}(x) d\mu_{\rho,n}(x) \right| \\ &\leq \frac{m+2}{m} \mathfrak{M}_k \mu_{\rho,n}([-1, 1]). \end{aligned}$$

As $\{\xi_1, \dots, \xi_m\} \subset [-1, 1]$, it is straightforward to see that $|\sigma_{\eta}| \leq \binom{m}{k} \leq 2^m$ for all $\eta = 0, \dots, m$. Therefore, for $i = 1, \dots, m$,

$$\begin{aligned} \left| \sum_{j=1}^m \varrho_{n,j} \sigma_{i,j} \right| &\leq \sum_{l=0}^i |\sigma_{i-l}| \left| \sum_{j=1}^m \varrho_{n,j} (-\xi_j)^l \right| \\ &\leq \frac{m2^m(m+2)}{m} \mathfrak{M}_k \mu_{\rho,n}([-1, 1]). \end{aligned} \tag{3.26}$$

Using the previous notation, we write

$$G_2(z) = \sum_{j=1}^m \frac{\varrho_{n,j}}{(z - \xi_j)} = \frac{P_{m-1}(z)}{\prod_{j=1}^m (z - \xi_j)} \text{ where } P_{m-1}(z) = \sum_{j=1}^m \varrho_{n,j} \prod_{\substack{i=1 \\ i \neq j}}^m (z - \xi_i).$$

From the classical formula of Viète,

$$\prod_{\substack{i=1 \\ i \neq j}}^m (z - \xi_i) = \sum_{i=0}^{m-1} (-1)^i \sigma_{i,j} z^{m-1-i},$$

see [141, Eq. (1.2.2)], and

$$P_{m-1}(z) = \sum_{j=1}^m \varrho_{n,j} \left(\sum_{i=0}^{m-1} (-1)^i \sigma_{i,j} z^{m-1-i} \right) = \sum_{i=0}^{m-1} (-1)^i \left(\sum_{j=1}^m \varrho_{n,j} \sigma_{i,j} \right) z^{m-1-i}.$$

Let $\mathfrak{M} = \max_{z \in K} |z|$. According to (3.26), for all $z \in K$,

$$\begin{aligned} |P_{m-1}(z)| &\leq \sum_{i=0}^{m-1} \left| \sum_{j=1}^m \varrho_{n,j} \sigma_{i,j} \right| |z|^{m-1-i}, \\ &\leq \frac{m2^m(\mathfrak{m}+2)\mathfrak{M}_k\mu_{\rho,n}([-1,1])}{\mathfrak{m}} \sum_{i=0}^{m-1} |z|^{m-1-i}, \\ &\leq \frac{\mathfrak{m}+2}{\mathfrak{m}} m^2 2^m \mathfrak{M}_k \mathfrak{M}_\rho \max\{\mathfrak{M}^{m-1}, 1\} \\ &\leq \frac{\mathfrak{m}+2}{\mathfrak{m}} 4(\ell-\ell^*)^2 4^{(\ell-\ell^*)} \mathfrak{M}_k \mathfrak{M}_\rho \max\{\mathfrak{M}^{2(\ell-\ell^*)-1}, 1\} = \mathcal{M}_k. \\ |G_2(z)| &= \frac{|L_{m-1}(z)|}{\prod_{j=1}^m |z - \xi_j|} \leq \frac{\mathcal{M}_k(m)}{\mathfrak{m}^m} \leq \frac{\mathcal{M}_k}{\min\{1, \mathfrak{m}^{2(\ell-\ell^*)}\}}. \end{aligned} \tag{3.27}$$

Finally, (3.25) and (3.27) establish the second assertion. $\square$

3.5 Proof of Theorem 3.1.3

Denote $R_n^{[k]}(z) = \frac{S_n^{[k]}(z)}{S_{n+k}(z)}$ and let $\hat{\mu}_k(z) = \int_{-1}^1 \frac{\hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x)$ be the k th Markov-type function associated to μ_ρ ($k \in \mathbb{N}$) as in (3.8). Note that $\hat{\mu}_k(z)$ is well defined and holomorphic in Ω_∞ ($\hat{\mu}_k \in \mathbb{H}(\Omega_\infty)$ for short) and $\hat{\mu}_k(\infty) = 0$.

For the remainder $(\hat{\mu}_k(z) - R_n^{[k]}(z))$, the following formulas take place.

Lemma 3.5.1

Let μ be a positive Borel measure supported on $[-1, 1]$ and S_n and $S_n^{[k]}$ defined as above. Then,

$$\hat{\mu}_{k,n}(z) - R_{n,1}^{[k]}(z) = S_{n+k,2}^+(z) (\hat{\mu}_k(z) - R_n^{[k]}(z)) = \mathcal{O}\left(\frac{1}{z^{2(n+1)+k-\ell-\ell^*}}\right), \tag{3.28}$$

where $\hat{\mu}_{k,n}(z) = \int_{-1}^1 \frac{\hat{Q}_{k-1}(x)}{z-x} d\mu_{\rho,n}(x)$.

Proof. From the definition of $S_n^{[k]}$, we get

$$\begin{aligned} S_n^{[k]}(z) &= \int_{-1}^1 \frac{S_{n+k}(z) - S_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) \\ &= S_{n+k}(z) \int_{-1}^1 \frac{\hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x) - \int_{-1}^1 \frac{S_{n+k}(x) \hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x) \\ &= S_{n+k}(z) \hat{\mu}_k(z) - \int_{-1}^1 \frac{S_{n+k}(x) \hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x). \end{aligned}$$

Then, we have

$$\widehat{\mu}_k(z) - R_n^{[k]}(z) = \int_{-1}^1 \frac{S_{n+k}(x) \hat{Q}_{k-1}(x)}{S_{n+k}(z)(z-x)} d\mu_\rho(x). \quad (3.29)$$

On the other hand, from orthogonality conditions of S_n

$$\begin{aligned} 0 &= \left\langle S_{n+k}(x), \frac{(S_{n-\ell+1}(z) - S_{n-\ell+1}(x)) \hat{Q}_{k-1}(x) \rho(x)}{z-x} \right\rangle_s \\ &= \int_{-1}^1 S_{n+k}(x) \frac{S_{n-\ell+1}(z) - S_{n-\ell+1}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x). \end{aligned}$$

Hence, it follows that

$$\int_{-1}^1 \frac{S_{n+k}(x) S_{n-\ell+1}(z)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) = \int_{-1}^1 \frac{S_{n+k}(x) S_{n-\ell+1}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x),$$

and from (3.29), we obtain

$$\begin{aligned} \widehat{\mu}_k(z) - R_n^{[k]}(z) &= \int_{-1}^1 \frac{S_{n+k}(x)}{S_{n+k}(z)} \frac{\hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x) \\ &= \int_{-1}^1 \frac{S_{n+k}(x) S_{n-\ell+1}(x)}{S_{n+k}(z) S_{n-\ell+1}(z)} \frac{\hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x) = \mathcal{O}\left(\frac{1}{z^{2(n+1)+k-\ell}}\right) \end{aligned}$$

The second equality in (3.28) is a direct consequence of the above equality. Finally, we compute

$$\begin{aligned} S_{n+k,2}^+(z) \widehat{\mu}_k(z) &= \int_{-1}^1 \frac{S_{n+k,2}^+(z) \hat{Q}_{k-1}(x)}{z-x} d\mu_\rho(x) \\ &= \int_{-1}^1 \frac{S_{n+k,2}^+(z) - S_{n+k,2}^+(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x) + \widehat{\mu}_{k,n}(z), \end{aligned} \quad (3.30)$$

and

$$\begin{aligned} S_{n+k,2}^+(z) R_n^{[k]}(z) &= \frac{S_{n+k,2}^+(z)}{S_{n+k}(z)} \int_{-1}^1 \frac{S_{n+k}(z) - S_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x), \\ &= \frac{(-1)^{\mathfrak{N}}}{S_{n+k,1}(z)} \int_{-1}^1 \frac{\hat{S}_{n+k}(z) - \hat{S}_{n+k}(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x), \\ &= \int_{-1}^1 \frac{S_{n+k,1}(z) S_{n+k,2}^+(z) - S_{n+k,1}(x) S_{n+k,2}^+(x)}{S_{n+k,1}(z)(z-x)} \hat{Q}_{k-1}(x) d\mu_\rho(x), \\ &= R_{n,1}^{[k]}(z) + \int_{-1}^1 \frac{S_{n+k,2}^+(z) - S_{n+k,2}^+(x)}{z-x} \hat{Q}_{k-1}(x) d\mu_\rho(x). \end{aligned}$$

The first equality in (3.28) now follows by subtracting the above expression from (3.30). $\square$

Proof of Theorem 3.1.3. Let K be any compact set on Ω_∞ and consider the level curve ℓ_τ defined by

$$\ell_\tau = \{z \in \mathbb{C} : |\varphi(z)| = \tau\}, \quad \text{where } \tau > 1 \text{ and } \varphi \text{ is as in (1.21).}$$

Since $\varphi(K)$ is a compact set, we can take τ sufficiently close to 1 such that $1 < \tau < \min |\varphi(K)|$ (remember that φ is the conformal map of the exterior of $[-1, 1]$ onto the exterior of the unit circle). From Lemma 3.4.2 and (3.21), the sequences $\{\widehat{\mu}_{k,n}\}$ and $\{R_{n,1}^{[k]}\}$ are uniformly bounded over ℓ_τ . Then, there exists a constant $\mathfrak{L}_\tau$, independent of n , such that for all $z \in \ell_\tau$

$$\left| \left(\widehat{\mu}_{k,n}(z) - R_{n,1}^{[k]}(z) \right) \varphi^{2(n+1)+k-\ell-\ell^*}(z) \right| \leq \mathfrak{L}_\tau \tau^{2(n+1)+k-\ell-\mathcal{N}}. \quad (3.31)$$

Taking into account that φ has a simple pole at ∞ , from (3.28), we have

$$\left(\left(\widehat{\mu}_{k,n} - R_{n,1}^{[k]} \right) \varphi^{2(n+1)+k-\ell-\ell^*} \right) \in \mathbb{H}(\Omega_\infty).$$

Now, from the maximum modulus principle the bound (3.31) also holds on K . Consequently, we have

$$\left| \widehat{\mu}_{k,n}(z) - R_{n,1}^{[k]}(z) \right| \leq \mathfrak{L}_\tau \left(\frac{\tau}{|\varphi(z)|} \right)^{2(n+1)+k-\ell-\ell^*} \quad z \in K.$$

Hence

$$\sup_{z \in K} \left| \widehat{\mu}_{k,n}(z) - R_{n,1}^{[k]}(z) \right| \leq \mathfrak{L}_\tau \left(\frac{\tau}{\min |\varphi(K)|} \right)^{2(n+1)+k-\ell-\ell^*} \xrightarrow{n \rightarrow \infty} 0,$$

which is equivalent to say that $R_{n,1}^{[k]}(z) \xrightarrow[n]{\Rightarrow} \widehat{\mu}_{k,n}(z) \quad K \subset \Omega_\infty$.

As above, $\Omega_\infty^* = \Omega_\infty \setminus \{c_1, c_2, \dots, c_N\}$. For the rest of the proof we assume that the compact set K is a subset of Ω_∞^* . From Lemma 3.3.3, there exists a constant $\mathfrak{L}_2 > 0$, independent of n , such that $\mathfrak{L}_2 \leq |S_{n+k,2}(z)|$ for all $z \in K$. Therefore, taking into account (3.28), we get

$$\sup_{z \in K} \left| \widehat{\mu}_k(z) - R_n^{[k]}(z) \right| \leq \frac{\mathfrak{L}_\tau}{\mathfrak{L}_2} \left(\frac{\tau}{\min |\varphi(K)|} \right)^{2(n+1)+k-\ell-\ell^*} \xrightarrow{n \rightarrow \infty} 0, \quad (3.32)$$

□

As a complement of Theorem 3.1.3, we have the following estimate for the degree of convergence (“speed”) of $R_n^{[k]}$ to $\widehat{\mu}_k$.

Corollary 3.5.0.1

Under the same hypotheses of Theorem 3.1.3, we have

$$\limsup_n \left\| \widehat{\mu}_k - R_n^{[k]} \right\|_K^{1/2n} \leq \|\varphi^{-1}\|_K < 1. \quad (3.33)$$

Proof. Taking the $2n$ -th root in (3.32), we get

$$\left\| \widehat{\mu}_k - R_n^{[k]} \right\|_K^{1/2n} \leq \left(\frac{\mathfrak{L}_\tau}{\mathfrak{L}_2} \right)^{1/2n} \left(\frac{\tau}{\min |\varphi(K)|} \right)^{(2(n+1)+k-\ell-\ell^*)/(2n)}. \quad (3.34)$$

Since $\tau < \min |\varphi(K)|$, (3.33) follows from (3.34). $\square$

Discrete-Continuous Jacobi-Sobolev and Fourier Series

4.1 Introduction

The convergence problem of the trigonometric Fourier series has been one of the main driving forces behind the mathematical analysis growth. For instance, the development of the theory of Hilbert spaces is closely related to the convergence problem of the Fourier series for square integrable 2π -periodic functions on $[-\pi, \pi]$,

$$\lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} \left| f(x) - \sum_{|k| \leq n} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iky} dy \right) e^{ikx} \right|^2 dx = 0.$$

In 1927, M. Riesz gave an analytic proof of the convergence of trigonometric Fourier series for L^p -integrable 2π -periodic functions on $[-\pi, \pi]$ using complex methods (see [147]). This proof was one of the sources to develop the main harmonic analysis tools such as the Hardy-Littlewood maximal functions and the singular integrals. In the proof, Riesz strongly used the conjugated function, which is the first singular integral considered. Later on A. P. Calderón and A. Zygmund studied singular integrals in higher dimensions using real analysis methods (see [25]).

Trough the Hilbert space theory the study of the L^2 convergence of the Fourier expansions in terms of classical orthogonal polynomials became a straightforward issue, since it only depends on the orthogonality. But we find a very different scenario in the case of the L^p convergence when $p \neq 2$. The study of L^p -convergence of Fourier series of classical orthogonal polynomials started in the 1940's with a series of papers (see [137, 138, 139, 140]). H. Pollard considered the mean convergence of the Legendre, Gegenbauer and Jacobi Fourier expansions extending the harmonic analysis techniques for Fourier series to this case (see Theorem 4.2.1). Pollard used a special decomposition of the Christoffel-Darboux kernel such that the L^p continuity of the Hilbert transform with respect to certain weights can be used. Later on B. Muckenhoupt revisited the convergence of Jacobi polynomials (see [126] and Theorem 4.2.2).

Additionally, H. Pollard proved in [139] by means of the asymptotic relation between the Hermite polynomials and the Gegenbauer polynomials (see [163, Eq. (5.3.4)]) that the Fourier expansions in terms of the Hermite polynomials only converge in L^p -norm when $p = 2$. Finally, due to the relation between the Hermite and the Laguerre polynomials, (see [163, Eq. (5.6.1)]), there exists also an anomalous behavior on the L^p -convergence of the Laguerre polynomials Fourier series.

As it would be expected, the orthogonal projection onto the usual $L^2(\mu)$ space does not have the required properties, which naturally leads to consider inner products of the form (1.9). Nonetheless, there are only a few general results about the

extremal polynomials with respect to the norm (1.10), for $p \neq 2$. Most of them, on the asymptotic behavior of the sequence of the corresponding extremal polynomials (see [40, 89, 92]).

From the anomalous behavior of the Fourier expansions in terms of the Hermite and Laguerre polynomials with respect to the L^p -convergence, it is fruitless to consider the convergence of the Fourier series of any of them in the Sobolev case. Therefore, due to the importance of the Jacobi polynomials in this topic, in the next section, we review some of their properties and also the main ideas of the proof of the L^p convergence of their Fourier series. In Section 4.3, we review what it is known about the convergence of the Fourier series of Jacobi-Sobolev polynomials.

Section 4.4 is dedicated to define the discrete-continuous Jacobi-Sobolev spaces and to study the necessary and sufficient conditions for its completeness. In the last section, we prove the convergence of the Fourier series in the norm of the discrete-continuous Jacobi-Sobolev spaces.

The content of this chapter is included in [41].

4.2 Fourier series of Jacobi polynomials

Consider the classical Jacobi polynomial $P_n^{(\alpha, \beta)}$ under the usual normalization given in Table 1.1 with parameters $\alpha, \beta > -1$, or equivalently, for $n, m = 0, 1, 2, \dots$

$$\langle P_n^{(\alpha, \beta)}(x), P_m^{(\alpha, \beta)}(x) \rangle_{\mu^{\alpha, \beta}} = \int_{-1}^1 P_n^{(\alpha, \beta)}(x) P_m^{(\alpha, \beta)}(x) d\mu^{\alpha, \beta}(x) = h_n^{(\alpha, \beta)} \delta_{n, m},$$

where $d\mu^{\alpha, \beta}(x) = w_{\alpha, \beta}(x)dx = (1-x)^\alpha(1+x)^\beta dx$, is the *Jacobi measure* (or beta distribution) and

$$h_n^{(\alpha, \beta)} = \frac{2^{\alpha+\beta+1}}{(2n+\alpha+\beta+1)} \frac{\Gamma(n+\alpha+1)\Gamma(n+\beta+1)}{\Gamma(n+1)\Gamma(n+\alpha+\beta+1)} \quad [163, Eq.(4.3.3)]. \quad (4.1)$$

Additionally,

$$\frac{dP_n^{(\alpha, \beta)}}{dx}(x) = \frac{(n+\alpha+\beta+1)}{2} P_{n-1}^{(\alpha+1, \beta+1)}(x) \quad [163, Eq.(4.21.7)]. \quad (4.2)$$

Also, $\{P_n^{(\alpha, \beta)}\}_{n \geq 0}$ satisfy the three term recurrence relation,

$$\begin{aligned} xP_n^{(\alpha, \beta)}(x) &= \frac{2(n+1)(n+\alpha+\beta+1)}{(\gamma_n+1)(\gamma_n+2)} P_{n+1}^{(\alpha, \beta)}(x) + \frac{\beta^2 - \alpha^2}{\gamma_n(\gamma_n+2)} P_n^{(\alpha, \beta)}(x) \\ &\quad - \frac{2(n+\alpha)(n+\beta)}{\gamma_n(\gamma_n+1)} P_{n-1}^{(\alpha, \beta)}(x), n \geq 0 \end{aligned}$$

with $\gamma_n = 2n + \alpha + \beta$, $P_{-1}^{(\alpha, \beta)} \equiv 0$ and $P_0^{(\alpha, \beta)} \equiv 1$.

For $1 \leq p \leq \infty$ consider the Banach space $\mathbf{L}^p(\mu^{\alpha, \beta})$ (See Definition 1.1.1). From (4.1)-(4.2), we have the following property for the k -th derivative of the orthonormal

Jacobi polynomials:

$$\begin{aligned} (P_n^{(\alpha,\beta)}(x))^{(k)} &= A_{n,k} P_{n-k}^{(\alpha+k,\beta+k)}(x), \quad \text{where} \\ A_{n,k}^{\alpha,\beta} &= \sqrt{\frac{\Gamma(n+1) \Gamma(n+\alpha+\beta+k+1)}{\Gamma(n-k+1) \Gamma(n+\alpha+\beta+1)}} \neq 0. \end{aligned}$$

Given $f \in \mathbf{L}^1(\mu^{\alpha\beta})$, consider the partial Fourier sum of f

$$\mathcal{F}_n^{\alpha,\beta}(f, x) = \sum_{k=0}^n \langle f, P_n^{(\alpha,\beta)} \rangle_{\mu^{\alpha\beta}} P_k^{(\alpha,\beta)}(x), \quad (4.3)$$

where $P_n^{(\alpha,\beta)}$ is the sequence of Jacobi orthonormal polynomials (i.e. orthonormal polynomials with respect to $\langle \cdot, \cdot \rangle_{\mu^{\alpha\beta}}$). The analysis of the convergence of Fourier-Jacobi series in $\mathbf{L}^p(\mu^{\alpha\beta})$ has a long history. The first result was obtained by H. Pollard in [139] for Gegenbauer polynomials and then in [140] for Jacobi polynomials,

Theorem 4.2.1 ([140, Th. A])

Let $\alpha, \beta \geq -1/2$,

$$M(\alpha, \beta) = 4 \max \left\{ \frac{\alpha+1}{2\alpha+3}, \frac{\beta+1}{2\beta+3} \right\} \quad \text{and} \quad m(\alpha, \beta) = 4 \min \left\{ \frac{\alpha+1}{2\alpha+1}, \frac{\beta+1}{2\beta+1} \right\}.$$

Then, for any $f \in \mathbf{L}^p(\mu^{\alpha\beta})$ with $M(\alpha, \beta) < p < m(\alpha, \beta)$, the Fourier-Jacobi expansion of f converges to f in $\mathbf{L}^p(\mu^{\alpha\beta})$, i.e.

$$\lim_{n \rightarrow \infty} \|f - \mathcal{F}_n^{\alpha,\beta}(f)\|_{\mathbf{L}^p(\mu^{\alpha\beta})} = \lim_{n \rightarrow \infty} \left(\int_{-1}^1 |f(x) - \mathcal{F}_n^{\alpha,\beta}(f, x)|^p d\mu^{\alpha\beta}(x) \right)^{\frac{1}{p}} = 0. \quad (4.4)$$

Furthermore, in [140, Th. B], Pollard also proved that the preceding result fails if $p < M(\alpha, \beta)$ or $p > m(\alpha, \beta)$. In [129], J. Newman and W. Rudin, study the case of the extreme points $p = M(\alpha, \beta)$ or $p = m(\alpha, \beta)$ proving that in such case the convergence also fails.

According to Banach-Steinhaus uniform boundedness principle (see [153, §5.8]), in order to prove (4.4), it is enough to prove the uniform $\mathbf{L}^p(\mu^{\alpha\beta})$ -boundedness of the partial sums, i.e. there exists a constant $C > 0$ such that

$$\|\mathcal{F}_n^{\alpha,\beta}(f)\|_{\mathbf{L}^p(\mu^{\alpha\beta})} \leq C \|f\|_{\mathbf{L}^p(\mu^{\alpha\beta})},$$

for all $f \in \mathbf{L}^p(\mu^{\alpha\beta})$ and for all $n \geq 0$.

B. Muckenhoupt extended Pollard's result for all $\alpha, \beta > -1$ (instead of $\alpha, \beta > -1/2$) in the Corollary of Theorem 1 of [126]. Muckenhoupt's proof uses Pollard's decomposition but now for the weights $w_{ap,bp}(x) = (1-x)^{ap}(1+x)^{bp}$, where $a, b \in \mathbb{R}$, such that

$$\left| a + \frac{1}{p} - \frac{\alpha+1}{2} \right| \leq \min \left\{ \frac{1}{4}, \frac{\alpha+1}{2} \right\} \quad \text{and} \quad \left| b + \frac{1}{p} - \frac{\beta+1}{2} \right| \leq \min \left\{ \frac{1}{4}, \frac{\beta+1}{2} \right\}. \quad (4.5)$$

Note that, if $a = \alpha/p$ and $b = \beta/p$, we get the Jacobi weights. It is easy to see that these conditions, for the case $\alpha, \beta > -1/2$, give Pollard's conditions $M(\alpha, \beta) < p < m(\alpha, \beta)$. The Muckenhoupt's proof is based on the following inequality, which we will use later.

Theorem 4.2.2 ([126, Th. 1])

Let $\alpha, \beta > -1$, $1 < p < \infty$ and $a, b \in \mathbb{R}$ such that (4.5) holds. Then there exists a constant, C , independent of f and n , such that

$$\left\| \mathcal{F}_n^{\alpha, \beta}(f) \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})} \leq C \|f\|_{\mathbf{L}^p(\mu^{\alpha\beta})},$$

where $\mathcal{F}_n^{\alpha, \beta}(f)$ is given by (4.3).

Additionally, in [126, Th. 2], Muckenhoupt also proved that condition (4.5) is also necessary.

4.3 Fourier series of Sobolev orthogonal polynomials

From now on, we will focus our attention on summarizing the main advances on the convergence of the Fourier-Sobolev series with respect to L^p -norms given by (1.10). The study of the convergence in norm (1.10) of the orthogonal polynomials Fourier series faces several problems. First of all, in general, the Sobolev polynomials do not satisfy a three-term recurrence relation and, therefore, there is not a Christoffel-Darboux formula for their Dirichlet-Szegő kernel. Consequently, the argument outlined above cannot be applied to this case. Additionally, since there are three different types of Sobolev inner products, we will need three different methods to study them separately. Let us consider the results that have been obtained so far in each case:

I. *Discrete case.* In [28], the authors give a complete characterization of the boundedness of the partial sum operators with respect to the inner product (1.9) where $\mathfrak{M}_0 = 1$, $\mu_0 = \mu^\alpha + M(\delta_{-1} + \delta_1)$ and μ^α is the Gegenbauer probability measure,

$$d\mu^\alpha(x) = \frac{\Gamma(2\alpha+2)}{2^{2\alpha+1}\Gamma^2(\alpha)}(1-x^2)^\alpha dx, \quad \alpha > -1/2,$$

and $\mu_1 = N(\delta_{-1} + \delta_1)$. Due to the lack of a Christoffel-Darboux formula for Sobolev orthogonal polynomials, the main tools involve a special case of a general weighted transplantation theorem, see [127, Th. 1.6] and the other one is a particular case of the multiplier result [127, Th. 1.10]. Using these tools, they prove (see [28, Th. 1.1]) the boundedness of the partial sum operators in the corresponding space for

$$\frac{4(\alpha+1)}{2\alpha+3} < p < \frac{4(\alpha+1)}{2\alpha+1}.$$

Then, using the density of the polynomials in the $\mathbf{L}^p(\mu^\alpha)$ space (see [149]) and the Banach-Steinhaus uniform boundedness principle, they obtain that, for these values

of p , the sequence of the partial sums $\{\mathcal{F}_n(f)\}_{n \geq 0}$ of f converges to the function f , in $\|\cdot\|_{L^p(\mu^\alpha)}$ (see [28, Cor. 1.3]).

II. *Continuous case.* The first article in this direction is due to F. Marcellán, Y. Quintana and A. Urieles [102]. They consider the inner product

$$\langle f, g \rangle_s = \int_{-1}^1 f(x)g(x)d\mu^{\alpha\beta}(x) + \int_{-1}^1 f'(x)g'(x)d\mu^{\alpha+1, \beta+1}(x).$$

In this case, it follows from (4.2) that the corresponding Jacobi-Sobolev polynomials are, up to a constant factor, the classical Jacobi polynomials and, therefore, they satisfy a three term-recurrence relation. In the paper, the authors actually only focus on the case of the Legendre-Sobolev polynomials, i.e. $\alpha = \beta = 0$. The existence of a recurrence relation for them (see [102, Eq. (3.35)]) implies that an explicit formula can be given for the Sobolev-Dirichlet-Szegő kernel corresponding to the n th partial sum. Unfortunately, the corresponding part for the derivatives generates a hypersingular transform which is not easy to handle without the requirement of a lot of regularity for the function. Consequently, the final result is not entirely satisfactory (see [102, Th. 3.1]).

The second article in this direction is due to Ó. Ciaurri and J. Mínguez [29]. They consider the general case of the Fourier series of Jacobi-Sobolev polynomials, $\alpha, \beta > -1$, with an arbitrary number order derivatives, i.e. orthogonal polynomials with respect to Sobolev-type inner product

$$\langle f, g \rangle_s = \sum_{k=0}^{\mathfrak{m}} \int_{-1}^1 f^{(k)}(x)g^{(k)}(x)d\mu^{\alpha+k, \beta+k}(x), \quad \alpha, \beta > -1, \quad \mathfrak{m} \geq 1.$$

Again, by (4.2), the corresponding Jacobi-Sobolev polynomials $\{S_n\}_{n \geq 0}$ are multiples of the classical Jacobi polynomials. They consider an alternative approach following basically what they had developed in [29], applying Muckenhoupt's multipliers and transplantation operators for Jacobi expansions (see [127, Th. 5.1, Th. 7.1 and Th. 1.10], [31]) and also Pollard's results [140].

They prove in [29, Th. 1], the boundedness of the partial sum operators $\mathcal{F}_n(f)$ in the corresponding space when

$$\max \left\{ \frac{4(\alpha + \mathfrak{m} + 1)}{2(\alpha + \mathfrak{m}) + 3}, \frac{4(\beta + \mathfrak{m} + 1)}{2(\beta + \mathfrak{m}) + 3} \right\} < p < \min \left\{ \frac{4(\alpha + \mathfrak{m} + 1)}{2(\alpha + \mathfrak{m}) + 1}, \frac{4(\beta + \mathfrak{m} + 1)}{2(\beta + \mathfrak{m}) + 1} \right\}.$$

Then, using the density of the polynomials $\{S_n\}_{n \geq 0}$ in the corresponding space, see [149] and the Banach-Steinhaus uniform boundedness principle, they obtain that this result is equivalent to the convergence of $\mathcal{F}_n(f)$ to f in the corresponding norm (see [29, Cor. 2]).

Let (μ_0, μ_1) be a coherent pair, see (1.24). The notion of coherent pairs of measures was introduced in [64] and it is closely related with the Sobolev inner product

$$\langle f, g \rangle_s = \int_a^b f(x)g(x)d\mu_0(x) + \int_a^b f'(x)g'(x)d\mu_1(x), \quad (4.6)$$

where $-\infty \leq a < b \leq \infty$. For a review of the study of the coherent pair of measures we recommend [106, §5]. The study of convergence of Fourier series of orthogonal polynomials with respect to the inner product (4.6) has been carried out in [30].

III: *Discrete-continuous case.* In this case, as far as we know, a general result about convergence in the L^p -norm of the Fourier series is still unknown. In [157], the author considers the case of the inner product

$$\langle f, g \rangle_s = \sum_{k=0}^{\mathfrak{m}-1} f^{(k)}(-1)g^{(k)}(-1) + \int_{-1}^1 f^{(\mathfrak{m})}(t)g^{(\mathfrak{m})}(t)(1-t)^\alpha dt.$$

He proved that the associated Fourier series converges uniformly in $[-1, 1]$.

We will consider a more general case in this direction in the next section.

4.4 Discrete-Continuous Jacobi-Sobolev Spaces

First, we need to consider the natural vector spaces for the problem. Let $p \geq 1$, $\mathfrak{m} \in \mathbb{N}$, $\alpha, \beta > -1$ and $\vec{c} = (c_0, c_1, \dots, c_{\mathfrak{m}-1}) \in \mathbb{R}^{\mathfrak{m}}$. Given a suitable function f , we define the Jacobi-Sobolev norm as:

$$\|f\|_{s,p} = \left(\sum_{k=0}^{\mathfrak{m}-1} |f^{(k)}(c_k)|^p + \int_{-1}^1 |f^{(\mathfrak{m})}(x)|^p d\mu^{\alpha,\beta}(x) \right)^{\frac{1}{p}}. \quad (4.7)$$

It is clear that $\|\cdot\|_{s,p}$ is a norm on the vector space of all polynomials $\mathbb{P}$ and we define the Jacobi-Sobolev discrete-continuous space $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$ as the completion of the normed space $(\mathbb{P}, \|\cdot\|_{s,p})$. In other words, $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$ is the collection of all equivalence class of Cauchy sequences in $(\mathbb{P}, \|\cdot\|_{s,p})$, under the following equivalence relation: $\{P_n\}_{n \geq 0}, \{Q_n\}_{n \geq 0}$ Cauchy sequences in $(\mathbb{P}, \|\cdot\|_{s,p})$ are equivalent $\{P_n\}_{n \geq 0} \sim \{Q_n\}_{n \geq 0}$ if and only if $\lim_{n \rightarrow \infty} \|P_n - Q_n\|_{s,p} = 0$.

On the other hand, let $U_{\vec{c},p}^{\alpha,\beta}$ be the vector space of functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying:

1. f has derivative at c_k of order k , $k = 1, 2, \dots, \mathfrak{m} - 1$.
2. f is $\mathfrak{m}$ times differentiable almost everywhere on $[-1, 1]$.
3. $f^{(\mathfrak{m})} \in \mathbf{L}^p(\mu^{\alpha,\beta})$.

Obviously, (4.7) is a seminorm on $U_{\vec{c},p}^{\alpha,\beta}$ and $\mathbb{P} \subset U_{\vec{c},p}^{\alpha,\beta}$. Let us write $f \sim g$ if and only if $\|f - g\|_{s,p} = 0$. It is a straightforward result that this is an equivalence relation on $U_{\vec{c},p}^{\alpha,\beta}$. Denote the vector space of the equivalence classes of this relation by $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$. Given $F \in \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ and if f is any element of F , then we define the norm in $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ as $\|F\|_{s,p} = \|f\|_{s,p}$. Thus, $(\mathbf{U}_{\vec{c},p}^{\alpha,\beta}, \|\cdot\|_{s,p})$ is a normed space.

In what follows, we will introduce some notions which will be useful in the sequel. Given $\vec{x} = (x_0, x_1, \dots, x_{m-1})$, $\vec{y} = (y_0, y_1, \dots, y_{m-1}) \in \mathbb{R}^m$, there exists a unique polynomial $\mathcal{A}_{\vec{x},\vec{y}}$ of degree at most $m - 1$, such that

$$\mathcal{A}_{\vec{x},\vec{y}}^{(k)}(x_k) = y_k, \quad k = 0, 1, \dots, m - 1.$$

The polynomial $\mathcal{A}_{\vec{x},\vec{y}}$ is known as the *Abel-Goncharov interpolation polynomial*, associated with $\vec{y}$ on $\vec{x}$ (see [1, Chap. 3], [33, §2.6], or [166, §8]). The Abel-Goncharov interpolation polynomial is a generalization of Taylor's polynomial, which corresponds to the case $x_{m-1} = x_{m-2} = \dots = x_0$.

The polynomial $\mathcal{A}_{\vec{x},\vec{y}}$ is given by the explicit expression

$$\mathcal{A}_{\vec{x},\vec{y}}(z) = \sum_{k=0}^{m-1} y_k \mathcal{G}_{\vec{x},k}(z),$$

where $\mathcal{G}_{\vec{x},0} \equiv 1$ and $\mathcal{G}_{\vec{x},k}$, for each $k = 1, \dots, m-1$, is the polynomial of degree k , generated by the k th iterated integral

$$\mathcal{G}_{\vec{x},k}(s) = \int_{x_0}^s \int_{x_1}^{s_1} \dots \int_{x_{k-2}}^{s_{k-2}} \int_{x_{k-1}}^{s_{k-1}} ds_k ds_{k-1} \dots ds_2 ds_1.$$

The polynomial $\mathcal{G}_{\vec{x},k}$ is called the *kth Goncharov's polynomial associated with $\vec{x}$* and satisfies

$$\mathcal{G}_{\vec{x},k}^{(\mathbf{v})}(x_{\mathbf{v}}) = \begin{cases} 0, & \text{if } \mathbf{v} \neq k, \\ 1, & \text{if } \mathbf{v} = k. \end{cases} \quad (4.8)$$

Proposition 4.4.1

Let $p \in [1, \infty)$ and f be an element of the equivalence class $F \in \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$, then there exists a sequence of polynomials $\{P_n\}_{n \geq 0}$ such that

$$\|P_n - f\|_{s,p} \longrightarrow 0,$$

for every $f \in F$. As a consequence, $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ can be identified as a normed subspace of $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$.

Proof. Let $f \in F \in \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$, then $f^{(\mathbf{m})} \in \mathbf{L}^p(\mu^{\alpha\beta})$. From the density of the vector space of continuous functions on $\mathbf{L}^p(\mu^{\alpha\beta})$ (see [153, Th. 3.14]) and Weierstrass' approximation theorem [20, §4.1], for all $n \in \mathbb{N}$, there exists a polynomial p_n such that

$$\left\| f^{(\mathbf{m})} - p_n \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})} < \frac{1}{n}.$$

Now, consider the vector $f(\vec{c}) = (f(c_0), f'(c_1), \dots, f^{(\mathbf{m}-1)}(c_{\mathbf{m}-1}))$ and the polynomial

$$P_n(s) = \mathcal{A}_{\vec{c},f(\vec{c})}(s) + \int_{c_0}^s \int_{c_1}^{s_1} \dots \int_{c_{\mathbf{m}-2}}^{s_{\mathbf{m}-2}} \int_{c_{\mathbf{m}-1}}^{s_{\mathbf{m}-1}} p_n(s_{\mathbf{m}}) ds_{\mathbf{m}} ds_{\mathbf{m}-1} \dots ds_2 ds_1.$$

It is clear that this polynomial satisfies $P_n^{(k)}(c_k) = f^{(k)}(c_k)$ for $k = 0, 1, \dots, \mathbf{m}-1$, and $P_n^{(\mathbf{m})} \equiv p_n$. Thus we have

$$\|f - P_n\|_{s,p} = \left\| f^{(\mathbf{m})} - P_n^{(\mathbf{m})} \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})} = \left\| f^{(\mathbf{m})} - p_n \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})} < \frac{1}{n},$$

and we get the first statement. Note that the sequence $\{P_n\}_{n \geq 0}$ does not depend on the function $f \in F$ chosen. Indeed, if $h \sim f$, then

$$\|P_n - h\|_{s,p} \leq \|P_n - f\|_{s,p} + \|f - h\|_{s,p} = \|P_n - f\|_{s,p} \longrightarrow 0.$$

The identification of $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ as a normed subspace of $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$ follows from the fact that $\{P_n\}_{n \geq 0}$ is a Cauchy sequence on $(\mathbb{P}, \|\cdot\|_{s,p})$

$$\|P_n - P_m\|_{s,p} \leq \|P_n - f\|_{s,p} + \|f - P_m\|_{s,p}.$$

If F is the equivalence class of f , then it can be identified with the equivalence class of $\{P_n\}_{n \geq 0}$ in $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$. $\square$

From now on, we will not distinguish between the equivalent class F and any representative of it f , and we will write $\mathbf{U}_{\vec{c},p}^{\alpha,\beta} \subset \mathbf{W}_{\vec{c},p}^{\alpha,\beta}$. It is worth to remark that although $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ is the function space of minimum requirements over f such that $\|f\|_{s,p}$ makes sense, this is not broad enough to describe every element of $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$, at least not for every combination of the parameters $p, \alpha, \beta, \mathfrak{m}$ and $\vec{c}$ (see Theorem 4.4.1).

In this section, we obtain all the combinations of the values p, α, β and $\vec{c}$ for which $\mathbf{W}_{\vec{c},p}^{\alpha,\beta} = \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$. The study of completeness of Sobolev spaces for the general vectorial measures $\vec{\mu}$, was given by J.M. Rodríguez et al. in [150, 149]. The Sobolev space defined in [150, 149], in the case of our specific vectorial measure $\vec{\mu} = (\delta_{c_0}, \delta_{c_1}, \dots, \delta_{c_{\mathfrak{m}-1}}, \mu^{\alpha,\beta})$, agrees with our space $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ in the essential case when $c_i \in [-1, 1], i = 0, 1, \dots, \mathfrak{m} - 1$.

We are going to prove the completeness of $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ by means of [150, Th. 5.1]. In this paper, the authors introduce several definitions which we will follow in order to use the theorem mentioned above. We also prove that for the complementary combinations of the parameters $\alpha, \beta, \mathfrak{m}, \vec{c}$, the space $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ is not complete and, consequently, $\mathbf{U}_{\vec{c},p}^{\alpha,\beta} \neq \mathbf{W}_{\vec{c},p}^{\alpha,\beta}$.

Definition 4.4.1 (see [76, Def. 1.4] and [150, Def. 2])

Let $p \geq 1$ and K be a closed interval on the real line. A weight function w defined on K satisfies condition $B_p(K)$ if $w^{-1} \in \mathbf{L}^{\frac{1}{p-1}}(K)$, and we write $w \in B_p(K)$.

If $A \subset \mathbb{R}$ and w is a weight function defined on A , then we say that $w \in B_p(A)$ when $w \in B_p(K)$ for every closed interval $K \subset A$.

Example 4.4.1

Consider the Jacobi weight function $w_{\alpha,\beta}(x) = (1-x)^\alpha(1+x)^\beta$, with $\alpha, \beta > -1$.

1. If $p = 1$, then

$$w_{\alpha,\beta} \in \begin{cases} B_1([-1, 1]), & \text{if } \alpha, \beta \in (-1, 0], \\ B_1((-1, 1]), & \text{if } \alpha \in (-1, 0], \beta \in (0, \infty), \\ B_1([-1, 1)), & \text{if } \alpha \in (0, \infty), \beta \in (-1, 0], \\ B_1((-1, 1)), & \text{if } \alpha, \beta \in (0, \infty). \end{cases}$$

2. If $p > 1$, then

$$w_{\alpha,\beta} \in \begin{cases} B_p([-1, 1]), & \text{if } \alpha, \beta \in (-1, p-1), \\ B_p((-1, 1]), & \text{if } \alpha \in (-1, p-1), \beta \in [p-1, \infty), \\ B_p([-1, 1)), & \text{if } \alpha \in [p-1, \infty), \beta \in (-1, p-1), \\ B_p((-1, 1)), & \text{if } \alpha, \beta \in [p-1, \infty). \end{cases}$$

Proposition 4.4.2

Let $\alpha, \beta > -1, p \geq 1, k = 0, 1, \dots, \mathfrak{m}$; and consider the vectorial weight

$$w(x) = (w_0(x), w_1(x), \dots, w_{\mathfrak{m}-1}(x), w_{\mathfrak{m}}(x)) = (0, 0, \dots, 0, w_{\alpha,\beta}(x)), \quad x \in [-1, 1], \quad (4.9)$$

$$\text{the interval } I_k = \begin{cases} (kp-1, (k+1)p-1], & \text{if } k = 0, 1, \dots, \mathfrak{m}-1, \\ (\mathfrak{m}p-1, \infty), & \text{if } k = \mathfrak{m}, \end{cases} \quad (4.10)$$

and the vectorial weight $\bar{w}^k = (\bar{w}_0^k, \bar{w}_1^k, \dots, \bar{w}_{\mathfrak{m}}^k)$, where

$$\bar{w}_j^k(x) = \begin{cases} \mathbb{1}_{[-1,0]}(x), & \text{if } j = 0, 1, \dots, \mathfrak{m}-k-1, \\ (1+x)^{\beta-(\mathfrak{m}-j)p} \mathbb{1}_{[-1,0]}(x), & \text{if } j = \mathfrak{m}-k, \dots, \mathfrak{m}-1, \\ w_{\alpha,\beta}(x), & \text{if } j = \mathfrak{m}, \end{cases}$$

and $\mathbb{1}_A$ denotes the characteristic function of the set $A \subset \mathbb{R}$.

If k is such that $\beta \in I_k$, then $\bar{w}^k$ is a right completion of w with respect to -1 (see [150, Def. 5]). Analogously, we can find a left completion of w with respect to 1 . Namely

$$\bar{w}_j^k(x) = \begin{cases} \mathbb{1}_{[0,1]}(x), & \text{if } j = 0, 1, \dots, \mathfrak{m}-k-1, \\ (1-x)^{\alpha-(\mathfrak{m}-j)p} \mathbb{1}_{[0,1]}(x), & \text{if } j = \mathfrak{m}-k, \dots, \mathfrak{m}-1, \\ w_{\alpha,\beta}(x), & \text{if } j = \mathfrak{m}, \end{cases}$$

where $\alpha \in I_k$.

Proof. It is straightforward to check $\bar{w}_{\mathfrak{m}}^k = w_{\mathfrak{m}}$ and there exists $\varepsilon = 1$ such that $\bar{w}_j^k = w_j$ for $x \notin [-1, -1 + \varepsilon] = [-1, 0]$. Then it only remains to prove that $\tilde{w}_j^k = \bar{w}_j^k - w_j = \bar{w}_j^k$, $0 \leq j \leq \mathfrak{m}-1$, satisfies

$$1. \quad \tilde{w}_j^k \in \mathbf{L}^1([-1, 0]).$$

$$2. \quad \Lambda_p(\tilde{w}_j^k, \bar{w}_{j+1}^k) < \infty \text{ where } \Lambda_p(u, v) := \sup_{-1 < r < 0} \left(\int_{-1}^r u \right) \|v^{-1}\|_{\mathbf{L}^{1/(p-1)}([-r, -1+\varepsilon])}.$$

The first statement comes from

$$\begin{aligned} \int_{-1}^0 \tilde{w}_j^k(x) dx &= \int_{-1}^0 \bar{w}_j^k(x) dx \\ &= \begin{cases} 1, & \text{if } j = 0, 1, \dots, \mathfrak{m}-k-1, \\ \int_{-1}^0 (1+x)^{\beta-(\mathfrak{m}-j)p} dx, & \text{if } j = \mathfrak{m}-k, \dots, \mathfrak{m}-1, \end{cases} < \infty, \end{aligned}$$

where the integral in the second case is finite because of $\beta - (\mathfrak{m} - j)p \geq \beta - kp > -1$. For the second statement we will split the proof in several cases.

- Case $p = 1$, $0 \leq j \leq \mathfrak{m} - k - 2$,

$$\begin{aligned}\Lambda_p(\tilde{w}_j^k, \bar{w}_{j+1}^k) &= \Lambda_p(\bar{w}_j^k, \bar{w}_{j+1}^k) = \sup_{-1 < r < 0} \left(\int_{-1}^r dx \right) \left\| \mathbb{1}_{[r,0]}^{-1} \right\|_{\mathbf{L}^\infty([r,0])} \\ &= \sup_{-1 < r < 0} (r+1) = 1.\end{aligned}$$

- Case $p = 1$, $j = \mathfrak{m} - k - 1$,

$$\begin{aligned}\Lambda_p(\tilde{w}_{\mathfrak{m}-k-1}^k, \bar{w}_{\mathfrak{m}-k}^k) &\leq \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} \left\{ \left(\int_{-1}^r dx \right) \sup_{r < x < 0} (1+x)^{-(\beta-k)} \right\} \\ &= \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} (r+1) = \max\{1, 2^{-\alpha}\},\end{aligned}$$

where we have used that $-\beta + k \geq -(k+1)p + 1 + k = 0$.

- Case $p = 1$, $\mathfrak{m} - k \leq j \leq \mathfrak{m} - 1$,

$$\begin{aligned}\Lambda_p(\tilde{w}_j^k, \bar{w}_{j+1}^k) &= \Lambda_p(\bar{w}_j^k, \bar{w}_{j+1}^k) = \sup_{-1 < r < 0} \left(\int_{-1}^r \bar{w}_j^k(x) dx \right) \left\| \left(\bar{w}_{j+1}^k \right)^{-1} \right\|_{\mathbf{L}^\infty([r,0])} \\ &\leq \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} \left\{ \int_{-1}^r \bar{w}_j^k(x) dx \sup_{r < x < 0} (1+x)^{-(\beta-(\mathfrak{m}-j)+1)} \right\} \\ &= \frac{\max\{1, 2^{-\alpha}\}}{\beta - (\mathfrak{m} - j) + 1} \sup_{-1 < r < 0} \left\{ (1+r)^{\beta-(\mathfrak{m}-j)+1} (1+r)^{-(\beta-(\mathfrak{m}-j)+1)} \right\} \\ &= \frac{\max\{1, 2^{-\alpha}\}}{\beta - (\mathfrak{m} - j) + 1},\end{aligned}$$

where we have used that $\beta - (\mathfrak{m} - j) + 1 > kp - (\mathfrak{m} - j) = k - (\mathfrak{m} - j) \geq 0$.

- Case $p > 1$, $0 \leq j \leq \mathfrak{m} - k - 2$,

$$\Lambda_p(\tilde{w}_j^k, \bar{w}_{j+1}^k) = \sup_{-1 < r < 0} \left(\int_{-1}^r dx \right) \left(\int_r^0 dx \right)^{p-1} = \sup_{-1 < r < 0} (r+1) (-r)^{p-1} \leq 1.$$

- Case $p > 1$, $j = \mathfrak{m} - k - 1$, $\beta < (k+1)p - 1$,

$$\begin{aligned}\Lambda_p(\tilde{w}_{\mathfrak{m}-k-1}^k, \bar{w}_{\mathfrak{m}-k}^k) &\leq \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} \left(\int_{-1}^r dx \right) \left(\int_r^0 (1+x)^{-\frac{\beta-kp}{p-1}} dx \right)^{p-1} \\ &= \frac{\max\{1, 2^{-\alpha}\} (p-1)^{p-1}}{(-\beta + (k+1)p - 1)^{p-1}} \sup_{-1 < r < 0} (r+1) \left(1 - (1+r)^{\frac{-\beta + (k+1)p - 1}{p-1}} \right)^{p-1} \\ &\leq \frac{\max\{1, 2^{-\alpha}\} (p-1)^{p-1}}{(-\beta + (k+1)p - 1)^{p-1}},\end{aligned}$$

where we have used that $-\beta + (k+1)p - 1 > 0$ in the last inequality and also to calculate the second integral.

- Case $p > 1$, $j = \mathfrak{m} - k - 1$, $\beta = (k + 1)p - 1$,

$$\begin{aligned}\Lambda_p(\tilde{w}_{\mathfrak{m}-k-1}^k, \bar{w}_{\mathfrak{m}-k}^k) &\leq \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} \left(\int_{-1}^r dx \right) \left(\int_r^0 (1+x)^{-\frac{\beta-kp}{p-1}} dx \right)^{p-1} \\ &= \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} (r+1) \left(\int_r^0 (1+x)^{-1} dx \right)^{p-1} \\ &= \sup_{-1 < r < 0} (r+1) (-\ln(r+1))^{p-1} < \infty,\end{aligned}$$

where we have used that, by L'Hôpital rule, $\lim_{x \rightarrow 0^+} x(\ln(1/x))^p = 0$.

- Case $p > 1$, $\mathfrak{m} - k \leq j \leq \mathfrak{m} - 1$,

$$\begin{aligned}\Lambda_p(\tilde{w}_j^k, \bar{w}_{j+1}^k) &\leq \max\{1, 2^{-\alpha}\} \sup_{-1 < r < 0} \int_{-1}^r \bar{w}_j^k(x) dx \left(\int_r^0 (1-x)^{-\frac{\beta-(\mathfrak{m}-j-1)p}{p-1}} dx \right)^{p-1} \\ &= \frac{\max\{1, 2^{-\alpha}\} (p-1)^{p-1}}{(\beta - (\mathfrak{m}-j)p + 1)^p} \sup_{-1 < r < 0} \left(1 - (1+r)^{\frac{\beta-(\mathfrak{m}-j)p+1}{p-1}} \right)^{p-1} \\ &= \frac{\max\{1, 2^{-\alpha}\} (p-1)^{p-1}}{(\beta - (\mathfrak{m}-j)p + 1)^p},\end{aligned}$$

where we have used that $\beta - (\mathfrak{m}-j)p + 1 > kp - (\mathfrak{m}-j)p \geq 0$.

□

Proposition 4.4.3

Denote by $\Omega^{(k)}$ the set of k -regular points of the vectorial weight (4.9) (see [150, Def. 6]).

- If $p > 1$, then for $\eta = 1, 2, \dots, \mathfrak{m}$;

$$\Omega^{(\mathfrak{m}-\eta)} = \begin{cases} [-1, 1], & \text{if } \alpha, \beta \in (-1, \eta p - 1), \\ (-1, 1], & \text{if } \alpha \in (-1, \eta p - 1), \beta \in [\eta p - 1, \infty), \\ [-1, 1), & \text{if } \alpha \in [\eta p - 1, \infty), \beta \in (-1, \eta p - 1), \\ (-1, 1), & \text{if } \alpha, \beta \in [\eta p - 1, \infty). \end{cases}$$

- If $p = 1$, then for $\eta = 1, 2, \dots, \mathfrak{m}$;

$$\Omega^{(\mathfrak{m}-\eta)} = \begin{cases} [-1, 1], & \text{if } \alpha, \beta \in (-1, \eta - 1], \\ (-1, 1], & \text{if } \alpha \in (-1, \eta - 1], \beta \in (\eta - 1, \infty), \\ [-1, 1), & \text{if } \alpha \in (\eta - 1, \infty), \beta \in (-1, \eta - 1], \\ (-1, 1), & \text{if } \alpha, \beta \in (\eta - 1, \infty). \end{cases}$$

Proof. From $w_{\alpha, \beta} \in B_p((-1, 1))$ it follows that y is $(\mathfrak{m} - 1)$ -regular for every $y \in (-1, 1)$ and the analysis of the regularity of $y = 1$ and $y = -1$ are analogous. Thus, we will only focus our attention on the point $y = -1$. First, we are going to prove that

if $p > 1$ and $\beta \in (-1, \eta p - 1) \subset \cup_{i=0}^{\eta-1} I_i$, where I_i is as in (4.10), then $y = -1$ is an $(\mathfrak{m} - \eta)$ -right regular point.

Take $0 \leq k \leq \eta - 1$ such that $\beta \in I_k$.

- Case $\beta \in (kp - 1, (k+1)p - 1)$,

$$\int_{-1}^0 \left(\bar{w}_{\mathfrak{m}-k}^k \right)^{-\frac{1}{p-1}} dx = \begin{cases} \int_{-1}^0 (x+1)^{-\frac{\beta-kp}{p-1}} dx, & \text{if } k \geq 1, \\ \int_{-1}^0 (1-x)^{-\frac{\alpha}{p-1}} (1+x)^{-\frac{\beta}{p-1}} dx, & \text{if } k = 0, \end{cases} < \infty,$$

where the integral in the first case is finite due to $\frac{\beta-kp}{p-1} < \frac{p-1}{p-1} = 1$. Note that $\mathfrak{m} - k \geq \mathfrak{m} - \eta + 1$.

- Case $\beta = (k+1)p - 1$,

$$\int_{-1}^0 \left(\bar{w}_{\mathfrak{m}-(k+1)}^k \right)^{-\frac{1}{p-1}} dx = 1.$$

Note that, as $\beta < \eta p - 1$, we get $k \leq \eta - 2$, so $\mathfrak{m} - (k+1) \geq \mathfrak{m} - \eta + 1$.

In both cases, we have proved that there exists $\mathfrak{m} - \eta < j \leq \mathfrak{m}$ such that $\bar{w}_j^k \in B_p([-1, 0])$, so $y = -1$ is $(\mathfrak{m} - \eta)$ -right regular.

Now, consider $p = 1$, and $\beta \in (-1, \eta - 1] = \cup_{k=0}^{\eta-1} I_k$. Take $0 \leq k \leq \eta - 1$ such that $\beta \in I_k$.

$$\sup_{-1 \leq x \leq 0} \left\{ \left(\bar{w}_{\mathfrak{m}-k}^k(x) \right)^{-1} \right\} = \begin{cases} \sup_{-1 \leq x \leq 0} \left\{ (x+1)^{-\beta+k} \right\}, & \text{if } k \geq 1, \\ \sup_{-1 \leq x \leq 0} \left\{ (1-x)^{-\alpha} (1+x)^{-\beta} \right\}, & \text{if } k = 0, \end{cases} < \infty.$$

Note that $\mathfrak{m} - k \geq \mathfrak{m} - \eta + 1$, so we have proved that there exists $\mathfrak{m} - \eta < j \leq \mathfrak{m}$ such that $\bar{w}_j^k \in B_p([-1, 0])$, so $y = -1$ is $(\mathfrak{m} - \eta)$ -right regular.

Now, we are going to prove that in the other case the point $y = -1$ is not $(\mathfrak{m} - \eta)$ -right regular. But first, we are going to prove the following auxiliary result. If w is a non-negative Lebesgue measurable function, then

$$\|w^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}([a,b])} \geq \frac{(b-a)^p}{\int_a^b w(x) dx}, \quad a < b. \quad (4.11)$$

This inequality comes from Hölder inequality

$$\begin{aligned} (b-a)^p &= \left(\int_a^b dx \right)^p = \left(\int_a^b w^{\frac{1}{p}} w^{-\frac{1}{p}} dx \right)^p \leq \left(\|w^{\frac{1}{p}}\|_{\mathbf{L}^p([a,b])} \|w^{-\frac{1}{p}}\|_{\mathbf{L}^{\frac{p}{p-1}}([a,b])} \right)^p \\ &= \|w\|_{\mathbf{L}^1([a,b])} \|w^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}([a,b])}. \end{aligned}$$

Let $\bar{w}$ be a right completion of w with respect to $y = -1$. Then, we will prove by induction that for $0 < s < \frac{\varepsilon}{\mathfrak{m}+1}$, (where ε is the given one for $\bar{w}$ in the definition of right completion) we have

$$\|\bar{w}_{\mathfrak{m}-k}^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}(J_{k,s})} \geq M_k s^{-\beta+(k+1)p-1}, \quad k = 0, 1, \dots, \mathfrak{m}, \quad (4.12)$$

where $J_{k,s} = [-1 + (\mathfrak{m} - k)s, -1 + (\mathfrak{m} - k + 1)s]$ and M_k is some positive constant independent of s . For $k = 0$ we have

- Case $p > 1, \beta \in [\eta p - 1, \infty)$,

$$\begin{aligned} \|\bar{w}_{\mathfrak{m}}^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}(J_{0,s})} &= \left(\int_{-1+\mathfrak{m}s}^{-1+(\mathfrak{m}+1)s} (1-x)^{-\frac{\alpha}{p-1}} (1+x)^{-\frac{\beta}{p-1}} dx \right)^{p-1} \\ &\geq \min \left\{ \frac{1}{2^\alpha}, \frac{1}{(2-\varepsilon)^\alpha} \right\} \left(\int_{-1+\mathfrak{m}s}^{-1+(\mathfrak{m}+1)s} (1+x)^{-\frac{\beta}{p-1}} dx \right)^{p-1} \\ &= M_0 s^{-\beta+p-1}, \end{aligned}$$

where

$$M_0 = \begin{cases} \min \left\{ \frac{1}{2^\alpha}, \frac{1}{(2-\varepsilon)^\alpha} \right\} \frac{(p-1)^{p-1}}{(\beta-p+1)^{p-1}} \left(\mathfrak{m}^{-\frac{\beta}{p-1}} - (\mathfrak{m}+1)^{-\frac{\beta}{p-1}} \right)^{p-1}, & \text{if } \beta > p-1, \\ \min \left\{ \frac{1}{2^\alpha}, \frac{1}{(2-\varepsilon)^\alpha} \right\} [\ln(\frac{\mathfrak{m}+1}{\mathfrak{m}})]^{p-1}, & \text{if } \beta = p-1, \end{cases} > 0.$$

- Case $p = 1, \beta \in (\eta - 1, \infty)$,

$$\begin{aligned} \|\bar{w}_{\mathfrak{m}}^{-1}\|_{\mathbf{L}^\infty(J_{0,s})} &= \sup_{-1+\mathfrak{m}s < x < -1+(\mathfrak{m}+1)s} \left\{ (1-x)^{-\alpha} (1+x)^{-\beta} \right\} \\ &\geq \min \{ 2^{-\alpha}, (2-\varepsilon)^{-\alpha} \} \sup_{-1+\mathfrak{m}s < x < -1+(\mathfrak{m}+1)s} \left\{ (1+x)^{-\beta} \right\} = M_0 s^{-\beta}, \end{aligned}$$

where $M_0 = \min \{ 2^{-\alpha}, (2-\varepsilon)^{-\alpha} \} \mathfrak{m}^{-\beta}$.

Now, suppose that (4.12) is true for $k-1$. Then from (4.11) it follows

$$\begin{aligned} \|\bar{w}_{\mathfrak{m}-k}^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}(J_{k,s})} &\geq \frac{s^p}{\int_{-1+(\mathfrak{m}-k)s}^{-1+(\mathfrak{m}-k+1)s} \bar{w}_{\mathfrak{m}-k}(x) dx} \\ &\geq \frac{s^p \|\bar{w}_{\mathfrak{m}-k+1}^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}(J_{k-1,s})}}{\int_{-1}^{-1+(\mathfrak{m}-k+1)s} \tilde{w}_{\mathfrak{m}-k}(x) dx \|\bar{w}_{\mathfrak{m}-k+1}^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}([-1+(\mathfrak{m}-k+1)s, -1+\varepsilon])}} \\ &\geq \frac{s^p \|\bar{w}_{\mathfrak{m}-k+1}^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}(J_{k-1,s})}}{\Lambda_p(\tilde{w}_{\mathfrak{m}-k}, \bar{w}_{\mathfrak{m}-k+1})} \geq \frac{M_{k-1} s^{-\beta+(k+1)p-1}}{\Lambda_p(\tilde{w}_{\mathfrak{m}-k}, \bar{w}_{\mathfrak{m}-k+1})}. \end{aligned}$$

Taking $M_k = \frac{M_{k-1}}{\Lambda_p(\tilde{w}_{\mathfrak{m}-k}, \bar{w}_{\mathfrak{m}-k+1})} > 0$ we conclude the proof of (4.12). Thus we have

$$\|\bar{w}_j^{-1}\|_{\mathbf{L}^{\frac{1}{p-1}}(J_{\mathfrak{m}-j,s})} \geq M_{\mathfrak{m}-j} s^{-\beta+(\mathfrak{m}-j+1)p-1}, \quad j = 0, 1, \dots, \mathfrak{m},$$

and from here it follows

- Case $p > 1, \beta \in [\eta p - 1, \infty)$,

$$\lim_{s \rightarrow 0} \left\| \bar{w}_j^{-1} \right\|_{\mathbf{L}^{\frac{1}{p-1}}([-1+js, -1+(j+1)s])} \neq 0, \quad j = \mathfrak{m} - \eta + 1, \dots, \mathfrak{m}.$$

- Case $p = 1, \beta \in (\eta - 1, \infty)$,

$$\lim_{s \rightarrow 0} \left\| \bar{w}_j^{-1} \right\|_{\mathbf{L}^\infty([-1+(\mathfrak{m}-j)s, -1+(\mathfrak{m}-j+1)s])} = \infty, \quad j = \mathfrak{m} - \eta + 1, \dots, \mathfrak{m}.$$

Thus $\bar{w}_j \notin B_p([-1, -1 + \varepsilon])$ for $j = \mathfrak{m} - \eta + 1, \dots, \mathfrak{m}$, so $y = -1$ is not $(\mathfrak{m} - \eta)$ -right regular. $\square$

Proposition 4.4.4

Let $\bar{\mu}_{[-1,1]} = (\mu_0, \dots, \mu_{\mathfrak{m}-1}, \mu^{\alpha\beta})$ be a vectorial measure, where

$$\mu_j = \begin{cases} \delta_{c_j}, & \text{if } c_j \in [-1, 1], \\ 0, & \text{if } c_j \notin [-1, 1] \end{cases}, \quad j = 0, 1, \dots, \mathfrak{m} - 1.$$

Then $\bar{\mu}_{[-1,1]}$ is a vectorial measure in $[-1, 1]$ of type 2 (see [149, Def. 11]) if and only if $c_j \in \Omega^{(j)}$, for every j such that $c_j \in [-1, 1]$.

Recall that $\Omega^{(k)}$ denotes the set of k -regular points of the vectorial weight (4.9) given explicitly in Proposition 4.4.3.

Proof. The condition $c_j \in \Omega^{(j)}$ is just the condition in the definition of strongly p -admissible vectorial measure (see [150, Def. 8]) and $\bar{\mu}_{[-1,1]}$ is finite (each component measure is finite). Then, we only have to verify that there exist real numbers $-1 \leq a_1 < a_2 < a_3 < a_4 \leq 1$ such that

1. $w_{\mathfrak{m}} \in B_p([a_1, a_4])$,
2. if $-1 < a_1$, then w_j is comparable to a non-decreasing weight in $[-1, a_2]$, for $0 \leq j \leq \mathfrak{m}$,
3. if $a_4 < 1$, then w_j is comparable to a non-decreasing weight in $[a_3, 1]$, for $0 \leq j \leq \mathfrak{m}$,

where $w_j = d\mu_j/dx$, $j = 0, 1, \dots, \mathfrak{m}$, see [149, Def. 1]. Therefore, we obtain $w_j \equiv 0$, $j = 0, 1, \dots, \mathfrak{m} - 1$; and $w_{\mathfrak{m}}(x) = w_{\alpha, \beta}(x)$. Since $w'_{\mathfrak{m}}(x) = (1-x)^{\beta-1}(1+x)^{\alpha-1}(\beta - \alpha - (\beta + \alpha)x)$ it follows that there exists $0 < \varepsilon < 1$ (depending on α and β) such that

$$w'(x) \begin{cases} > 0 \text{ for } x \in (-1, -1 + \varepsilon], & \text{if } \beta > 0, \\ < 0 \text{ for } x \in [1 - \varepsilon, 1), & \text{if } \alpha > 0. \end{cases}$$

Thus, from Example 4.4.1, the above three conditions are easily verified taking $a_2 = -1 + \varepsilon$, $a_3 = 1 - \varepsilon$ and

$$\begin{aligned} \bullet \text{ Case } p > 1 & \begin{cases} a_1 = -1, a_4 = 1, & \text{if } \alpha, \beta \in (-1, p-1), \\ a_1 = -1 + \frac{\varepsilon}{2}, a_4 = 1, & \text{if } \alpha \in (-1, p-1), \beta \in [p-1, \infty), \\ a_1 = -1, a_4 = 1 - \frac{\varepsilon}{2}, & \text{if } \alpha \in [p-1, \infty), \beta \in (-1, p-1), \\ a_1 = -1 + \frac{\varepsilon}{2}, a_4 = 1 - \frac{\varepsilon}{2}, & \text{if } \alpha, \beta \in [p-1, \infty). \end{cases} \\ \bullet \text{ Case } p = 1 & \begin{cases} a_1 = -1, a_4 = 1, & \text{if } \alpha, \beta \in (-1, 0], \\ a_1 = -1 + \frac{\varepsilon}{2}, a_4 = 1, & \text{if } \alpha \in (-1, 0], \beta \in (0, \infty), \\ a_1 = -1, a_4 = 1 - \frac{\varepsilon}{2}, & \text{if } \alpha \in (0, \infty), \beta \in (-1, 0], \\ a_1 = -1 + \frac{\varepsilon}{2}, a_4 = 1 - \frac{\varepsilon}{2}, & \text{if } \alpha, \beta \in (0, \infty). \end{cases} \end{aligned}$$

□

Proposition 4.4.5

If $\eta \in \mathbb{Z}_+$, then there exist $\eta + 1$ positive real numbers $\{\lambda_{\eta,k}\}_{k=0}^{\eta}$ such that

$$\left[\frac{1}{(1 \pm x) \ln(1 \pm x)} \right]^{(\eta)} = \frac{(\mp 1)^{\eta}}{(1 \pm x)^{\eta+1} \ln(1 \pm x)} \sum_{k=0}^{\eta} \frac{\lambda_{\eta,k}}{\ln^k(1 \pm x)} \quad (4.13)$$

Proof. We will prove it by induction on η . The case $\eta = 0$ is trivial. Suppose the formula (4.13) holds for $\eta = n$ and we are going to prove it for $\eta = n + 1$.

$$\begin{aligned} \left(\frac{1}{(1 \pm x) \ln(1 \pm x)} \right)^{(n+1)} &= \left(\frac{(\mp 1)^n}{(1 \pm x)^{n+1} \ln(1 \pm x)} \sum_{k=0}^n \frac{\lambda_{n,k}}{\ln^k(1 \pm x)} \right)' \\ &= (\mp 1)^n \sum_{k=0}^n \left(\frac{\lambda_{n,k}}{(1 \pm x)^{n+1} \ln^{k+1}(1 \pm x)} \right)' \\ &= \frac{(\mp 1)^{n+1}}{(1 \pm x)^{n+2} \ln(1 \pm x)} \sum_{k=0}^{n+1} \frac{(n+1)\lambda_{n,k} + k\lambda_{n,k-1}}{\ln^k(1 \pm x)}, \end{aligned}$$

where $\lambda_{n,-1} = \lambda_{n,n+1} = 0$. Now the proof is completed taking $\lambda_{n+1,k} = (n+1)\lambda_{n,k} + k\lambda_{n,k-1}$, for $k = 0, 1, \dots, n+1$. □

Proposition 4.4.6

Let $p \geq 1$, $\eta \in \mathbb{N}$, $\alpha, \beta > -1$, $a \in (-1, 1)$ and

$$\alpha \begin{cases} \geq \eta p - 1, & \text{if } p > 1, \\ > \eta p - 1, & \text{if } p = 1, \end{cases}$$

then the functions

$$\begin{aligned}\phi_\eta(x) &= \begin{cases} \left[\ln \left(\ln \left(\frac{1}{1-x} \right) \right) \right]^{(\eta)}, & \text{if } \alpha = \eta p - 1, \\ \left[\ln \left(\frac{1}{1-x} \right) \right]^{(\eta)}, & \text{if } \alpha > \eta p - 1, \end{cases} \quad x \in [a, 1), \\ \phi_{-\eta}(x) &= \begin{cases} \left[\ln \left(\ln \left(\frac{1}{1+x} \right) \right) \right]^{(\eta)}, & \text{if } \alpha = \eta p - 1, \\ \left[\ln \left(\frac{1}{1+x} \right) \right]^{(\eta)}, & \text{if } \alpha > \eta p - 1, \end{cases} \quad x \in (-1, a],\end{aligned}$$

are continuous, satisfy $\phi_\eta \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[a,1]})$, $\phi_{-\eta} \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[-1,a]})$ and

$$\lim_{x \rightarrow \pm 1} \int_a^x \int_a^{x_1} \cdots \int_a^{x_{\eta-1}} \phi_{\pm\eta}(x_\eta) dx_\eta dx_{\eta-1} \cdots dx_1 = \infty.$$

Proof. The first and the last statements are straightforward and the proofs of $\phi_\eta \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[a,1]})$ and $\phi_{-\eta} \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[-1,a]})$ are analogous, so we will only prove $\phi_\eta \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[a,1]})$. Using (4.13) we obtain

• Case $\alpha = \eta p - 1$, $p > 1$,

$$\begin{aligned}\|\phi_\eta\|_{\mathbf{L}^p(\mu^{\alpha\beta}|_{[a,1]})}^p &= \int_a^1 \left| \left[\ln \left(\ln \left(\frac{1}{1-x} \right) \right) \right]^{(\eta)} \right|^p (1-x)^{\eta p - 1} (1+x)^\beta dx \\ &= \int_a^1 \left| \frac{1}{(1-x)^\eta \ln(1-x)} \sum_{k=0}^{\eta-1} \frac{\lambda_{\eta-1,k}}{\ln^k(1-x)} \right|^p (1-x)^{\eta p - 1} (1+x)^\beta dx \\ &\leq \max\{(1+a)^\beta, 2^\beta\} \int_a^1 \frac{1}{\ln^p(\frac{1}{1-x})} \left| \sum_{k=0}^{\eta-1} \frac{(-1)^k \lambda_{\eta-1,k}}{\ln^k(\frac{1}{1-x})} \right|^p d \ln \left(\frac{1}{1-x} \right) \\ &\leq \max\{(1+a)^\beta, 2^\beta\} \int_{\ln(\frac{1}{1-a})}^\infty \frac{1}{y^p} \left| \sum_{k=0}^{\eta-1} \frac{(-1)^k \lambda_{\eta-1,k}}{y^k} \right|^p dy < \infty,\end{aligned}$$

where the last integral is convergent by the limit comparison test

$$\lim_{y \rightarrow \infty} \frac{\frac{1}{y^p} \left| \sum_{k=0}^{\eta-1} \frac{(-1)^k \lambda_{\eta-1,k}}{y^k} \right|^p}{\frac{1}{y^p}} = \lambda_{\eta-1,0}^p > 0.$$

• Case $\alpha > \eta p - 1$, $p \geq 1$,

$$\begin{aligned}\int_a^1 |\phi_\eta(x)|^p d\mu^{\alpha\beta}(x) &= \int_a^1 \left| \left[\ln \left(\frac{1}{1-x} \right) \right]^{(\eta)} \right|^p (1-x)^\alpha (1+x)^\beta dx \\ &\leq \frac{\max\{(1+a)^\beta, 2^\beta\}}{\alpha - \eta p + 1} (1-a)^{\alpha - \eta p + 1}.\end{aligned}$$

□

We are ready to prove the completeness of the spaces $\mathbf{U}_{c,p}^{\alpha,\beta}$.

Theorem 4.4.1

Let $p \geq 1$, $\alpha, \beta > -1$. Then $\mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$ is a complete space if and only if one of the following statements hold

- $\mathfrak{m} = 1$.
- $\mathfrak{m} \geq 2$, $p > 1$ and for $\eta = 1, \dots, \mathfrak{m} - 1$,

$$c_{\mathfrak{m}-\eta} \in \begin{cases} \mathbb{R}, & \text{if } \alpha, \beta \in (-1, \eta p - 1), \\ \mathbb{R} \setminus \{-1\}, & \text{if } \alpha \in (-1, \eta p - 1), \beta \in [\eta p - 1, \infty), \\ \mathbb{R} \setminus \{1\}, & \text{if } \alpha \in [\eta p - 1, \infty), \beta \in (-1, \eta p - 1), \\ \mathbb{R} \setminus \{-1, 1\}, & \text{if } \alpha, \beta \in [\eta p - 1, \infty). \end{cases}$$

- $\mathfrak{m} \geq 2$, $p = 1$, and for $\eta = 1, \dots, \mathfrak{m} - 1$,

$$c_{\mathfrak{m}-\eta} \in \begin{cases} \mathbb{R}, & \text{if } \alpha, \beta \in (-1, \eta - 1], \\ \mathbb{R} \setminus \{-1\}, & \text{if } \alpha \in (-1, \eta - 1], \beta \in (\eta - 1, \infty), \\ \mathbb{R} \setminus \{1\}, & \text{if } \alpha \in (\eta - 1, \infty), \beta \in (-1, \eta - 1], \\ \mathbb{R} \setminus \{-1, 1\}, & \text{if } \alpha, \beta \in (\eta - 1, \infty). \end{cases}$$

In addition, in the affirmative case we obtain $\mathbf{U}_{\vec{c}, p}^{\alpha, \beta} = \mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$.

Note that there are not restrictions over c_0 in any case.

Proof. Let $\{f_n\}_{n \geq 0} \subset \mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$ be a Cauchy sequence. From Proposition 4.4.1 there exists a sequence of polynomials $\{P_n\}_{n \geq 0}$ such that $\|P_n - f_n\|_{s, p} < \frac{1}{n}$.

Consider the space $V^{k, p}([-1, 1], \vec{\mu}_{[-1, 1]})$ defined in [149, Def. 9], where the measure $\vec{\mu}_{[-1, 1]}$ is as in Proposition 4.4.4 and whose norm is given by

$$\|f\|_{\vec{s}, p} = \left(\sum_{\substack{k=0 \\ c_k \in [-1, 1]}}^{\mathfrak{m}-1} \left| f^{(k)}(c_k) \right|^p + \int_{-1}^1 \left| f^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha, \beta}(x) \right)^{\frac{1}{p}}.$$

From Proposition 4.4.4, [150, Remark 4 in Def. 16] and [150, Th. 5.1] this space is complete. So, as the restriction of P_n to $[-1, 1]$ belongs to $V^{k, p}([-1, 1], \vec{\mu}_{[-1, 1]})$ and

$$\|P_m - P_n\|_{\vec{s}, p} \leq \|P_m - P_n\|_{s, p} \leq \frac{1}{m} + \|f_m - f_n\|_{s, p} + \frac{1}{n},$$

there exists a function $\tilde{f} \in V^{k, p}([-1, 1], \vec{\mu}_{[-1, 1]})$ such that

$$\|P_n - \tilde{f}\|_{\vec{s}, p} \longrightarrow 0. \quad (4.14)$$

On the other hand, since

$$\begin{aligned} \left| P_m^{(j)}(c_j) - P_n^{(j)}(c_j) \right|^p &\leq \sum_{k=0}^{\mathfrak{m}-1} \left| P_m^{(k)}(c_k) - P_n^{(k)}(c_k) \right|^p \\ &\quad + \int_{-1}^1 \left| P_m^{(\mathfrak{m})}(x) - P_n^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha\beta}(x) = \|P_m - P_n\|_{s,p}, \end{aligned}$$

it follows that $\{P_n^{(j)}(c_j)\}_{n \geq 0}$, $j = 0, 1, \dots, \mathfrak{m}-1$, is a Cauchy sequence in $\mathbb{R}$, then there exist values $f_{c_k}^k$ such that $P_n^{(k)}(c_k) \rightarrow f_{c_k}^k$, $k = 0, 1, \dots, \mathfrak{m}-1$.

Note that, from (4.14) we obtain $\tilde{f}^{(k)}(c_k) = f_{c_k}^k$, for $c_k \in [-1, 1]$, where the derivative of $\tilde{f}$ at ± 1 , as usual, means the corresponding usual side derivative.

Consider the vector $f(\vec{c}) = (f_{c_0}^0, f_{c_1}^1, \dots, f_{c_{\mathfrak{m}-1}}^{\mathfrak{m}-1})$ and the function

$$f(x) = \begin{cases} \tilde{f}(x), & \text{if } x \in [-1, 1], \\ \mathcal{A}_{\vec{c}, f(\vec{c})}(x), & \text{if } x \in \mathbb{R} \setminus [-1, 1], \end{cases}$$

which belongs to $\mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$ and satisfies

$$\begin{aligned} \|P_n - f\|_{s,p}^p &= \sum_{c_k \notin [-1, 1]} \left| P_n^{(k)}(c_k) - f^{(k)}(c_k) \right|^p + \|P_n - f\|_{\tilde{s}, p}^p \\ &= \sum_{c_k \notin [-1, 1]} \left| P_n^{(k)}(c_k) - f_{c_k}^k \right|^p + \|P_n - \tilde{f}\|_{\tilde{s}, p}^p \longrightarrow 0. \end{aligned}$$

Finally, from $\|f_n - f\|_{s,p} \leq \|f_n - P_n\|_{s,p} + \|P_n - f\|_{s,p} \longrightarrow 0$, we conclude that $\mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$ is a complete space. Thus, as $\mathbb{P} \subset \mathbf{U}_{\vec{c}, p}^{\alpha, \beta} \subset \mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$ and $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$ is the completion of $\mathbb{P}$ we obtain $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta} = \mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$.

The second part of the proof is to show that the conditions over c_k are also necessary. Then $\mathfrak{m} \geq 2$ and let us consider the case $c_{\mathfrak{m}-\eta} = 1$ ($\alpha \geq \eta p - 1$ if $p > 1$ or $\alpha > \eta - 1$ if $p = 1$), the other case, when $c_{\mathfrak{m}-\eta} = -1$, is analogous. Take the sequence of functions

$$f_n(x) = \mathbb{1}_{[a, b_n]}(x) \int_a^x \int_a^{x_1} \cdots \int_a^{x_{\mathfrak{m}-1}} \phi_\eta(x_{\mathfrak{m}}) dx_{\mathfrak{m}} dx_{\mathfrak{m}-1} \cdots dx_1,$$

where ϕ_η is the function defined in Proposition 4.4.6 and the values a and b_n , $n \geq 0$, are taken as $-1 < a < b_0 < 1$ and $\{b_n\}_{n \geq 0}$ is a non-decreasing sequence with 1 as limit and such that $c_k \notin \{a\} \cup [b_0, 1)$ for $k = 0, 1, \dots, \mathfrak{m}-1$. Since $\phi_\eta \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[a, 1]})$ it follows that $f_n^{(\mathfrak{m})} \in \mathbf{L}^p(\mu^{\alpha\beta})$, so $f_n \in \mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$ for $n \in \mathbb{N}$. Note also that, from the condition on a and b_n , f_n vanishes on neighborhoods of each $c_j \notin (-1, 1)$, and f_n does not depend on n over neighborhoods of each $c_j \in (-1, 1)$. Now we will prove that f_n is a Cauchy sequence

$$\begin{aligned} \|f_m - f_n\|_{s,p}^p &= \sum_{k=0}^{\mathfrak{m}-1} \left| f_m^{(k)}(c_k) - f_n^{(k)}(c_k) \right|^p + \int_{-1}^1 \left| f_m^{(\mathfrak{m})}(x) - f_n^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha\beta}(x), \\ &= \left| \int_{b_n}^{b_m} |\phi_\eta(x)|^p d\mu^{\alpha\beta}(x) \right|. \end{aligned}$$

Since $\phi_\eta \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[a,1]})$, the sequence $\int_{b_n}^1 |\phi_\eta(x)|^p d\mu^{\alpha\beta}(x)$ tends to 0 when $n \rightarrow \infty$, so it is a Cauchy sequence on $\mathbb{R}$. Consequently, f_n is a Cauchy sequence on $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$. On the other hand, if $f \in \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ is a function such that $f_n \rightarrow f$ in $\mathbf{U}_{\vec{c},p}^{\alpha,\beta}$, then

$$\sum_{k=0}^{\mathfrak{m}-1} \left| f_n^{(k)}(c_k) - f^{(k)}(c_k) \right|^p + \int_{-1}^1 \left| f_n^{(\mathfrak{m})}(x) - f^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha\beta}(x) = \|f_n - f\|_{s,p}^p \rightarrow 0.$$

Hence, $f_n^{(k)}(c_k) \rightarrow f^{(k)}(c_k)$ for $k = 0, 1, \dots, \mathfrak{m}-1$ and

$$\int_a^{b_n} \left| \phi_\eta(x) - f^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha\beta}(x) \rightarrow 0.$$

Since $f \in \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ we get $\phi_\eta - f^{(\mathfrak{m})} \in \mathbf{L}^p(\mu^{\alpha\beta}|_{[a,1]})$, so

$$\int_a^1 \left| \phi_\eta(x) - f^{(\mathfrak{m})}(x) \right|^p d\mu^{\alpha\beta}(x) = 0,$$

and, as a consequence, $f^{(\mathfrak{m})} \equiv \phi_\eta$ a.e. on $[a, 1]$. Thus we can write for $x \in [a, 1]$

$$f^{(\mathfrak{m}-\eta)}(x) = \int_a^x \int_a^{x_1} \dots \int_a^{x_{\eta-1}} \phi_\eta(x_\eta) dx_\eta dx_{\eta-1} \dots dx_1 + \mathcal{Q}_{\eta-1}(x), \quad x \in [a, 1],$$

where $\mathcal{Q}_{\eta-1}$ is a polynomial of degree at most $\eta - 1$. Now using that $c_{\mathfrak{m}-\eta} = 1$ we obtain that f has derivative of order $\mathfrak{m} - \eta$ on $[a, 1]$ and the function $f^{(\mathfrak{m}-\eta)}$ is the derivative of $f^{(\mathfrak{m}-\eta-1)}$ on the closed interval $[a, 1]$ (note that $\eta \leq \mathfrak{m} - 1$). From Darboux's theorem [152, Th. 5.12] it follows that $f^{(\mathfrak{m}-\eta)}([1 - \varepsilon, 1])$ is an interval for every sufficiently small $\varepsilon > 0$ and we get the contradiction $\lim_{x \rightarrow 1^-} f^{(\mathfrak{m}-\eta)}(x) = \infty$ (see Proposition 4.4.6). $\square$

Corollary 4.4.1.1

- If $\mathfrak{m} = 1$ and $p \geq 1$, then $\mathbf{W}_{c_0,p}^{\alpha,\beta} = \mathbf{U}_{c_0,p}^{\alpha,\beta}$, $\forall c_0 \in \mathbb{R}$.
- If $\mathfrak{m} \geq 2$ and $p > 1$, then

$$\mathbf{W}_{\vec{c},p}^{\alpha,\beta} = \mathbf{U}_{\vec{c},p}^{\alpha,\beta}, \quad \forall \vec{c} \in \mathbb{R}^{\mathfrak{m}}, \quad \text{if and only if } \alpha, \beta \in (-1, p-1).$$

- If $\mathfrak{m} \geq 2$ and $p = 1$, then

$$\mathbf{W}_{\vec{c},p}^{\alpha,\beta} = \mathbf{U}_{\vec{c},p}^{\alpha,\beta}, \quad \forall \vec{c} \in \mathbb{R}^{\mathfrak{m}}, \quad \text{if and only if } \alpha, \beta \in (-1, 0].$$

4.5 Sobolev-type orthogonal polynomials and Fourier series

On $\mathbb{P}$, let us consider the Sobolev-type inner product, related to the norm (4.7) with $p = 2$

$$\langle f, g \rangle_s := \sum_{k=0}^{\mathfrak{m}-1} f^{(k)}(c_k) g^{(k)}(c_k) + \int_{-1}^1 f^{(\mathfrak{m})}(x) g^{(\mathfrak{m})}(x) d\mu^{\alpha,\beta}(x). \quad (4.15)$$

Let $P_{\vec{c},n}^{(\alpha,\beta)}$ be the $\mathfrak{m}$ -th iterated integral of $P_n^{(\alpha,\beta)}$ (the n th Jacobi orthonormal polynomial), normalized by the conditions

$$\left(P_{\vec{c},n}^{(\alpha,\beta)}\right)^{(k)}(c_k) = 0, \quad k = 0, 1, \dots, \mathfrak{m}-1, \quad (4.16)$$

or equivalently,

$$P_{\vec{c},n}^{(\alpha,\beta)}(s) = \int_{c_0}^s \int_{c_1}^{s_1} \cdots \int_{c_{\mathfrak{m}-2}}^{s_{\mathfrak{m}-2}} \int_{c_{\mathfrak{m}-1}}^{s_{\mathfrak{m}-1}} P_n^{(\alpha,\beta)}(s_{\mathfrak{m}}) ds_{\mathfrak{m}} ds_{\mathfrak{m}-1} \cdots ds_2 ds_1.$$

Theorem 4.5.1

The polynomials

$$s_n(x) = \begin{cases} \mathcal{G}_{\vec{c},n}(x), & \text{if } 0 \leq n \leq \mathfrak{m}-1, \\ P_{\vec{c},n-\mathfrak{m}}^{(\alpha,\beta)}(x), & \text{if } \mathfrak{m} \leq n, \end{cases} \quad (4.17)$$

constitute a family of orthonormal polynomials with respect to (4.15) on $\mathbf{W}_{\vec{c},2}^{\alpha,\beta}$, which is also complete.

The algebraic and analytical properties of the polynomials $\{s_n\}_{n \geq 0}$ have been studied in [136].

Proof. First, note that, for each $n \geq 0$, s_n is a polynomial of degree n and from (4.8) and (4.16)

$$s_n^{(k)}(c_k) = \begin{cases} 1, & \text{if } n = k, \\ 0, & \text{if } n \neq k, \end{cases} \quad \text{for } 0 \leq k \leq \mathfrak{m}-1. \quad (4.18)$$

Now, we are going to prove that $\{s_n\}_{n \geq 0}$ is an orthonormal polynomial sequence with respect to (4.15).

If $m, n \geq \mathfrak{m}$, then from (4.17) and (4.18)

$$\langle s_m, s_n \rangle_s = \int_{-1}^1 p_{m-\mathfrak{m}}^{(\alpha,\beta)}(x) p_{n-\mathfrak{m}}^{(\alpha,\beta)}(x) d\mu^{\alpha,\beta}(x) = \begin{cases} 1, & \text{if } n = m, \\ 0, & \text{if } n \neq m. \end{cases}$$

Otherwise, we obtain that $s_m^{(\mathfrak{m})} s_n^{(\mathfrak{m})} \equiv 0$ and from (4.18)

$$\langle s_m, s_n \rangle_s = \sum_{k=0}^{\mathfrak{m}-1} s_m^{(k)}(c_k) s_n^{(k)}(c_k) = \begin{cases} 1, & \text{if } n = m, \\ 0, & \text{if } n \neq m. \end{cases}$$

The completeness is a direct consequence of the fact that $\mathbf{W}_{\vec{c},p}^{\alpha,\beta} = \overline{\mathbb{P}}$. $\square$

The next step is to define rigorously what means the Fourier series of $f \in \mathbf{W}_{\vec{c},p}^{\alpha,\beta}$, where $p \geq 1$, $\mathfrak{m} \in \mathbb{N}$, $\alpha, \beta > -1$ and $\vec{c} = (c_0, c_1, \dots, c_{\mathfrak{m}-1}) \in \mathbb{R}^{\mathfrak{m}}$.

For $f \in \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ we denote

$$\mathcal{F}_n(f, x) := \sum_{k=0}^n \langle f, s_k \rangle_s s_k(x).$$

Under the assumptions of Theorem 4.4.1, $\mathbf{W}_{\vec{c},p}^{\alpha,\beta} = \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ but, in general, we only know that $\mathbf{W}_{\vec{c},p}^{\alpha,\beta} = \overline{\mathbf{U}_{\vec{c},p}^{\alpha,\beta}}$. However, if $f \in \mathbf{W}_{\vec{c},p}^{\alpha,\beta} \setminus \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$, then f can be identified as a Cauchy sequence $\{f_n\}_{n \geq 0} \subset \mathbf{U}_{\vec{c},p}^{\alpha,\beta}$ with norm $\|f\|_{s,p} = \lim_{n \rightarrow \infty} \|f_n\|_{s,p}$.

The existence of this limit is guaranteed since $\{\|f_n\|_{s,p}\}_{n \geq 0}$ is a Cauchy sequence in $\mathbb{R}$. On the other hand, if $\{f_n\}_{n \geq 0}$ is a Cauchy sequence in $\mathbf{W}_{\vec{c},p}^{\alpha,\beta}$ we get

$$\|f_n - f_m\|_{s,p}^p = \sum_{k=0}^{\mathfrak{m}-1} \left| f_n^{(k)}(c_k) - f_m^{(k)}(c_k) \right|^p + \left\| f_n^{(\mathfrak{m})}(x) - f_m^{(\mathfrak{m})}(x) \right\|_{\mathbf{L}^p(\mu^{\alpha,\beta})}^p.$$

Therefore, the sequences $\{f_n^{(k)}(c_k)\}_{n \geq 0}$, $k = 0, 1, \dots, \mathfrak{m}-1$, and $\{f_n^{(\mathfrak{m})}\}_{n \geq 0}$ are also Cauchy sequences on the complete spaces $\mathbb{R}$ and $\mathbf{L}^p(\mu^{\alpha,\beta})$, respectively. Thus, the following limits exist

$$f_{c_k}^k = \lim_{n \rightarrow \infty} f_n^{(k)}(c_k), \quad k = 0, 1, \dots, \mathfrak{m}-1; \quad \text{and} \quad f^{(\mathfrak{m})} = \lim_{n \rightarrow \infty} f_n^{(\mathfrak{m})}.$$

Then, we get

$$\begin{aligned} \|f\|_{s,p}^p &= \lim_{n \rightarrow \infty} \|f_n\|_{s,p}^p = \lim_{n \rightarrow \infty} \sum_{k=0}^{\mathfrak{m}-1} \left| f_n^{(k)}(c_k) \right|^p + \left\| f_n^{(\mathfrak{m})} \right\|_{\mathbf{L}^p(\mu^{\alpha,\beta})}^p \\ &= \sum_{k=0}^{\mathfrak{m}-1} \left| f_{c_k}^k \right|^p + \|f^{(\mathfrak{m})}\|_{\mathbf{L}^p(\mu^{\alpha,\beta})}^p. \end{aligned}$$

From the proof of Theorem 4.4.1, we see that a function f satisfying

$$f^{(k)}(c_k) = f_{c_k}^k, \quad \text{for } k = 0, 1, \dots, \mathfrak{m}-1, \quad \text{and} \quad f^{(\mathfrak{m})} \equiv f^{(\mathfrak{m})} \text{ a.e. on } [-1, 1]$$

does not always exist. But even so, we can understand the value $f_{c_k}^k$ as the k th derivative of the equivalent class f at c_k and the function $f^{(\mathfrak{m})} \in \mathbf{L}^p(\mu^{\alpha,\beta})$ as its $\mathfrak{m}$ -th derivative a.e. on $[-1, 1]$. Then, in a natural way, we can define for every $f \in \mathbf{W}_{\vec{c},p}^{\alpha,\beta}$

$$\langle f, s_n \rangle_s = \sum_{k=0}^{\mathfrak{m}-1} f_{c_k}^k s_n^{(k)}(c_k) + \int_{-1}^1 f^{(\mathfrak{m})}(x) s_n^{(\mathfrak{m})}(x) d\mu^{\alpha,\beta}(x),$$

and now the Fourier series of f makes sense for all $f \in \mathbf{W}_{\vec{c},p}^{\alpha,\beta}$.

From (4.17)-(4.18), it is straightforward to prove that if $f \in \mathbf{W}_{\vec{c},p}^{\alpha,\beta}$, then

$$\langle f, s_n \rangle_s = \begin{cases} f^{(n)}(c_n), & \text{if } 0 \leq n \leq \mathfrak{m}-1, \\ \langle f^{(\mathfrak{m})}, P_{n-\mathfrak{m}}^{(\alpha,\beta)} \rangle_{\mu^{\alpha,\beta}}, & \text{if } \mathfrak{m} \leq n. \end{cases} \quad (4.19)$$

Theorem 4.5.2

Let $\alpha, \beta > -1$,

$$M(\alpha, \beta) = 2 \frac{\max\{\alpha, \beta, -\frac{1}{2}\} + 1}{\max\{\alpha, \beta, -\frac{1}{2}\} + \frac{3}{2}} \quad \text{and} \quad m(\alpha, \beta) = 2 \frac{\max\{\alpha, \beta, -\frac{1}{2}\} + 1}{\max\{\alpha, \beta, -\frac{1}{2}\} + \frac{1}{2}}. \quad (4.20)$$

If $M(\alpha, \beta) < p < m(\alpha, \beta)$, then, for any $f \in \mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$, its Fourier expansion with respect of the polynomials $\{s_n\}_{n \geq 0}$, converges to f in $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$, i.e.

$$\lim_{n \rightarrow \infty} \|f - S_n(f, \cdot)\|_{s, p} = 0.$$

It is not difficult to verify that the definition of $M(\alpha, \beta)$ and $m(\alpha, \beta)$ in (4.20) is the same as in Theorem 4.2.1 (when $\alpha, \beta \geq -1/2$) and in (4.5) (with $a = \alpha/p$ and $b = \beta/p$).

Let $\alpha, \beta > -1$, $\gamma = \max\{\alpha, \beta\}$,

$$M(\gamma) = 2 \frac{\max\{\gamma, -\frac{1}{2}\} + 1}{\max\{\gamma, -\frac{1}{2}\} + \frac{3}{2}} \quad \text{and} \quad m(\gamma) = 2 \frac{\max\{\gamma, -\frac{1}{2}\} + 1}{\max\{\gamma, -\frac{1}{2}\} + \frac{1}{2}}.$$

(γ, p) is said to be a convergence pair if $M(\gamma) < p < m(\gamma)$. In Figure 4.1, the entire shaded region $\mathcal{U}$ (light and dark) is the set of all pairs (γ, p) of convergence. The dark region $\mathcal{U}_0$, at the left, is the set of (γ, p) such that $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta}$ is a space of functions; i.e. $\mathbf{W}_{\vec{c}, p}^{\alpha, \beta} = \mathbf{U}_{\vec{c}, p}^{\alpha, \beta}$, for all $\vec{c} \in \mathbb{R}^m$ (see Corollary 4.4.1.1).

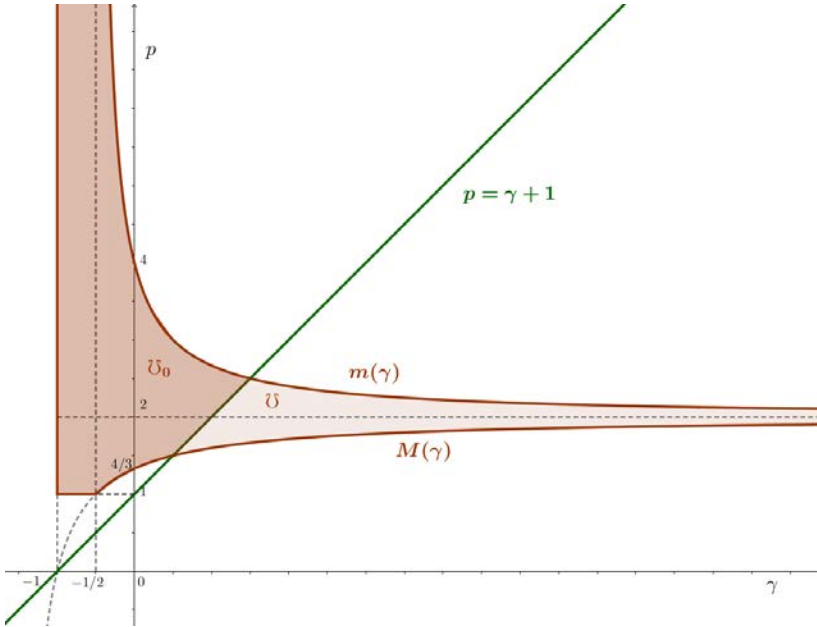


Figure 4.1: The open regions $\mathcal{U} = \{(\gamma, p) : M(\gamma) < p < m(\gamma)\}$ and $\mathcal{U}_0 = \{(\gamma, p) \in \mathcal{U} : p > \gamma + 1\}$.

Proof of Theorem 4.5.2. Given $f \in \mathbf{L}^p(\mu^{\alpha, \beta})$, let $\mathcal{F}_n^{\alpha, \beta}(f)$ be the n th partial sum of the Fourier-Jacobi series of f

$$\mathcal{F}_n^{\alpha, \beta}(f, x) = \sum_{k=0}^n \langle f, p_k^{(\alpha, \beta)} \rangle_{\mu^{\alpha, \beta}} p_k^{(\alpha, \beta)}(x).$$

Now, given $f \in \mathbf{W}_{\tilde{c},p}^{\alpha,\beta}$, from (4.19)

1.- If $0 \leq k \leq \mathfrak{m}-1$, then $\mathcal{F}_n^{(k)}(f, c_k) = \begin{cases} 0, & \text{if } n < k, \\ f^{(k)}(c_k), & \text{if } n \geq k. \end{cases}$

2.- If $n \geq \mathfrak{m}$, then $\mathcal{F}_n^{(\mathfrak{m})}(f) = \mathcal{F}_{n-\mathfrak{m}}^{\alpha,\beta}(f^{(\mathfrak{m})})$.

From Theorem 4.2.2, with $a = \alpha/p$ and $b = \beta/p$, there exists a constant C , independent on n and f , such that

$$\left\| \mathcal{F}_{n-\mathfrak{m}}^{\alpha,\beta}(f^{(\mathfrak{m})}) \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})} \leq C \left\| f^{(\mathfrak{m})} \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})}.$$

We first prove that there exists a constant C_1 , independent on n and f , such that

$$\|\mathcal{F}_n(f)\|_{s,p} \leq C_1 \|f\|_{s,p}, \quad \text{for all } f \in \mathbf{W}_{\tilde{c},p}^{\alpha,\beta}. \quad (4.21)$$

If $n \geq \mathfrak{m}$, then

$$\begin{aligned} \|\mathcal{F}_n(f)\|_{s,p}^p &= \sum_{k=0}^{\mathfrak{m}-1} \left| \mathcal{F}_n^{(k)}(f, c_k) \right|^p + \int_{-1}^1 \left| \mathcal{F}_n^{(\mathfrak{m})}(f, x) \right|^p d\mu^{\alpha,\beta}(x) \\ &= \sum_{k=0}^{\mathfrak{m}-1} \left| f^{(k)}(c_k) \right|^p + \left\| \mathcal{F}_{n-\mathfrak{m}}^{\alpha,\beta}(f^{(\mathfrak{m})}) \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})}^p \\ &\leq \sum_{k=0}^{\mathfrak{m}-1} \left| f^{(k)}(c_k) \right|^p + C^p \left\| f^{(\mathfrak{m})} \right\|_{\mathbf{L}^p(\mu^{\alpha\beta})}^p \leq \max\{1, C^p\} \|f\|_{s,p}^p. \end{aligned}$$

If $0 \leq n \leq \mathfrak{m}-1$, then

$$\begin{aligned} \|\mathcal{F}_n(f)\|_{s,p}^p &= \sum_{k=0}^{\mathfrak{m}-1} \left| \mathcal{F}_n^{(k)}(f, c_k) \right|^p + \int_{-1}^1 \left| \mathcal{F}_n^{(\mathfrak{m})}(f, x) \right|^p d\mu^{\alpha,\beta}(x) \\ &\leq \sum_{k=0}^{\mathfrak{m}-1} \left| f^{(k)}(c_k) \right|^p \leq \|f\|_{s,p}^p. \end{aligned}$$

This yields formula (4.21) with $C_1 = \sqrt[p]{\max\{1, C^p\}}$.

Given $f \in \mathbf{W}_{\tilde{c},p}^{\alpha,\beta}$ and $\varepsilon > 0$, then there exists $P \in \mathbb{P}$, such that $\|f - P\|_{s,p} < \frac{\varepsilon}{C_1+1}$, where C_1 is given by (4.21). Note that for all n greater than or equal to the degree of P we get $P \equiv \mathcal{F}_n(P)$ and, as a consequence,

$$\begin{aligned} \|\mathcal{F}_n(f) - f\|_{s,p} &\leq \|\mathcal{F}_n(f) - P\|_{s,p} + \|P - f\|_{s,p} = \|\mathcal{F}_n(f) - \mathcal{F}_n(P)\|_{s,p} + \|P - f\|_{s,p} \\ &= \|\mathcal{F}_n(f - P)\|_{s,p} + \|P - f\|_{s,p} \leq (C + 1) \|P - f\|_{s,p} < \varepsilon. \end{aligned}$$

Therefore, $\lim_{n \rightarrow \infty} \|f - \mathcal{F}_n(f)\|_{s,p} = 0$. □

Conclusions and Future Research

5.1 Conclusions

Let $p > 1$, $\mathfrak{m} \in \mathbb{N}$, and let $\vec{\mu} = (\mu_0, \mu_1, \dots, \mu_{\mathfrak{m}})$ be a k -standard vectorial measure (see Definition 1.3.1). Given a suitable function f , we define the Sobolev p -norm as

$$\|f\|_{s,p} = \left(\sum_{k=0}^{\ell} \int |f^{(\ell)}(x)|^p d\mu_k(x) \right)^{\frac{1}{p}}. \quad (5.1)$$

We say that $\hat{S}_n$ is the n th extremal polynomial with respect to $\|\cdot\|_{s,p}$, if $\hat{S}_n$ is the unique monic polynomial of degree n , such that

$$\|\hat{S}_n\|_{s,p} = \inf_{P_n(x)=x^n+\dots} \|P_n\|_{s,p}.$$

For $p = 2$, the norm (5.1) can be generated by a Sobolev inner product, as in Definition 1.3.2, i.e. $\|\cdot\|_{s,2} = \sqrt{\langle \cdot, \cdot \rangle_s}$. It is obvious that in this case, the n th extremal polynomial with respect to $\|\cdot\|_{s,2}$ is the n th monic orthogonal polynomial with respect to $\langle \cdot, \cdot \rangle_s$. As it was introduced in Section 1.3, the Sobolev inner products are classified essentially into three classes: Continuous, Discrete and Discrete-continuous, and each of these classes is characterized by the measures involved in them. Here, we studied various related problems of each of those classes, in chapters 2, 3 and 4.

The Sobolev orthogonal polynomials have been extensively studied, in the last three decades, as a natural extension of the standard orthogonal polynomials. For details about this subject, we refer the reader to the surveys [57, 108, 110, 106, 121] and the introductory chapter of this dissertation.

Below, we list the most relevant results that have been achieved from chapter 2 to 4:

Chapter 2. Sobolev Extremal Polynomials with respect to singular measures.

As far as we know, on Sobolev extremal polynomials with respect to the norm (5.1), when not necessarily $p = 2$, the only works are [89, 93]. Both papers, deal with Sobolev norm defined by sequentially dominated vectorial measures (see Section 1.4.3). In the first part of the chapter, we present a general characterization of the extremal polynomials with respect to (5.1), and the second part is dedicated to the study of extremal polynomials with respect to a non-sequentially dominated Sobolev norm.

- 2.1. As it is known, if $p = 2$, $\|\cdot\|_{s,2} = \sqrt{\langle \cdot, \cdot \rangle_s}$, where the inner product $\langle \cdot, \cdot \rangle_s$ is defined in (1.9). From the Pythagorean Theorem (see Theorem 1.2.1), S_n is the n th monic extremal polynomial with respect to (5.1) if and only if $\langle S_n, P \rangle_s = 0$, for all $P \in \mathbb{P}_{n-1}$.

In Theorem 2.2.1 (or Theorem 2.1.2 for $\mathfrak{m} = 1$), we proved that a monic polynomial L_n is the n th extremal polynomial if and only if

$$\langle P, S_n \rangle_{s,p} = 0, \quad \forall P \in \mathbb{P}_{n-1};$$

$$\text{where } \langle P, Q \rangle_{s,p} := \sum_{k=0}^{\mathfrak{m}} \int P^{(k)} \operatorname{sgn}(Q^{(k)}) |Q^{(k)}|^{p-1} d\mu_k, \quad \forall P, Q \in \mathbb{P}.$$

This characterization of the extremal polynomials with respect to the norm (5.1), is an extension of the analogous theorem for L^p spaces (see [20, §2.2 and Ex. 7-h]).

- 2.2. In Theorem 1.2.3, we determined the location of the critical points of the Sobolev extremal polynomials with respect to the norm $\|\cdot\|_{s,p}$, for a mutually singular standard vectorial measure $\vec{\mu} = (\mu_0, \mu_1)$ (i.e. $k = 0$, $\mathfrak{m} = 1$ in Definition 1.3.1 and $\mathbf{C}_h(\operatorname{supp}(\mu_1)) \cap \mathbf{C}_h(\operatorname{supp}(\mu_0))^\circ = \emptyset$). Here, the norm may correspond to the discrete or continuous case. We proved that the zeros of L'_n are simple and contained in the interior of the convex hull of the support of the measures μ_0 and μ_1 . As a consequence, in Corollary 2.3.0.1 we also describe a set that bounds the zeros of L_n . This result extends [92, Th. 1.3] to the extremal case and it was conjectured in [55, Conj. 1 and Conj. 2] for the orthogonal case ($p = 2$) and arbitrary configurations of the supports of the measures μ_0 and μ_1 .
- 2.3. Under some additional conditions of regularity over $\vec{\mu} = (\mu_0, \mu_1)$, which are not so restrictive but it forces us to consider only the continuous case, we also found the n th root asymptotic behavior of the corresponding sequence of derivatives of L_n in Theorem 2.4.1. Finally, we proved the regular asymptotic zero distribution of the derivatives of L_n in Theorem 2.1.3 and of L_n itself in Theorem 2.4.2. The latter one is governed by the balayage of the equilibrium measure of $\operatorname{supp}(\mu_0) \cup \operatorname{supp}(\mu_1)$ onto a certain region which was explicitly described. These results are extensions of [55, Th. 1] and [55, Th. 2] for the extremal case.

Chapter 3. Sobolev-type Orthogonality and Rational Approximation.

In this chapter, we studied the sequence of polynomials orthogonal with respect to the discrete Sobolev inner product

$$\langle f, g \rangle_s = \int f(x)g(x)d\mu(x) + \sum_{j=1}^{\mathcal{N}} \sum_{k=0}^{\ell_j} M_{j,k} f^{(k)}(c_j) g^{(k)}(c_j), \quad (5.2)$$

and its application to rational approximation of a class of holomorphic functions.

- 3.1. In Theorem 3.1.2, we proved that if the inner product $\langle \cdot, \cdot \rangle_s$ is sequentially-ordered (see Definition 3.1.1), the n th orthogonal polynomial has at least $n - \ell^*$ simple zeros on $\mathbf{C}_h(\text{supp}(\mu))^\circ$, where ℓ^* denotes the number of terms ($M_{j,k} \neq 0$) in the discrete part of (5.2). We gave two examples of non sequentially-ordered inner products, for which the theorem is not longer true. This theorem generalizes the following results [6, Prop. 3.2], [34, Th. 4.1] [117, Th. 5.1] [119, Th. 4.1] [3, Th. 2.2] and [68, Prop. 4.11] all of them for the case of only one mass point appears in the inner product (i.e. $\mathcal{N} = 1$).
- 3.2. In (3.7), we introduced the notion of associated polynomials to the orthogonal polynomials with respect to (5.2). Some of their algebraic properties are studied in Section 3.2.
- 3.3. The classical Markov's Theorem is one of the most relevant result in Approximation Theory (see [130, Chap. 2, §6], [134, §34] or [163, §3.5]). This result is closely related to several branches of mathematics, such as the moment problem, the theory of probability, the continued fractions or quadrature formulas, among others. For more details on history and extensions, we refer the reader to the surveys [17, 63] and [88] for the extensions in multipoint Padé approximation. For the Sobolev case, the only extension of Markov's Theorem appears in [96, Th. 5.5], for the particular case of Sobolev discrete inner product with only a derivative of first order in $c_1 = 0$.
Here, we stated the Markov's Theorem for Sobolev polynomials orthogonal with respect to sequentially ordered inner products $\langle \cdot, \cdot \rangle_s$, where the measure μ is supported on $[-1, 1]$, $\mu \in \mathbf{M}(0, 1)$ and the mass points $c_j \notin [-1, 1]$ (see Theorem 3.1.3).

Chapter 4. Discrete-Continuous Jacobi-Sobolev Spaces and Fourier Series.

Let $p \geq 1$, $\mathfrak{m} \in \mathbb{N}$, $\alpha, \beta > -1$ and $\vec{c} = (c_0, c_1, \dots, c_{\mathfrak{m}-1}) \in \mathbb{R}^{\mathfrak{m}}$. Given a suitable function f , following (1.10), we considered the Jacobi-Sobolev discrete-continuous norm

$$\|f\|_{s,p} = \left(\sum_{k=0}^{\ell-1} \left| f^{(k)}(c_k) \right|^p + \int_{-1}^1 \left| f^{(\ell)}(x) \right|^p d\mu^{\alpha,\beta}(x) \right)^{\frac{1}{p}}, \quad (5.3)$$

we have denoted by $U_{\vec{c},p}^{\alpha,\beta}$ the space of functions associated to this norm and the completeness with respect to the norm (5.3) of the linear space of the polynomials by $W_{\vec{c},p}^{\alpha,\beta}$.

- 4.1. We presented a necessary and sufficient condition for $W_{\vec{c},p}^{\alpha,\beta} = U_{\vec{c},p}^{\alpha,\beta}$ (see Theorem 4.4.1 and Corollary 4.4.1.1). As a consequence, we obtained, under these conditions, that $U_{\vec{c},p}^{\alpha,\beta}$ is complete and the vector space of polynomials is a dense subspace. In this sense, the results contained in section 4.4, complement those obtained in [150, 149, 151].
- 4.2. The convergence problem of the Fourier series has been one of the main driving forces behind the mathematical analysis growth. In the Sobolev case of orthogonality, convergence of the partial sums of the Fourier series in $\|\cdot\|_{s,p}$ norm, has been studied for the cases continuous or discrete in [102, 28, 29, 30]. In the discrete-continuous case, the only references are [158, 157, 159], where the authors study problems related to uniform convergence and mixed Jacobi series. In Theorem 4.5.2, we determined the conditions for the convergence of such partial sums in norm (5.3). This theorem, extends the classical result of Pollard and Muckenhoupt for Discrete-Continuous Sobolev norms it and complements those results reached for the continuous and discrete case.

5.2 Proposed problems

Based upon the results of this work, we consider the following problems for future research:

Open Problem 1

Study algebraic and analytic properties of the extremal polynomials with respect to norms of type (5.1) with respect to general classes of measures, in the following cases:

- 1.1. *Sobolev norms with $p = 1$ or $p = \infty$.*
- 1.2. *The norm is non sequentially dominated, with compact support, and under more complex situation than two mutually singular measures (as in Chapter 2).*
- 1.3. *Norms with respect to measures with unbounded support.*

Open Problem 2

Given an inner product as (5.2), in Section 3.2 we introduce the notion of associated polynomials and we study some properties. About these families of polynomials it is interesting:

- 2.1. *Study the properties of their zeros and its possible orthogonality with respect to a certain bilinear form.*
- 2.2. *Establish the connection between these families of polynomials and the space of solutions of difference equation deduced from the $(2\ell + 1)$ -term recurrence relation (3.9).*

Open Problem 3

Extending the Markov's Theorem for discrete Sobolev products, not necessarily sequentially ordered.

Open Problem 4

The Markov Theorem version for a measure with unbounded support, is known as the Stieltjes theorem (see [17, §3]).

- 4.1. *In the proof of Markov's Theorem for Sobolev inner product, the comparative asymptotics (1.21) is fundamental. But, an analogous result for measures with unbounded support, there not exists.*

Assume that, in the inner product (5.2), the support of μ is unbounded and $c_j \in (\mathbb{C} \setminus \text{sup } \mu)$ for $j = 1, \dots, N$. Determine the relative asymptotic behavior between the sequence of orthogonal polynomials with respect to (5.2) and the sequence of orthogonal polynomials with respect to μ .

- 4.2. *Prove an analogous of Markov Theorem for Sobolev orthogonality (Theorem 3.1.3) if the measure μ is supported on $\mathbb{R}_+$ or $\mathbb{R}$, i.e. an extended Stieltjes Theorem.*

Open Problem 5

The results in Chapter 4 and [158, 157, 159], are the starting point to the study of the Fourier expansions and approximation by polynomials in discrete-continuous Sobolev spaces. The orthogonal polynomials with respect to this inner product have been used implicitly in spectral methods for boundary values problems. The error estimate for the spectral method requires to estimate the error of polynomial approximation in the Sobolev space.

- 5.1. *Find these estimates in the discrete continuous case and apply them to problem solving in other fields.*
- 5.2. *To study problems analogous to those discussed in this dissertation, for Fourier series of orthogonal polynomials with respect to Sobolev inner products, non necessarily discrete-continuous.*

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